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Study of the performance of the first silicon modules of the future CMS tracker for HL-LHC

Dissertation presented for the award of the academic degree of
Master [120] in physics

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List of Acronyms

2S	Strip–Strip Module
ASIC	Application-Specific Integrated Circuit
bPOL12V	12V Buck Point-of-Load DC-DC Converter
BSM	Beyond the Standard Model
CBC	CMS Binary Chip
CIC	Concentrator Integrated Circuit
CMN	Common-Mode Noise
CMS	Compact Muon Solenoid
FEH	Front-End Hybrid
HL-LHC	High-Luminosity Large Hadron Collider
L1T	Level-1 Trigger
LHC	Large Hadron Collider
lpGBT	Low-Power Gigabit Transceiver
MPA	Macro Pixel ASIC
POH	Power Optical Hybrid
PS	Pixel–Strip Module
ROH	Readout Optical Hybrid
SEH	Service Hybrid
SPS	Super Proton Synchrotron
SSA	Short Strip ASIC
TB2S	Tracker Barrel 2S
TBPS	Tracker Barrel PS
TEDD	Tracker Endcap Double-Disc
VTRx+	Versatile Transceiver Plus

OT	Outer Tracker
NBD	Negative Binomial Distribution
CERN	European Organization for Nuclear Research
CFRP	Carbon Fiber Reinforced Polymer
LINAC	LINear ACcelerator
MIP	Minimum Ionizing Particle
RMS	Root Mean Squared
DAQ	Data Acquisition System
ECAL	Electromagnetic Calorimeter
HCAL	Hadronic Calorimeter

Chapter 1

Introduction

1.1 Physics motivation

The Large Hadron Collider (LHC) is a particle accelerator that speeds up protons in a ring to very high energies by using high magnetic fields to bend the particle beam and electric fields to accelerate it. During Run 3, proton-proton collisions at the LHC are operated at a centre-of-mass energy of 13.6 TeV [1] [2]. Before injection into the LHC, protons are accelerated by linear accelerators (LINACs) and the Super Proton Synchrotron (SPS), reaching an energy of 450 GeV [1].

The LHC's greatest discovery so far is the Higgs boson in 2012. Nowadays, physicists have only measured a few of its properties with limited accuracy. To get a further degree of precision about Higgs boson physics, the High-Luminosity Large Hadron Collider (HL-LHC) is being built to maximize the physics potential of the LHC by delivering an order of magnitude more data than the original design (about 3,000 fb^{-1} of integrated luminosity over 10 years [3] [2]), allowing several notable advancements :

- The measurement of Higgs couplings to fermions and bosons with a 1 to 2% precision in constrast with 10% today [4].
- Tests of whether the Higgs mechanism behaves exactly as predicted by the Standard Model.

One of the most important objectives of the HL-LHC physics programme is the first direct measurement of Higgs self-coupling via Higgs pair production (HH) [3]. This process is substantially rarer than single Higgs boson production, hence the need for a large amount of data. The Higgs self-coupling is important to measure precisely because it determines the shape of the Higgs potential which is crucial for understanding electroweak symmetry breaking and the stability of the universe.

Moreover, the HL-LHC will enable searches for physics beyond the Standard Model (BSM) by probing for deviations in fundamental coupling strengths. Indeed, many BSM scenarios predict rare decays or high-mass particles (e.g supersymmetry, dark matter mediators or heavy Z bosons) which are only accessible at a higher luminosity. This increase in luminosity leads to more collisions which in turn gives a better statistical reach for such rare events.

The luminosity for a collider experiment is given by:

$$L = f\frac{N_1 N_2}{4\pi\sigma_x\sigma_y} \tag{1.1}$$

where N_1 and N_2 are the number of particles per bunch in each beam, f is the bunch-collision frequency and σ_x, σ_y are the RMS beam sizes in horizontal and vertical directions at the interaction point [5]. An interesting quantity to obtain from the luminosity is the event rate $= L \cdot \sigma$, with σ being the cross section of a specific process. Therefore, one can change L to get higher event rate but σ is given by nature.

HL-LHC makes several improvements in order to increase the luminosity. Firstly, equation 1.1 shows that the number of protons per bunch and the number of bunches per beam both affect the luminosity. So, the HL-LHC will use beams with more protons per bunch which is achieved through upgrades to the injector chain. Secondly, to focus the beams more tightly at the collision points and decrease σ_x and σ_y, HL-LHC will utilize new Nb_3Sn superconducting quadrupole magnets near Atlas and CMS [1]. This type of magnet provides a higher magnetic field strength and have a larger aperture allowing smaller beam size which leads to a higher collision probability per bunch crossing and an increase in luminosity as shown in equation 1.1. In the end, the HL-LHC will reach instantaneous peak luminosities of $5 \times 10^{34} cm^{-2} s^{-1}$ with center of mass energy of 14 Tev.

From a technical point of view, current LHC detectors (e.g CMS tracker) and accelerators will reach their radiation and performance limits by the late 2020s. The HL upgrade which includes better magnets, higher beam luminosity and improved detectors ensures that the LHC remains up and running for years to come.

1.2 Scope and objectives

One of the endcaps of the upgraded CMS Outer Tracker is being assembled at UCLouvain and is scheduled for delivery to CERN in 2028 [1] as part of the CMS Phase-2 upgrade programme. During this project, I investigated the common-mode noise observed in the silicon modules that will be installed in the tracker endcap. The main objective of this thesis is to characterize the effect of common-mode noise on the two Outer Tracker module technologies: the Strip-Strip (2S) modules and the Pixel-Strip (PS) modules. Measurements were performed in two experimental configurations. First, individual modules were characterized on the burn-in test bench. These measurements provide a reference for identifying noise components associated with the module itself and its local services. Second, modules were measured after integration on a dee structure during the sector tests. In this configuration, several modules share mechanical, electrical and cooling services, more closely reproducing the conditions expected in the final tracker endcap. The analysis aims to quantify common-mode effects in the strip and macro-pixel sensors by studying event-level occupancy distributions. The measured distributions are compared with statistical models incorporating independent channel noise, common-mode fluctuations and correlations between front-end components of the modules.

Chapter 2 introduces the CMS detector and the operating principles of silicon sensors. Chapter 3 goes deeper into the CMS detector and describes the mechanical and electronic architecture of the 2S and PS modules. Then, chapter 4 presents the experimental setup, calibration procedure and statistical framework. Chapter 5 reports the results obtained for burn-in and sector measurements. Finally, chapter 6 summarizes the main findings, discusses the limitations of the analysis and identifies possible follow-up measurements.

Chapter 2

CMS description and silicon sensors

2.1 CMS detector description and upgrade overview

At the LHC, there are four different detectors placed at each interaction points including the famous Compact Muon Solenoid (CMS). CMS is built like an onion of concentric detector layers around the interaction point where each layer measures different properties of the particles that emerge from a collision as represented in Figure 2.1.

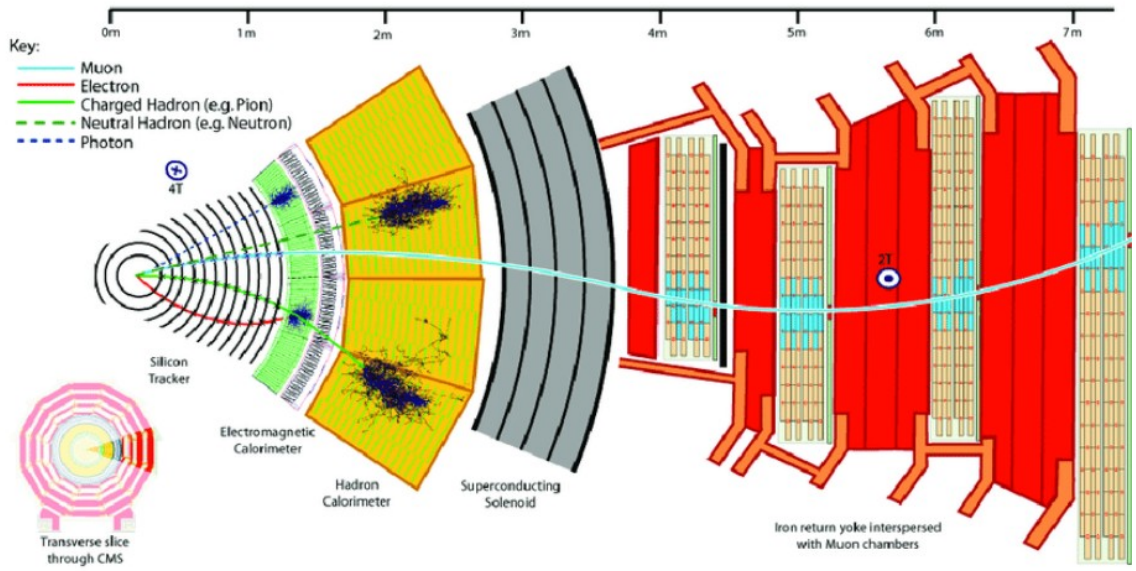


Figure 2.1: Transverse slice view through the CMS detector [6].

2.1.1 Silicon tracker

The silicon tracker is made of pixel and strip sensors which measures charged particle trajectories within 10μm [1]. It makes use of the electron-hole pair resulting from the ionization of a passing charged particle. The Bethe-Block formula describes the average energy loss only by ionization of a charged particle per unit distance as it travels through the silicon:

$$-\left\langle \frac{dE}{dx} \right\rangle = Kz^2 \frac{Z}{A} \frac{1}{\beta^2} \left[\frac{1}{2} \ln \left(\frac{2m_e c^2 \beta^2 \gamma^2 T_{\max}}{I^2} \right) - \beta^2 - \frac{\delta(\beta\gamma)}{2} \right] \tag{2.1}$$

with $K = 4\pi N_A r_e^2 m_e c^2$, z the charge of the incident particle, $T_{\max}$ the maximum kinetic energy that can be transmitted to a free electron in a single collision, I the mean excitation energy, Z the atomic number, A the atomic mass, N_A Avogadro's number, m_e the electron mass, c the speed of light, r_e the classical electron radius, and δ the density–effect correction, $\beta = v/c$ where v is the speed of the incident particle and $\gamma = \frac{1}{\sqrt{1-\beta^2}}$ [7]. This formula can be represented graphically as in Figure

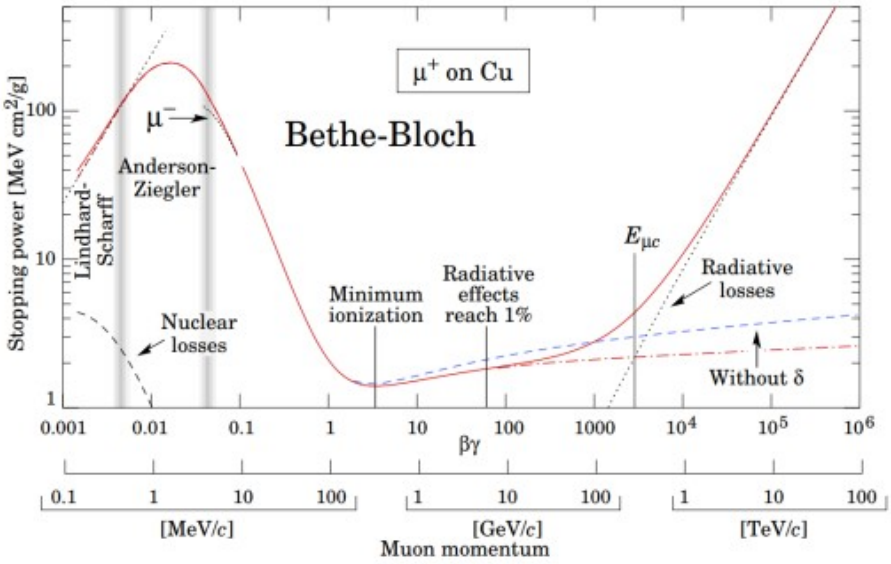


Figure 2.2: The stopping power $\langle dE/dX \rangle$ of copper for penetrating muons depending on the muon momentum [8].

2.2. The Bethe formula is valid in the range $0.1 < \beta\gamma < 1000$, as it only accounts for energy losses due to ionization and excitation of bound electrons, which lead to the emission of low-energy photons following atomic de-excitation. For $\beta\gamma < 0.1$, the electrons can no longer be considered stationary, while for $\beta\gamma > 1000$, radiative corrections can no longer be neglected [7]. Also, the minimum of the curve at approximately $\beta\gamma \simeq 3$ corresponds to the minimum energy that a particle can deposit in a medium. Particles at this energy are known as minimum ionizing particles (MIPs). In order to be able to detect the passage of such particles, the detector noise must be kept well below this energy.

Once the particles have lost their energy in the different layers of the silicon tracker, a signal is generated corresponding to the position measurement and physicists can identify the points compatible with the targeted particle trajectory by track finding (Figure 2.4). To deduce the particle track from the detector's measurements, one can use the fact that the particle's velocity component is perpendicular to the magnetic field $\mathbf{B}$ which forces the particle to follow a three-dimensional helical path (Figure 2.3). In the transverse plane, this projects as a circular motion described by [9]:

$$\mathbf{F} = q\mathbf{v_T} \times \mathbf{B} = \frac{m\mathbf{v_T}^2}{R} \tag{2.2}$$

Because of the perpendicularity between $\mathbf{v_T}$ and $\mathbf{B}$ in the CMS, one have :

$$F = qv_T \cdot B = \frac{mv_T^2}{R} \rightarrow p_T = qB \cdot R \xrightarrow{\text{in convenient units}} p_T[Gev/c] = 0.3B[T] \cdot R[m] \tag{2.3}$$

It measures the transverse momentum p_T by using the radius of curvature R. In other words, one can deduce the transverse momentum p_T from the bending of the trajectory. The greater the curvature of the trajectory, the lower the particle's energy.

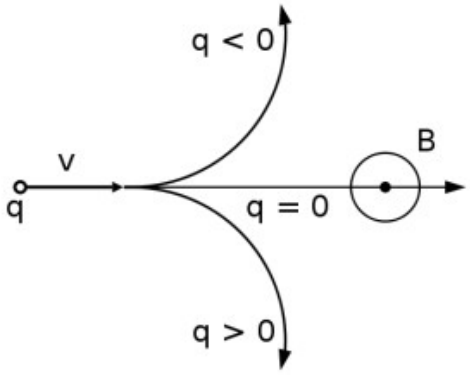


Figure 2.3: Motion of a charged particle under the influence of a perpendicular magnetic field [10].

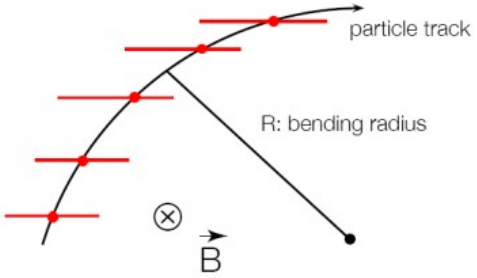


Figure 2.4: Red layers of position sensitive silicon detectors are placed inside the magnetic field region to measure the trajectory of the incoming particle [11].

2.1.2 Calorimeters

Calorimeters measure particle energy by stopping incident particles and absorbing their secondary shower products (Figure 2.5). To prevent energy leakage and maintain resolution, calorimeters are bulky and highly hermetic. Unlike silicon trackers, calorimeter relative energy resolution improves with increasing energy, parameterized as:

$$\frac{\sigma_E}{E} = \frac{a}{\sqrt{E}} \oplus \frac{b}{E} \oplus c, \tag{2.4}$$

where $\oplus$ represents quadratic addition of the stochastic term (a), noise term (b), and constant instrumental term (c) [9].

Electromagnetic Calorimeter (ECAL)

The ECAL measures electrons and photons using a homogeneous detector design (e.g., $PbWO_4$ scintillating crystals), where the same material acts as both absorber and active medium. High-energy electromagnetic showers are driven by alternating bremsstrahlung ($e \to e\gamma$) and pair production ($\gamma \to e^+e^-$) [9]. This cascade continues until particle energies drop below the critical energy E_C, where ionization dominates and the shower dissipates.

Hadronic Calorimeter (HCAL)

Particles penetrating the ECAL reach the HCAL, which measures hadronic jets using a sampling design with alternating absorber and active layers. Hadronic showers develop through inelastic nuclear collisions, producing secondary hadrons and invisible energy losses. To account for these fluctuations in cascade compositions, compensation techniques are used to balance the detector response between the electromagnetic and purely hadronic components of the shower [9].

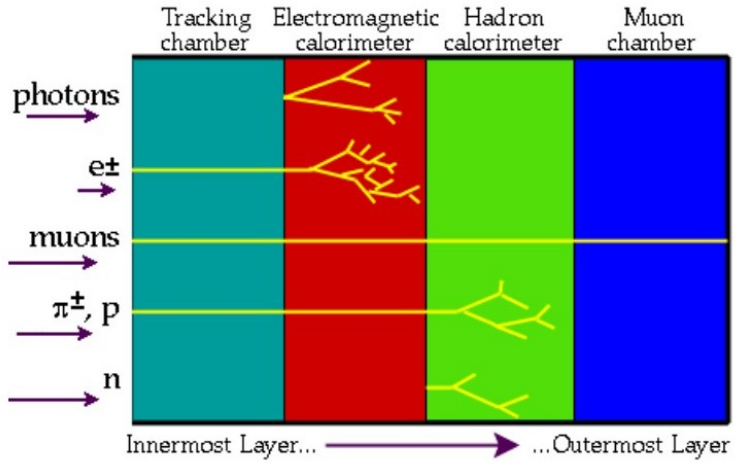


Figure 2.5: A generic particle physics detector layout. The arrangement illustrates the response of successive detector subsystems (from the innermost tracking chamber to the outermost muon chamber) to various particle species. Photons and electrons are absorbed in the electromagnetic calorimeter, charged hadrons and protons interact primarily in the hadron calorimeter, neutrons penetrate further before showering in the hadron calorimeter, and high-energy muons pass through all inner layers to be measured in the muon chambers [12].

2.1.3 Superconducting solenoid

The superconducting solenoid provides a strong magnetic field (3.8T) to bend charged-particle tracks so that their momentum can be measured by the silicon tracker. Such a powerful magnetic field can be generated thanks to the zero resistance property of a superconducting material [9]. The current inside the solenoid of CMS can then flow endlessly without any resistance and produce a high magnetic field thanks to Ampere's law.

2.1.4 Muon chambers

Muons traverse the silicon tracker and calorimeters while typically depositing only a limited amount of energy. Because most muons are not stopped in these inner detector systems, dedicated muon chambers are installed in the outermost part of CMS. Four types of detectors are employed: drift tubes, cathode strip chambers, resistive plate chambers and gas electron multiplier chambers [9]. All four detectors operate with gas where passing muons ionize the gas and release free electrons. The amplification and collection of these electrons depend on the specific detector technology. The track of each particle is reconstructed by fitting a curve to the signals recorded in the four muon stations, allowing the determination of the momentum of muons.

2.1.5 Trigger and data acquisition system (DAQ)

At the LHC nowadays, proton bunches collide every 25 ns so that the frequency of collision f is 40 MHz [1]. Each event produces data of the order of 1 MB [13]. Therefore, every event produced by collisions cannot be store because the dead

time of the acquisition system is determined by computer which can only treat 100 MB/s. It means that the storage capacity is only 100 events out of the 40000000 every second. Consequently, data from the CMS detector is routed through the trigger system, which determines whether an event is of sufficient physics interest to be retained. Selected events are then forwarded to the Data Acquisition (DAQ) system for assembly and storage.

Trigger

In order to reduce the amount of data that is to be stored, it is essential to have a multilevel trigger that has the purpose of conserving all the interesting events. It is composed of the hardware-based level-1 trigger which apply loose selection cuts to reduce the output rate to 100kHz (current, pre-upgrade figure; see chapter 3 for the HL-LHC target) because it has to treat all the events produced within a few micro seconds [13]. The subsequent software-based high level trigger (HLT) is a lot more selective in its cuts because it has a lot more time to run. In the end of this triggers loop, the final output rate amounts to 100 Hz [13]. Also, an analog pipeline is positioned just before the DAQ system which allows to hold the data while waiting for the response of the level-1 trigger. The trigger selection logic must be synchronized with the event data stream emerging from the analog pipeline.

Data acquisition system (DAQ)

The DAQ system has several tasks :

- Readout detector electronics : collects and digitizes data fragments for each accepted event.
- Event Builder : switches combines event fragments from all sub-detectors into a complete “event record".
- Event filtering : uses HLT software farms to receive built events and apply final software trigger to make the final event selection decisions.
- Storage : sends accepted events to permanent storage.

2.1.6 CMS tracker upgrade overview

All the detectors described until now needs to be upgraded to compensate for the increase in luminosity of the upcoming HL-LHC. The CMS phase 2 upgrade, happening between 2026 and 2030, will be dedicated to that purpose. Specifically, the inner and Outer Tracker (OT) will be the most prone to improvement since they are located near the beam crossing points of the LHC where radiation damages are the strongest. Indeed, the growth in luminosity leads to an increase of particle density meaning that pileup will increase as well up to 200 collisions per bunch crossing on average [1]. While more collisions mean more opportunities to study rare physics processes, they also create an extremely chaotic environment. As a result, the detector must distinguish particles originating from hundreds of overlapping collisions. This requires unprecedented precision in tracking performance which is why CMS is upgrading its silicon tracker system.

2.2 Basic principle of silicon sensors

Silicon is an indirect semiconductor that acts as an insulator at low temperatures and shows a measurable conductivity at higher temperature. Semiconductors are widely used in collider experiments as silicon microstrip or silicon pixel detectors near the primary interaction point. This is the case because silicon sensors offer a very good intrinsic resolution compared to gaseous detectors. Indeed, for every 3.6 eV released by a particle crossing the medium, one electron-hole pair is produced whereas about 30eV is required to ionize a gas molecule for gaseous detectors [14]. Also, silicon sensors are prone to implement in HEP because of the high density of silicon. As a matter of fact, the average energy loss of a minimum ionizing particle in silicon is 390eV/μm thus creating 108 e-h pairs/μm for 300μm thick silicon [14] [1]. It is especially interesting for its favorable energy band gap and the possibility to engineer the gap properties by adding impurities inside the material. This last process is called doping.

2.2.1 Fundamental properties of silicon

As for every semiconductor, silicon has a forbidden region in its energy band structure rendering impossible the transition of electrons and holes to states in this region. At low temperature and without any impurities, electrons are stuck in the valence band and the conduction band is empty. If the temperature goes higher or an ionizing particle is penetrating silicon, an electron can get excited to the conduction band leaving a hole behind them in the valence band. This phenomenon is called electron-hole pair generation. Then the excited electron inevitably transition back to the lowest energy state and recombine with the hole in the valence band by emitting radiations. This process is called recombination. The band gap of silicon is an indirect bandgap which means that the lowest conduction band energy does not lie at the same position in k-space as the highest valence band energy. Therefore, recombination happens by emitting not only a photon of energy E_{gap} =1.12eV but also a phonon to preserve the energy and momentum conservation laws [14]. However, in a real device, the recombination rate is completely dominated by the additional levels introduced in the band gap by either the doping of the material or the natural crystal defects. Consequently, the lifetime τ_L of a excited state is completely determined by the impurities, dopant atoms or crystal dislocation :

$$\tau_L = \frac{1}{\sigma v_{th} N_t} \tag{2.5}$$

with v_{th} the thermal velocity ($\approx 10^7\,\mathrm{cm/s}$), the charge carrier cross-section σ ($\approx 10^{-15}\,\mathrm{cm}^2$), and impurity/trap concentration N_t [14]. Therefore, recombination is very unlikely in ideal semiconductors without impurities meaning that direct band-to-band (EC $\Leftrightarrow$ EV) generation or recombination is technically negligible and thus τ_L can range up to milliseconds. The role played by the impurities, acting not only as traps but also as generation centers for electron–hole pairs are described by the Shockley–Read–Hall theory [15].

Only a very small number of electrons are thermally excited from the valence to the conduction band meaning that intrinsic silicon is a poor conductor at room

[1] However, because the energy-loss distribution in a thin sensor is asymmetric, the most probable deposited charge is smaller and is commonly approximated by about 80 e-h pairs/μm.

temperature. Therefore, it is essential to enhance the electrical property of silicon for its usage in all modern electronic devices. This can be done through doping, a process that introduces impurity-induced energy states within the band gap that modify the carrier concentration. Indeed, these states enhance the probability of promoting electrons or generating holes. n-type silicon is obtained by incorporating type V dopants such as arsenic. These atoms substitute for silicon in the lattice and form four covalent bonds, analogous to Si, while their fifth valence electron is only weakly bound. As a result, they create a donor energy level located just below the conduction band (see Figure 2.6). Conversely, p-type silicon is produced by adding type III dopants (e.g. boron, aluminum or gallium). When these atoms replace a silicon atom, they provide only three valence electrons for bonding resulting in the creation of a hole in the lattice. This leads to the formation of an acceptor energy level situated slightly above the valence band (see Figure 2.7).

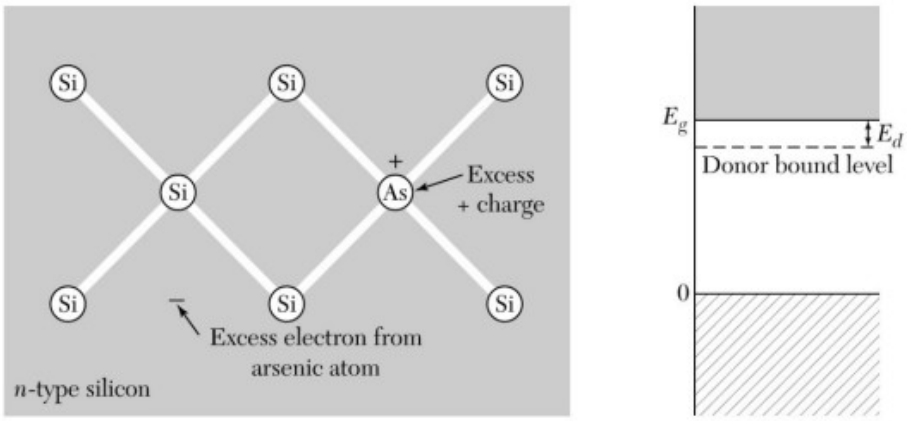


Figure 2.6: Atomic lattice representation and electronic band scheme of n-type silicon. Left: A silicon crystal lattice in which a tetravalent silicon atom is replaced by a pentavalent arsenic (As) dopant, introducing an excess electron and a positive fixed donor ion. Right: The corresponding energy band diagram showing the discrete donor bound level located at an energy offset E_d just below the conduction band edge within the band gap E_g [16].

The conductivity and resistivity of doped silicon materials is defined as :

$$\sigma = e(\mu_e N_d + \mu_h N_a) \quad \text{and} \quad \rho = \frac{1}{\sigma} \tag{2.6}$$

where e is the electron charge, μ_e and μ_h are the mobilities of electrons and holes, respectively, with mobility $\mu = \dfrac{v_D}{E}$ (E the electric field, v_D the drift velocity), N_d and N_a are the densities of electrons and holes respectively [14]. By referring to equation 2.6, it is now obvious that the doping of silicon directly increases the densities of electrons or holes leading to a rise of the conductivity. Microscopically, mobility μ is given by :

$$\mu_{e,h} = \frac{e\,\tau_s}{m^*_{e,h}} \tag{2.7}$$

with $m^*_{e,h}$ the effective mass of the electron/hole and τ_s the time between two scattering processes encompassing collision with crystal defect, intentionally introduced impurities and phonons [14]. The mobilities in silicon are quite high resulting in

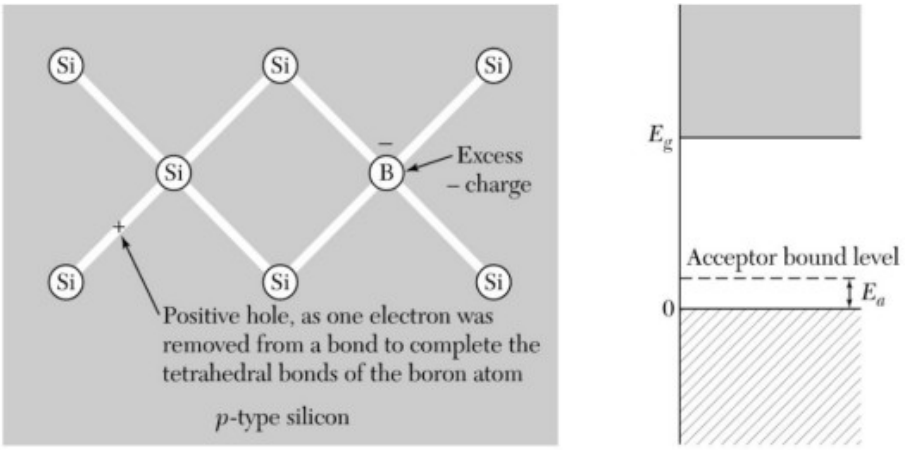


Figure 2.7: Atomic lattice representation and electronic band scheme of p-type silicon. Left: A silicon lattice doped with a trivalent boron (B) atom, which creates a localized negative acceptor charge and a mobile positive hole by capturing an electron to complete its covalent bonding. Right: The corresponding energy band diagram illustrating the acceptor bound level positioned at an energy offset E_a just above the valence band edge [16].

a 10 ns collection/readout time in silicon sensors of several 100μm thickness which explains why silicon is well suited for its use as HEP detectors [14].

An additional important feature in an energy band diagram is the Fermi energy E_F and it can be regarded as the energy where exactly half of the available levels are occupied. Therefore, the Fermi level is shifted depending on the impurities since they add levels near the conduction or valence band. According to the principle of detailed balance, the concentrations of free electrons in the conduction band and free holes in the valence band have to be constant in thermal equilibrium [15] and therefore the Fermi energy must be constant everywhere as well. The value of E_F is very important whenever a contact between two materials exists because it defines the relative levels of all the other bands. This property is clearly observed in p-n junction as will be shown in the following section.

2.2.2 p-n junction

If only intrinsic silicon substrates were used in sensors, the resulting signal would be lost in the number of free charge carriers because there would be only about $\sim 2 \cdot 10^4$ generated e-h pairs induced by ionizing particles out of the $\sim 10^9$ free charge carriers [14]; hence the need to deplete the silicon volume of free charge carriers. This can be achieved using p and n type silicon in a reverse biased p-n junction configuration, therefore exploiting the basic diode properties.

A p–n junction is formed by bringing into contact two semiconductor regions with different doping types : a p-type region, containing a high concentration of holes as majority carriers, and a n-type region, containing a high concentration of electrons as majority carriers. When the p and n type semiconductors are joined, free majority charge carriers begin to diffuse across the junction to the opposite regions leaving behind ionized donor atoms on the n-side and ionized acceptor atoms on the p-side. This movement of charge carriers results from the property that the Fermi energy must stay constant in thermal equilibrium resulting in $E_{fn} = E_{fp} = E_F$ (Figure

2.8). Therefore, electrons diffuse from the n-side toward the p-side and holes diffuse from the p-side toward the n-side. The diffusion current $j_{diffusion}$ is described by Fick's first law :

$$j_{diffusion} = -eD\nabla n \tag{2.8}$$

equalizing the carrier density according to its gradient ∇n with the diffusion constant D [14]. As these carriers cross the interface, they recombine with majority carriers on the opposite side. Since the remnant dopant atoms are fixed in the crystal lattice, they generate a region depleted of mobile charge carriers, known as the depletion region [15]. This region contains positively charged donor ions $+qN_d$ on the n-side and negatively charged acceptor ions $-qN_a$ on the p-side. Therefore, spatial separation of charges produces an internal electric field E, pointing from the n-side toward the p-side. As the electric field grows, it establishes a built-in potential V_{bi}, which eventually becomes large enough to prevent further diffusion of electrons and holes. As a result a dynamic equilibrium is created : the drift current caused by the internal electric field compensates the diffusion current at the p-n junction [15].

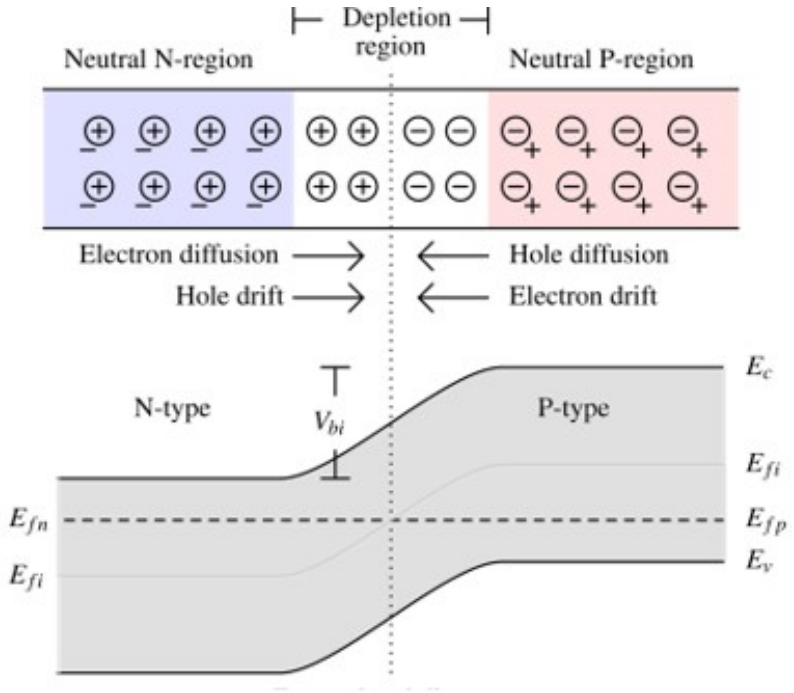


Figure 2.8: Space charge distribution and energy band diagram of a p-n junction in thermal equilibrium. Top: Spatial arrangement of ionized donors and acceptors across the neutral n-region, the central depletion region, and the neutral p-region, alongside the directions of carrier drift and diffusion. Bottom: Corresponding energy band profile showing the conduction band (E_c), valence band (E_v), intrinsic Fermi levels (E_{fi}), quasi-Fermi levels (E_{fn}, E_{fp}), and the built-in potential barrier V_{bi} [15].

For now, the whole system is completely defined by the energy barriers which are fully defined by the doping concentrations. However, there is a limitation in the length of the depletion region because V_{bi} is on the order of a few to some hundreds of millivolts resulting in a depletion region of some tenths of micrometers. Therefore, an additional technique is needed to increase the depleted volume. This approach involves putting the p-n junction in a reverse bias mode [14]. When an external voltage V_{ext} is applied such that the n-side is connected to the positive terminal,

the junction becomes reversed biased. This external voltage increases the intrinsic potential barrier as :

$$V = V_{\text{bi}} + |V_{\text{ext}}| \tag{2.9}$$

As a result, in this reverse bias regime, the depletion region widens and majority carrier injection is suppressed. Only a very small reverse saturation current flows due to minority carriers.

2.2.3 Silicon detectors

Silicon detectors are the primary technology used for tracking charged particles in modern high-energy physics experiments. Their purpose is to measure the position, momentum and sometimes the energy loss of particles produced in collisions. They operate in reverse bias mode and are placed very close to the interaction point, where particle densities and radiation levels are extreme. They operate in reverse bias mode because the depletion region acts as an active volume in which e-h pairs generation by an incoming ionizing particle can generate signals. Indeed, an ionizing particle crossing the silicon loses energy mainly by ionization, producing an electron–hole pairs ($\sim$80 pairs/μm for silicon) that can be collected at the electrodes [14]. Under the electric field, electrons drift toward the anode and holes drift toward the cathode. This motion induces a measurable current pulse on readout electrodes. The typical signal size is a few femto-coulombs and still has to be amplified.

It is important to have a depletion region as large as possible because charge carriers created in it can be collected at the junction inducing a signal, while charge carriers created in the non-depleted zone recombine with free majority carriers or with the generation partner and the signal is lost. One can understand how to increase the length of the depletion region w through equation 2.10.

$$w = \sqrt{2\epsilon\mu\rho V_{\text{ext}}} \tag{2.10}$$

with ϵ the permittivity and ρ the electrical resistivity of the semiconductor. V_{ext} has to be sufficiently strong compared to V_{bi} such that the full volume is depleted [15] [14]. In order to know the value of V_{ext} which would deplete the detector of all the charge carriers, one can use equation 2.10 and replace w by the full sensor thickness L ($200\mu m$ for the cms tracker) giving:

$$V_{\text{FD}} = \frac{L^2}{2\epsilon\mu\rho} \tag{2.11}$$

The full-depletion voltage V_{FD} is the minimum reverse-bias voltage required for the depletion region to extend through the full sensor thickness. It is distinct from the breakdown voltage, which defines the upper operating limit. To run over-depleted the operation voltage V_{ext} is set higher than V_{FD} while maintaining an adequate safety margin below breakdown. Indeed, it is essential to be in an overdepletion regime since V_{FD} increases over time because of radiation damage which induce further impurity level in the band gap.

As mentioned earlier, impurities in the band gap enable Shockley–Read–Hall (SRH) generation of carriers which is the main source of emission/transition in silicon detectors [14]. These transitions create a current called leakage current I_{leak} even when no particle is detected. Leakage current is undesirable because of several

factors. First, it increases the electronic noise which reduces signal-to-noise ratio (SNR) and limits the minimum detectable energy. As a result, higher noise makes it harder to distinguish a real particle signal from fluctuations. Second, leakage current contributes to additional power dissipation as follows [14]:

$$P = I_{\text{leak}} \cdot V_{\text{ext}} \tag{2.12}$$

This causes heating of the detector modules which increases the demand on the cooling system. Moreover, excess heating increases the rate of transition through phonons between energy levels in silicon which in turn increases leakage even further therefore creating a positive feedback loop. Radiations also contribute to this feedback loop by creating defects in silicon that increase SRH trap density which again increase leakage current exponentially. Finally, a high leakage current restricts how much reverse bias voltage can be applied before breakdown. Indeed, without sufficient bias, the sensor may not fully deplete consequently leading to a lower charge collection efficiency and a reduced energy resolution. Therefore, leakage current must be controlled to prevent detector failure.

In experiments carried out at the LHC, leakage current grows with radiation dose and temperature and once it exceeds safe operational thresholds, modules must be replaced. Consequently, leakage current directly determines the detector lifetime which will come to an end by the late 2020s for the LHC; hence the craving of physicists to carry out the HL upgrade as soon as possible.

Isolation

This short section puts forward two types of electrical isolation performed on detectors: the SiO_2 layer and p-type implant structures. These techniques prevent the formation of inversion channels at the Si-SiO_2 interface, ensuring proper isolation between adjacent electrodes and low leakage current.

First, an additional a thin layer of SiO_2, which serves as an insulator, has to be implemented on the silicon sensors because pure silicon is too chemically reactive to be used singly in detectors [17]. Another benefit of the Si-SiO_2 interface is its excellent electrical properties. Indeed, this interface has a very low density of states in the band gap meaning that it does not introduce much recombination centers. Therefore, the surface of the detector follows the classical MOS structure which consists of a highly doped silicon strip covered by a SiO_2 insulation layer with metal strips on top to be connected to the readout electronics, called the AC-coupling [1]. This capacitive coupling facilitates the use of charge amplifying chip.

Second, it is essential to be in a surface depletion regime which can be achieved by using p-type implant structures (p-stop, p-spray) or applying a high enough reverse bias voltage that it extends the depletion region to the surface. Indeed, depletion must extend not only through the bulk but also across the surface where electrodes and oxide interfaces exist. At a Si-SiO_2 interface, undesired fixed positive oxide charges accumulate which attract electrons at the silicon surface, forming an inversion layer. If this layer is not depleted, it acts like an unwanted conductive sheet that connects strips/pixels electrically which can easily decrease the inter-electrode resistance by several orders of magnitude [14] [18]. Without proper surface depletion between pixels/strips, charges diffuse between electrodes and noise increases dramatically.

Working principle of a silicon tracking device

The silicon outer tracker used for CMS is a n-in-p detector as represented in Figure 2.9. For this type of sensor, ionizing particles goes through the p^{++} backplane and penetrate through a fully depleted silicon p-doped slice where they create e-h pairs in the p-bulk silicon material. The + refers to the amount of doping ; the bulk has the lowest doping, the n^+ strips have a higher doping and the p^{++} back plane has the highest doping concentration. Then, the generated electrons and holes drift along the electric field, created by the bias voltage V_{FD}, to the n^+ doped strips, capacitively coupled to the Al readout strips, and the p^{++} back plane respectively [14]. The Al readout strips are DC connected to the charge amplifier of the readout chip and then the custom ASIC amplifies and shapes the signal. Additionally, the implantation of p^+-stop is necessary in order to isolate the individual strips. If that was not the case, the n^+ strips would be short-circuited by the electron accumulation layer created by the attraction of positive fixed oxide charge on the SiO_2 and Si-SiO_2 interface. Finally, the position is reconstructed from which n^+ strips collect charge and how much charge each strip collects.

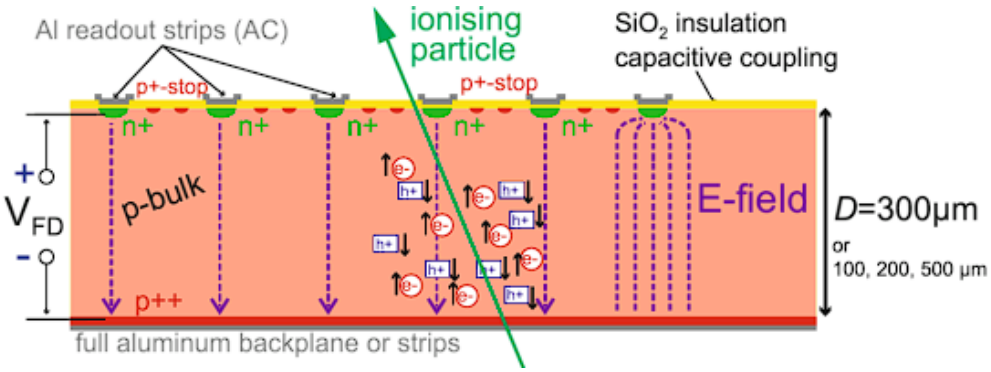


Figure 2.9: Working principle of an n-in-p AC-coupled silicon strip detector. The thickness D is equal to 200 μ m for the modules of the CMS phase-2 upgrade [14].

An important parameter to optimize for the design of those strip detectors is the distance between two adjacent strips called the strip pitch. Indeed, a signal spreading over lots of strips could lead to a loss of resolution. So a smaller strip pitch leads to a better resolution. However, in reality, a charge created by a particle also diffuses in the silicon and spreads over multiple strips. The position can then be estimated from the relative charge collected by neighboring strips. The resulting resolution generally improves with increasing signal-to-noise ratio [14].

Additionally, three different rings with a specific purpose are needed around silicon sensors: the bias ring, the guard ring and the edge ring. The bias ring is used to apply the operating voltage to the detector. The negative high voltage is applied to the backplane while the bias ring is held at ground potential. Each strip is connected to the bias ring through a bias resistor. The bias ring ensures that all strips are at the same electrical potential by providing homogeneous biasing across the sensor. Therefore, it prevents variations in strip response caused by potential differences which would degrade the detector performance [14]. Then, the guard ring surrounds the active area of the sensor and is placed just outside the outermost strips. Its purpose is to ensure a smooth and uniform electric field near the edges in order to minimize edge effects, such as field distortions. As a result, this ring

helps maintaining uniform charge collection and avoids premature breakdown at the edges. Lastly, crystal defects can occur at the edges if the sensor gets damaged. Indeed, this damage introduces energy levels in the forbidden band gap which act as recombination centers and generate leakage currents, especially under high electric fields. To avoid this issue, a highly doped p^{++} silicon implant located at the very periphery of the sensor, called the edge ring, is used to fix the electric potential at the sensor edge. Therefore, it prevents high electric fields from extending into the damaged region thereby strongly reducing leakage currents [19].

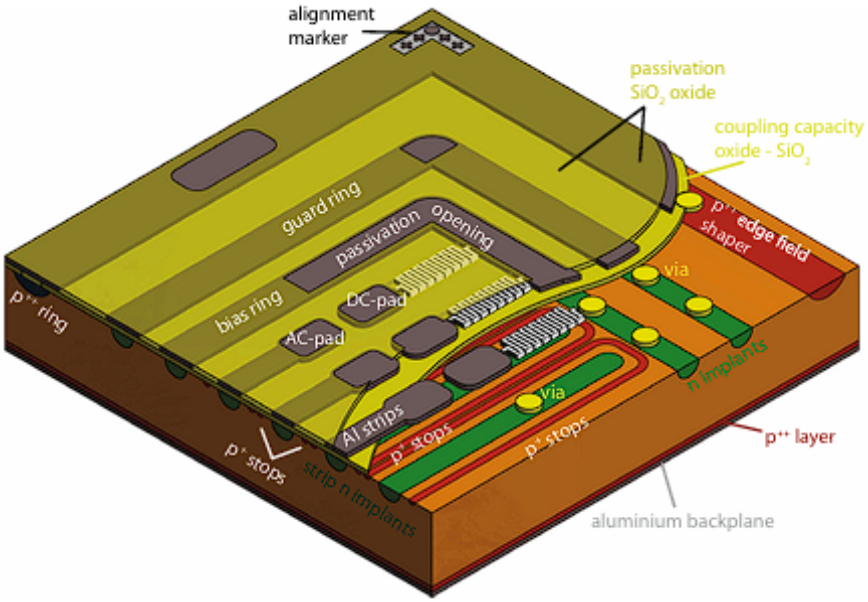


Figure 2.10: 3D schematic of an n-in-p sensor showing the 3 different rings : the bias, guard and edge rings [14].

Chapter 3

CMS Outer Tracker module designs

This chapter describes the two types of silicon modules used in the outer tracker region of CMS for the HL-LHC upgrade.

3.1 Overview of the Outer Tracker modules

The outer tracker, presented in Figure 3.1, is composed of six cylindrical barrel layers covering the region of $|z| < 1200$mm and five endcap double discs in the region of $1200 < |z| < 2700$mm [1]. Two types of modules will be installed between $r = 21$cm and $r = 112$cm: strip-strip (2S) modules, assembled from two strip sensors, and pixel-strip (PS) modules, assembled from a strip sensor and a macro-pixel sensor [1]. Both module types are designed to be self-contained, carrying their own components for power conversion, opto-electrical conversion and control electronics.

The 2S modules include identical strip sensors of size 10×10 cm^2 and of thickness $200\,\mu$m, manufactured with 1.8mm and 4mm sensor spacings, and are placed at $r > 600$mm. The PS modules have a sensor size of 10×5 cm^2 and are constructed

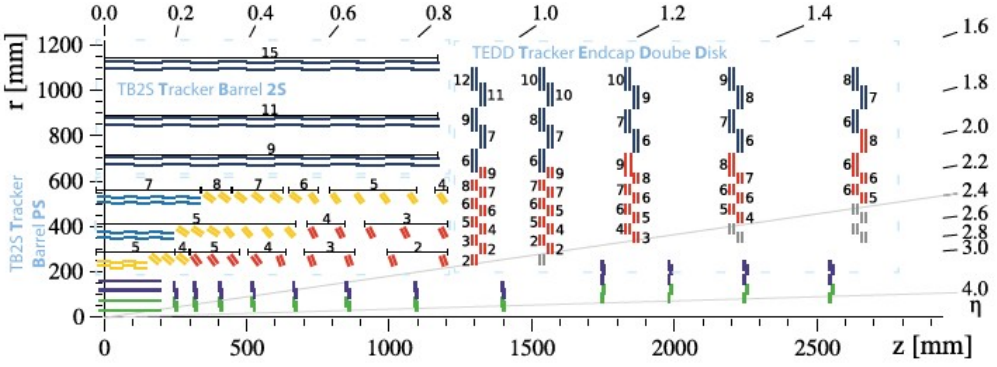


Figure 3.1: Sketch of one quarter of the CMS Tracker layout. The interaction point is located at the point (0,0). The sensor spacing is colour coded with dark blue for 1.8mm, light blue for 1.6mm, yellow for 2.6mm and red for 4mm spacing; grey modules ($|\eta| > 2.5$) do not participate in the trigger. The numbers represent the "number of strips in the stub acceptance windows", tuneable during running, as explained in Section 3.2.1 [14].

with 1.6mm, 2.6mm and 4mm sensor spacings[1], placed closer to the interaction

[1]For the tracker endcap double disk in Louvain-La-Neuve, only the 4mm spacing is implemented.

point at 200mm $< r <$ 600mm to keep the occupancy at an acceptable level since the shorter strips of a PS module give a finer granularity and a faster, more localized readout [1]. This way only a small fraction of channels is hit in each event even at very high collision rates. In both module types, the sensor spacing is achieved by gluing the two sensors onto spacers made of Al-CF, whose high thermal conductivity and low coefficient of thermal expansion allows for good heat conduction from the sensor and hybrids.

The three outermost Outer Tracker barrel layers, called TB2S, are built using 2S modules mounted on a ladder structure and are designed to closely resemble the current tracker implementation. The inner three OT layers, called TBPS, use PS modules on a different mechanical structure: at small z, so-called planks hold the PS modules, while at higher z the modules are mounted on rings to achieve a tilt that spares some modules and improves trigger performance at larger z [1].

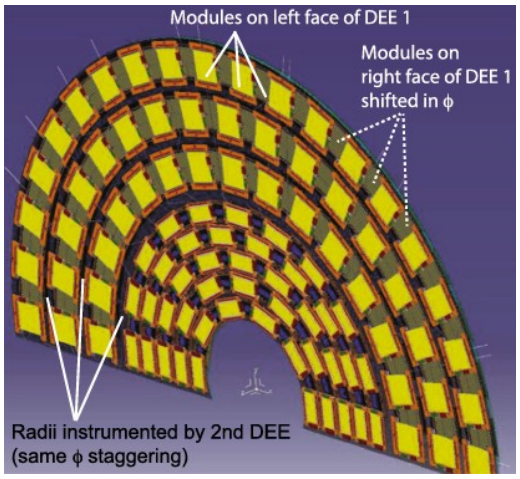


Figure 3.2: Illustration of a single dee in the TEDD region [14].

In the endcap region, called TEDD, modules are mounted on the front and back sides of half-disks called dees, supporting both 2S and PS modules and including CO_2 cooling loops. Four dees form one double disk, and each TEDD is made of five double disks, the first two (TEDD1) larger and more densely populated than the other three (TEDD2). Within a double disk, one disk covers one set of radii while the other covers the radii missing from the first, together forming a hermetic layer without modules clashing.

The HL-LHC Outer Tracker also have on-detector electronic architecture designed to handle unprecedented collision pileup. Operating at 40 MHz, the system first has to buffer the data to accommodate trigger latencies up to 12.5 µs and achieves high-efficiency readout at Level-1 accept rates reaching 750 kHz [1]. To do that, the front-end electronics are tasked with performing local trigger processing and data reduction directly on the module, all while strictly adhering to severe constraints on radiation tolerance, power dissipation, bandwidth, and latency.

This event selection process is achieved thanks to two classes of radiation-hard ASIC: a sensor-level front-end readout chip, which differs by module type, and a Concentrator Integrated Circuit (CIC) which aggregates the front-end outputs and interfaces with the module's optical and slow-control services. The 2S module's front-end chip is the CBC, performing binary strip readout, clustering, and on-chip inter-sensor correlation to generate trigger stubs. The PS module instead combines two front-end chips, the SSA for its strip sensor and the MPA for its macro-pixel sensor, whose combination enables finer spatial resolution and correlates hits between the two sensor types. Concretely, the front-end electronics of either module type must be able to:

- Read out a large number of strip and pixel channels.

- Operate at the LHC bunch crossing rate (40 MHz).
- Withstand high radiation levels.
- Locally correlate hits between the two sensors.
- Provide fast binary hit information.
- Form trigger stubs compatible with the Level-1 track trigger (see Section 3.2.1).
- Transmit data efficiently under high trigger rates.

While this global concept, e.g a front-end readout stage feeding a common CIC, is shared by both module types, its detailed implementation differs substantially, reflecting their different sensor layouts, and is described separately for 2S and PS modules in Section 3.2.

Each hybrid also hosts the module's power and communication chain: DC-DC converters, the low-power Gigabit Transceiver (lpGBT), and the Versatile Transceiver (VTRX+), alongside the front-end and concentrator ASICs. A flexible hybrid circuit, folded around a spacer, establishes the connection from the sensors to the ASICs via wire bonding. In both module types, a 12 V low-voltage input is first DC-DC converted to 2.55 V by the bPOL12V ASIC to feed the VTRX+, with further DC-DC conversion stages (bPOL2.5V) then supplying the remaining electronics [1].

3.2 Description of module structure (sensor, readout electronics, mechanical support)

2S module

As shown in Figure 3.4, a 2S module consists of two strip sensors, two front-end hybrids (FEH), one service hybrid (SEH), and two spacers. Each FEH, on either side of the sensor, carries 8 CBC chips (16 in total), each reading 254 channels, connected via the flexible hybrid to the two strip sensors (2×1016 strips of length 5cm and pitch $90\,\mu$m). The CBC chips are connected alternately to the upper and lower sensor, enabling the stub-finding process described in Section 3.2.1. The SEH combines the readout block (lpGBT and VTRx+) with the power block e.g two DC-DC converters. Each DC-DC converter is made of a bPOL12V delivering 2.55 V to the VTRx+ and bPOL2.5V, and a bPOL2.5V delivering 1.25 V to the CBCs, CIC, and lpGBT (Figure 3.3). The readout data generated by the 16 CBCs is aggregated, formatted, and buffered by the single CIC before transmission off the module. Because the power block is not located at the center of the SEH, the wire length to each FEH differs. Consequently, a difference of potential appears between the two sides. A ground balancer connecting the two FEH is therefore used to further reduce the resulting current flow and the noise [21].

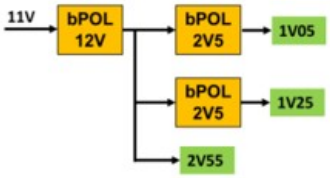


Figure 3.3: Two-stage DC-DC conversion [20].

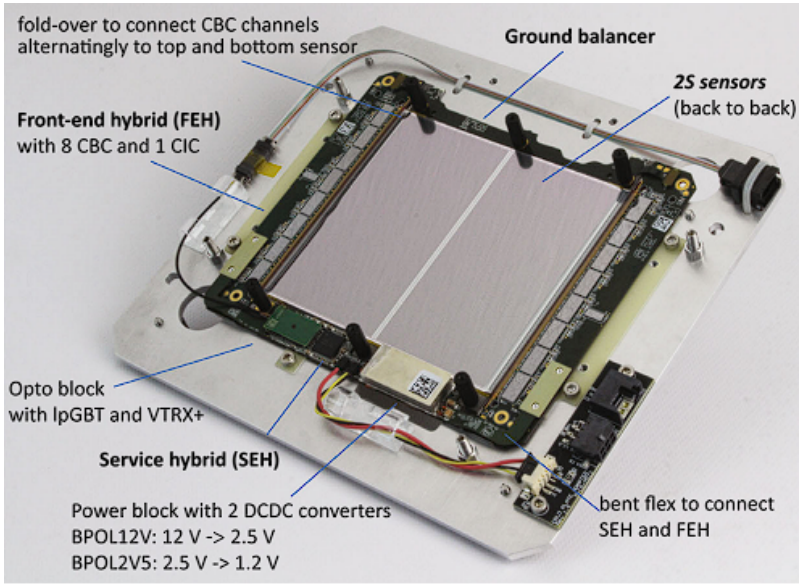


Figure 3.4: Photograph of a fully assembled 2S tracker module. Key components are explicitly labeled, including the back-to-back 2S silicon sensors, the front-end hybrid (FEH) populated with 8 CBC readout chips and 1 CIC concentrator, the service hybrid (SEH), the opto-block housing the LpGBT and VTRx+, the power block featuring dual DC-DC converters (BPOL12V and BPOL2V5), the ground balancer, and the flexible interconnects [14].

PS module

On the front face of the PS module (Figure 3.5), 16 SSA ASICs, each comprising 120 channels, are wire-bonded to the PS-S strip sensor, giving a total of 1920 strips of length 2.5 cm and pitch 100 μm. On the back, 16 MPA ASICs are bump-bonded to the PS-p macro-pixel sensor, giving 30720 macro-pixels of size 1.5 mm and pitch 100 μm [1]. Processed signals from the SSAs are routed via the bent flexible hybrid to the MPAs, which handle the p_T stub finding described in Section 3.2.1. Unlike the 2S module, whose two hybrid sides are functionally identical, the PS module's front and back sides read out two different sensor technologies and, as Chapter 5 discusses, show correspondingly different noise behaviour.

The pixelated sensor improves the module's z-vertex resolution, at the cost of a higher power dissipation than the 2S module. This explains why the PS module is built on a thermally conductive baseplate, unlike the 2S module whose produced heat is routed to the cooling system through the Al-CF spacers alone. This thermally conductive baseplate enhances heat spreading and is coupled to the cooling system through carbon foam interfaces and CO_2 cooling pipes placed under the pixel sensor.

The PS module implements two FEH (left and right), as for the 2S module, but splits the service-hybrid functionality across two separate hybrids placed opposite each other in Figure 3.5: a Power Hybrid (POH) and a Read-Out Hybrid (ROH), rather than the single combined SEH of the 2S module. POH is made a first bPOL2.5V which delivers 1.25 V to the lpGBT and the analog MPA and SSA circuits, while a second bPOL2.5V delivers 1.05 V to the CIC concentrator and the digital MPA and SSA circuits. The VTRx+ transceiver is powered separately at 2.55 V directly from the bPOL12V output stage (Figure 3.3). The same CIC design

used in the 2S module is implemented here, now aggregating data from the SSA and MPA chips instead of the CBC.

Table 3.1 summaries the key structural differences between the two module types introduced above.

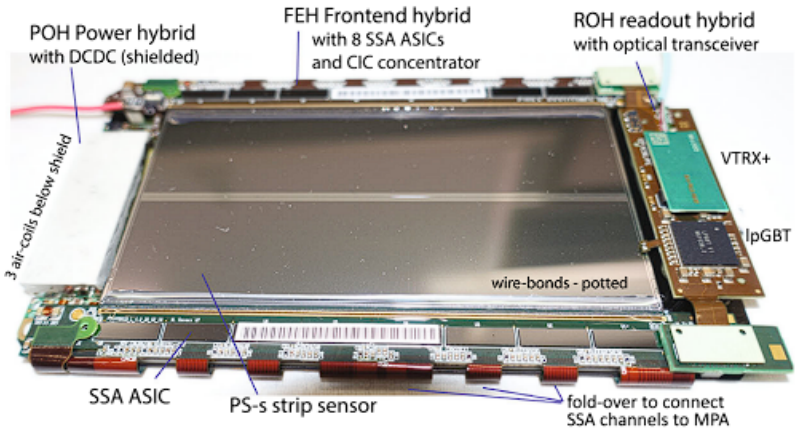


Figure 3.5: Photograph of a fully assembled PS tracker module. The layout displays the PS-s strip sensors, front-end hybrids with SSA ASICs and a CIC concentrator, the readout hybrid (ROH) equipped with optical transceivers (VTRx+ and LpGBT), the shielded power hybrid (POH) with DC–DC converters and the fold-over flexible circuits linking the SSA channels to the MPA [14].

Table 3.1: Key specifications and structural comparison between the 2S (Strip-Strip) and PS (Pixel-Strip) modules for the CMS Phase-2 Outer Tracker Upgrade.

Parameter / Specification	**2S Module**	**PS Module**
Sensor Topology	Strip–Strip	Pixel–Strip
Active Sensor Dimensions	10×10 cm^2	10×5 cm^2
Sensor Spacings (Δz)	1.8 mm, 4.0 mm	1.6 mm, 2.6 mm, 4.0 mm
Front-End ASICs	CBC	SSA & MPA
Data Aggregation ASIC	CIC	CIC
Strip/Pixel Dimensions	5 cm $\times$ 90 μm	Strips: 2.5 cm $\times$ 100 μm
		Pixels: 1.5 mm $\times$ 100 μm
Capacitance Profile	Higher strip capacitance	Lower capacitance (short-strip/pixel)

3.2.1 Level-1 trigger

The module concept, and in fact the full layout of the tracker, was strongly shaped by the requirement to contribute to the Level-1 trigger. In the OT, modules discriminate between high- and low-p_T particles, and only the reduced data from high-p_T tracks (typically $p_T > 2$GeV [1]) is transmitted for each bunch crossing which enables the tracker to cope with the higher pileup that comes with increased luminosity. The remainder of this section explains how this discrimination is achieved.

Modules in the OT incorporate two narrowly spaced sensors made of parallel strips or pixels. The spacing between them is an important design parameter, since

a larger spacing is needed in the endcaps than in the barrel to achieve the same discriminating power; hence the different sensor spacings introduced above. The front-end chips correlate the signals received from both sensors by using their curvature in the magnetic field of CMS in order to be able to discriminate track momenta. High-p_T tracks pass within a narrow band of strips or macro-pixels in the upper sensor, called the acceptance window, while low-p_T tracks miss it since they are more strongly bent (Figure 3.6). The acceptance window's size is tunable via the front-end chip settings, which is essential since it must increase with the radius of the barrel for particles of the same p_T [22] [23]. A pair of correlated hits in the lower and upper sensor is called a stub. As introduced above, this correlation is performed by the CBC for 2S modules and by the MPA (using data from the SSA) for PS modules.

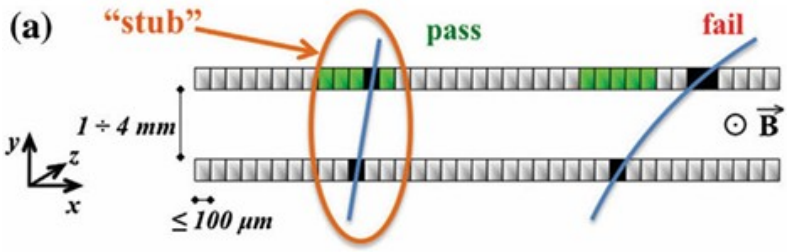


Figure 3.6: Illustration of the working principle of p_T discrimination. The two horizontal lines represent the two silicon sensors in a module, with the squares within depicting the pitch. The blue curves show the tracks of two different particles, and the dark squares are the strips or macro-pixels where a signal is detected. The channels shown in green represent the "acceptance window" defining an accepted "stub" [1].

3.2.2 Description of modules electronics

Beyond the general architecture introduced above, this section details the front-end and readout electronics of each module type, starting with the CBC (2S modules) before turning to the SSA and MPA (PS modules).

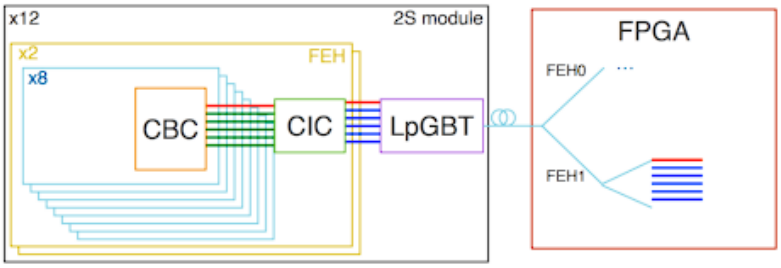


Figure 3.7: Illustration of the front-end electronic on a 2S module and the back-end electronic with the FPGA. Each colour-coded to represent a different data or control path. The labels x12, x2, and x8 indicate multiplicities e.g how many identical links or channels are aggregated at each stage [24].

For 2S modules, the initial stage of signal processing is performed by the CBC, illustrated in Figure 3.7. This chip processes the charge deposited by ionizing particles in the strips, converting it into a binary signal that indicates the presence of

a hit. Additionally, it buffers data from multiple events to account for the Level-1 trigger latency of approximately 12.8 μs [1]. As noted in Section 3.2, a single CBC reads out 254 strips, equally divided between the top and bottom sensors, with 16 CBCs distributed across each module.

As shown in Figure 3.7, the CBCs transmit their data to the CIC via six differential lines: five carry stub data for the L1 trigger, and the sixth transmits full raw data upon reception of a trigger. The CIC collects, buffers, and formats these incoming streams from multiple CBCs, synchronizing and preparing them for high-speed transmission over the lpGBT/VTRX+ optical link introduced in Section 3.1, which supports simultaneous data readout, timing distribution, control, and monitoring.

Figure 3.8 shows the CBC's analog front-end, connected to every single strip. It includes a pre-amplifier and post-amplifier to enhance the signal, followed by an adjustable offset compensating for variations in amplifier response and threshold levels between strips. The processed signal is then fed into an inverting comparator, which applies a threshold V_{cth}, configurable and typically set uniformly across the module, to determine whether the signal corresponds to a valid hit.

Without the hysteresis adjustment of the comparator, an analog signal sitting close to V_{cth} would enable high-frequency electronic noise to make the output of the comparator oscillate rapidly between "0" and "1" which would create multiple false digital hits from a single particle. Hysteresis avoids this so-called "noise chattering" via a Schmitt-trigger mechanism [25] with two distinct threshold levels:

1. A rising threshold (V_{high}) where the input signal must rise above this threshold to trigger a digital high pulse.

2. A falling threshold (V_{low}) where the signal must drop below this lower threshold to reset the comparator once it has been triggered.

The gap between these two thresholds is set and adjusted manually but it comes with a trade-off. On one hand, too little hysteresis enables high-frequency noise to trigger false double-hits which can saturate the 40 MHz trigger data path with useless data. On the other hand, too much hysteresis slows the comparator's reset resulting in an increase in time-walk which could assign a hit to the wrong bunch crossing. The chip must therefore be tuned so that noise oscillations are fully suppressed while time-walk stays within its required limit (typically under 16ns [26]).

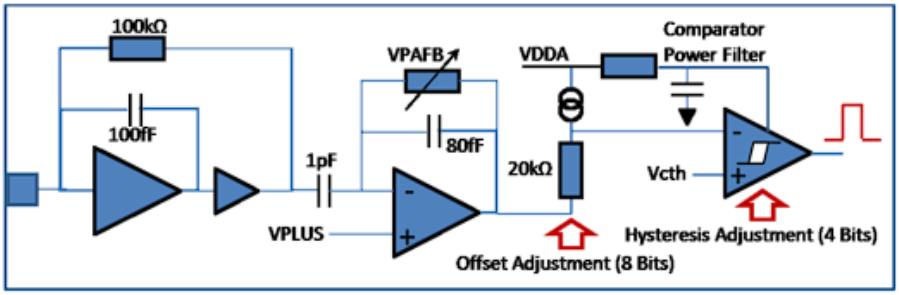


Figure 3.8: CBC analog front-end connected to every single strip of the sensors constituting the 2S module [26].

For PS modules, the overall architecture (Figure 3.9) differs from the 2S module mainly at the front-end level. Instead of the CBC, 16 SSAs process the strip signals

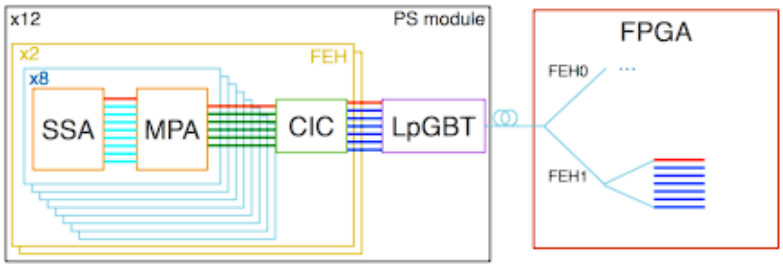


Figure 3.9: Same as Figure 3.7 but for a PS module [24].

and transmit high-bandwidth data to 16 MPAs, which perform hit selection and correlate signals between the two sensor types to generate stubs. The downstream chain (i.e CIC, lpGBT, VTRX+, and DC-DC conversion) otherwise follows the same design already described for the 2S module, now aggregating SSA/MPA rather than CBC data.

3.2.3 Electrical connections between sensors and chips

As shown in Figure 3.10, both module designs rely on Carbon Fiber Reinforced Polymer (CFRP) supports for mechanical stiffness. Additionally, a flexible hybrid circuit is laminated onto the CFRP and routed around the module's edge to distribute power and signals between the sensors and the front-end electronics.

In the 2S module, the two strip sensors are electrically connected via wires to the front-end hybrid hosting the CBC readout chips which are themselves wire-bonded to their corresponding hybrid. While the hybrid provides the routing between the sensor strips and the CBCs, an additional flexible circuit connects the module electronics to the opposite hybrid and the service electronics. Moreover, a Kapton layer serves as a high-voltage insulator between the sensors and the mechanical elements.

In the PS module, the strip sensor electrically connects to the hybrids hosting the SSA chips using the same approach as the 2S design. Conversely, the macro-pixel sensor is bump-bonded directly to the MPA readout chips, which are then wire-bonded to the front-end hybrids. Unlike the 2S module, where the CIC sits on the top hybrid, the PS module's CIC is mounted exclusively on the bottom hybrid in order to minimize the length of the strip wire bonds.

Ultimately, both module designs integrate flexible hybrids, CFRP supports, and dedicated spacers to achieve precise mechanical alignment, efficient readout connectivity, and effective thermal control. The specific impacts of this shared architecture on the noise performance of each module will be explored in Chapters 4 and 5.

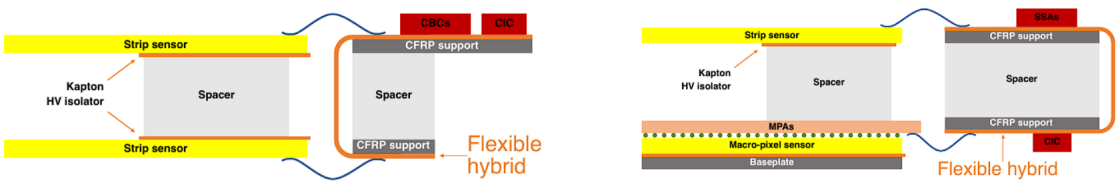


Figure 3.10: Sketches of the front-end hybrid assembly and electrical connectivity of the 2S module (left) and PS module (right) of the outer tracker [1].

Chapter 4

Experimental methodology & noise framework

This chapter details the experimental set-ups utilized for data acquisition, alongside the statistical framework developed to characterize common mode noise.

4.1 Experimental setup

Each 2S or PS module studied in my thesis was characterized on a dedicated burn-in test bench (Section 4.1.1) before being integrated into a larger mechanical structure, namely a dee (Section 4.1.2). The test procedure followed the same sequence for every module: the module was first connected to the readout system through its optical and electrical harnesses, then placed under controlled environmental conditions, and finally operated through a standardized sequence of calibration and noise characterization runs.

In both tests, the modules were operated inside a climate chamber capable of regulating both temperature and relative humidity, which was used to reproduce the thermal conditions the Outer Tracker modules will experience in the HL-LHC detector, well below room temperature. Temperature and humidity were logged for the full duration of every run, so that any run in which the environmental conditions drifted outside the operating window could be identified and, if necessary, excluded from the analysis.

I had the chance to contribute to the CP3 team across both experimental set-ups. I gained hands-on experience in the handling of 2S modules from their unboxing and inspection to burn-in thermal testing. I also joined in the integration of 2S and PS modules onto the dee structure, assimilating the precision positioning, mechanical fixation, and electrical and optical readout connectivity required for optimal detector performance. For instance, I learned how to apply the thermal paste between PS module baseplates and the dee's cooling circuit to ensure low thermal impedance. Additionally, I helped route flexible hybrid cables, verified optical link functionality, and performed post-mounting readout and power connectivity checks to ensure no degradation occurred during mechanical assembly.

The rest of this section describes the physical setup and the operating sequence; the noise-related quantities extracted from the data it produces are defined in Section 4.3.

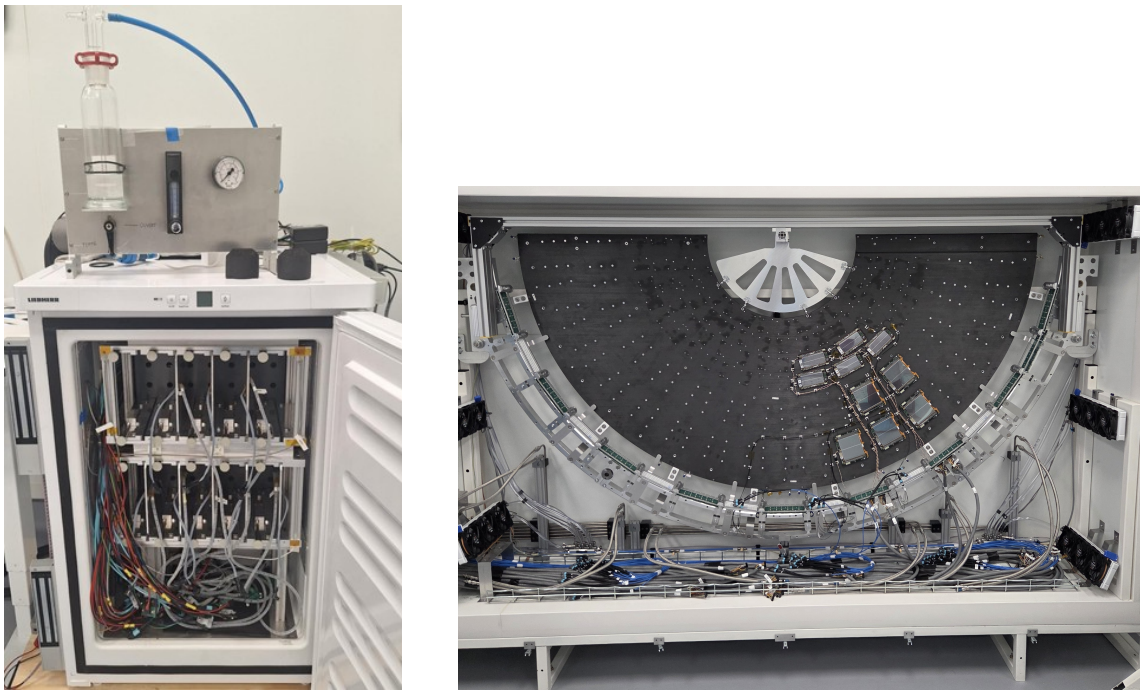
4.1.1 Burn-in setup

The burn-in setup in Figure 4.1 is specifically engineered to thermally stress silicon modules and identify early component failures. It utilizes a modified commercial refrigerator capable of housing up to ten modules on an internal metal shelf, with electrical and optical services routed from an adjacent technical room. A Julabo refrigerated circulator [27] cools the internal environment down to -35°C to achieve the necessary thermal stress.

4.1.2 Dee setup

Unlike the isolated module testing performed in the burn-in phase, the dee setup in Figure 4.1 is designed for large-scale integration testing. It centers around a substantial cold box built to house an entire half-disk support structure, known as a dee. This integrated environment is essential for verifying post-integration module health, confirming proper grounding, validating thermal interfaces, and evaluating noise interference between neighboring modules.

Because powering and cooling a fully populated dee simultaneously would overwhelm available resources, evaluations are conducted as "sector tests", which limit operation to a maximum of two cooling loops at a time. The environment is cooled to -35°C using a dual approach: liquid CO_2 cools the loops, while a Julabo circulator pumps silicon oil to cool the ambient air via internal fans. In order to prevent condensation on the module surface and connectors, the cold box is continuously flushed with dry air and monitored by a network of temperature, humidity, and specialized CO_2 sensors to rapidly detect any loop leaks.



(a) **(b)**

Figure 4.1: The two setups in the clean room to test the performance of the modules. On the left, the burn-in fridge for individual module characterization, and on the right the cold box for testing integrated modules on the dee.

4.1.3 Operating sequence

A test began by connecting the module's optical link (used for the high speed data and fast-control signals) and its electrical low voltage and high voltage services to the back-end readout system. A dry-air line was then switched on and left running for the whole duration of the test. This mitigation is especially important once the climate chamber is driven to sub-zero temperatures, as is done to emulate the operating conditions expected during HL-LHC running. Only once a stable dry-air flow was established were the low voltage power supply and the micro-controller (Arduino-based) used for slow-control and interlock monitoring switched on. This ordering (optical/electrical connection, then dry air, then power) was followed systematically to avoid powering a module in a humid environment and to guarantee that temperature data was available before any bias voltage was applied to the sensor.

4.1.4 Electrical test equipment

Each module was read out through the back-end electronics chain used throughout the Outer Tracker system: an FC7-based back-end board [28] provided the fast-control, DAQ link, and slow-control configuration to the module's front-end and service hybrids, while low-voltage operational power and high-voltage sensor bias were delivered by dedicated external power supplies. The on-detector ASICs (CBCs for 2S modules, SSAs and MPAs for PS modules, and the CIC concentrator common to both) were configured over I^2C through this digital link. A separate, independent low-voltage supply and an Arduino-based monitoring unit provided interlock and temperature read-back, decoupled from the main power path, so that the module could be powered down automatically if a temperature or current threshold was exceeded. All configuration, calibration and data-taking sequences were driven by the standard Outer Tracker DAQ software stack, which also produced the ROOT histogram files analyzed throughout this chapter and Chapter 5.

4.2 Impact of the set-up on noise

The set-up in which 2S and PS modules are operated plays a critical role in their measured noise performance, primarily through differences in grounding topology, electromagnetic shielding, and inter-module coupling.

In the burn-in setup, modules are evaluated in relative isolation where the internal metal rack and the surrounding three-layer walls (metal, insulator, metal) act as an effective Faraday cage to suppress external RF interference that would otherwise modulate the ASIC baselines across entire readout chips. Additionally, this set-up achieves exceptional stability by employing a low-noise CAEN power supply [29][30][31] tied directly to the building's main ground, hence preventing power supply return ground loops that typically inject CMN. Therefore, noise measurements in this regime predominantly reflect the intrinsic electronic baseline of the front-end ASICs, with only minimal contributions from CMN.

In contrast, operating modules integrated onto a dee inside the cold box exposes them to a significantly more complex environment representative of final detector conditions. In this configuration, measured noise levels are subject to shared power

return lines, ground loops across the extensive mechanical support structure, capacitive cross-talk from operating neighboring modules during multi-loop sector tests, and external electromagnetic interference pickup along extended power and optical service lines [32][33]. Additionally, the high-pressure liquid CO_2 cooling flow, the circulating thermal oil or even internal air-circulation fans can introduce subtle thermal gradients that affect sensor noise. Systematically comparing noise characteristics between the isolated burn-in setup and the integrated dee environment is therefore essential to decouple true module-level defects from system-induced noise introduced during detector assembly.

4.3 Noise measurements

The front-end ASICs of both 2S and PS modules (CBC, SSA and MPA) use a binary readout: for every channel, the analog pulse height is compared to a single programmable threshold (VCth for the CBC, an equivalent threshold DAC for the SSA/MPA) and only a hit/no-hit bit is transmitted, with no pulse-height information. Because of this, the noise of a channel cannot be measured directly from a distribution of pulse heights and must instead be inferred from how the hit probability of that channel responds to a scan of the threshold. This gives the so-called S-curve.

In the absence of an injected signal, random channels can still fire occasionally because of pedestal fluctuations. This phenomenon is best described by random noise superimposition on the baseline which occasionally pushes the signal above threshold. In order to study this random noise, one can generate an S-curve by scanning the threshold for a fixed number of triggers and recording the fraction of times a channel fires (Figure 4.2). The midpoint of this curve defines the channel's pedestal, which is the threshold where the hit probability is exactly 50%. Furthermore, the width of the S-curve is directly related to the channel's intrinsic noise. A narrower curve indicates lower noise because a small threshold step around the pedestal is sufficient to shift the hit probability from 0 to 1. Conversely, a noisier channel produces a broader transition which require a much larger threshold excursion to achieve the same change in the channel's hit probability.

In practice, the pedestal and noise of every channel were extracted with a binary search algorithm rather than a full linear threshold scan: at each step of the search, 1000 triggers were taken at 100 kHz and the measured hit occupancy was compared to a target value of 56 %, and the threshold was adjusted up or down accordingly until convergence [34]. This target occupancy, chosen slightly above 50 % rather than exactly at it, together with the number of triggers used at each step, sets the statistical precision with which the pedestal is located. The channel noise is then obtained from the same procedure by relating the width of the region over which the occupancy changes appreciably to the standard deviation of the underlying Gaussian noise distribution assumed for each channel. Since real threshold scans would be prohibitively slow to repeat over all channels of a module and over many operating conditions, this binary-search calibration is what is actually run before every data-taking sequence, and it is what produces the per-channel pedestal and noise values used throughout this chapter and the next.

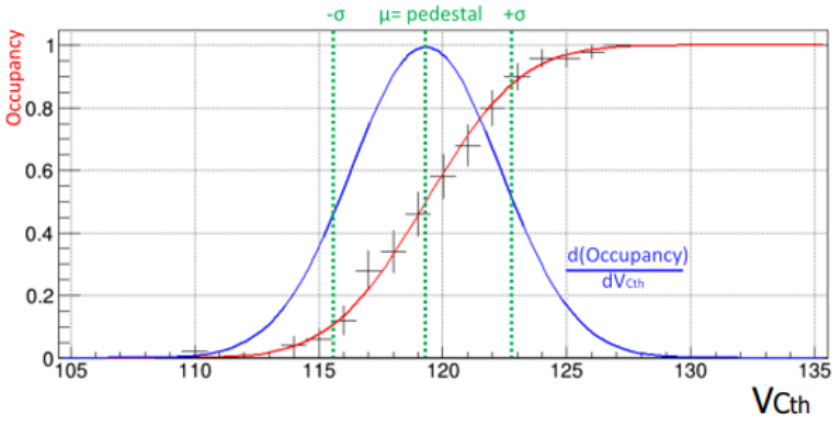


Figure 4.2: Illustration of the S-curve method: measured hit occupancy of a single channel as a function of the applied threshold τ. The curve is centered on the channel's pedestal and its width around the 50 % occupancy point encodes the channel's random noise σ; a noisier channel produces a wider, shallower transition between 0 and 100 % occupancy.

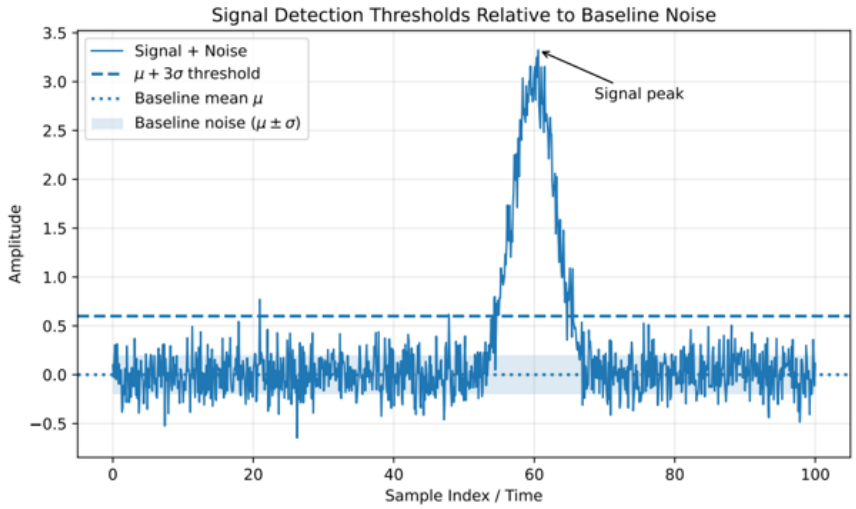
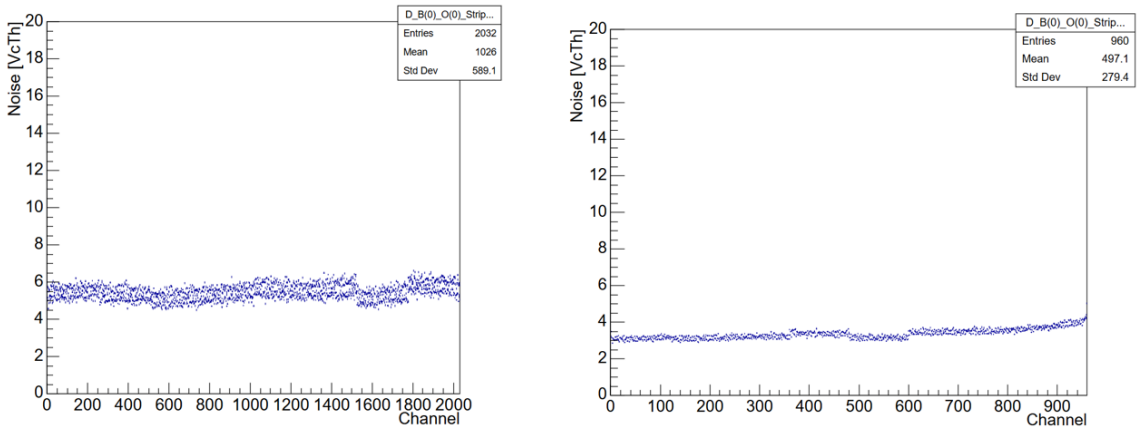


Figure 4.3: Schematic representation of a threshold-crossing measurement, illustrating the baseline mean μ, noise range $\mu \pm \sigma$, and a $\mu + 3\sigma$ detection threshold for all channels of a module. A hit is recorded when the fluctuating analog signal exceeds a threshold set relative to the baseline mean and noise width σ. This illustration is conceptual and does not represent the output of the binary-search calibration.



(a) **(b)**

Figure 4.4: Per-strip noise distribution measured across channels for (a) a 2S module and (b) a PS module. The noise values (expressed in VCth DAC units) are extracted via binary-search calibration. The plot demonstrates stable and uniform channel performance, with individual strip noise clustering consistently around 5 to 6 VCth across all 2032 readout channels for (a) and around 3 VCth across all 960 readout channels for (b).

Comparing the per-strip noise obtained this way between 2S and PS modules (Figures 4.4a and 4.4b) shows a systematically different noise level between the two module types, which is attributed to the different input capacitance seen by the front-end amplifier in each case: 2S modules read out long strip sensors with a correspondingly larger strip capacitance than the short strips (SSA) and macro-pixels (MPA) of PS modules, and since the equivalent noise charge of a front-end amplifier increases with the detector capacitance at its input, 2S channels are expected (and observed) to be noisier than PS channels.

4.3.1 Common mode noise measurement

Beyond the random, channel-by-channel noise described above, silicon strip modules are also affected by CMN: a fluctuation that shifts the baseline of many channels simultaneously. The source of this kind of noise can result from a shared pick-up on the power or ground return path, or a capacitive coupling common to all channels served by the same front-end chip. Because it is fully correlated across channels, common-mode noise cannot be measured from a single channel's S-curve. Indeed, by construction, a per-channel threshold scan integrates over many triggers and is insensitive to a fluctuation that is common to all channels within a given trigger. It must instead be inferred from the statistics of how many channels of a group fire together, event by event, once every channel's threshold is set at a common, low working point.

4.3.2 Common mode noise measurements for 2S modules

The following section is strongly inspired from [35].

Consider a group of n channels, each independently firing with the same per-channel probability $p(\tau)$ at a chosen threshold τ (measured in units of single-channel

noise σ above the pedestal). In the absence of common-mode noise, the number of channels firing in a given event, N_e follows a binomial distribution,

$$N_e \sim \mathrm{Bin}\big(n,\, p(\tau)\big), \tag{4.1}$$

with mean $\bar{N}_e = n\,p(\tau)$ and variance $\mathrm{Var}(N_e) = n\,p(\tau)\big(1 - p(\tau)\big)$.

Common-mode noise is modeled as an event-by-event random shift of the effective threshold τ seen by every channel in the group, drawn from a Gaussian distribution of width Γ:

$$g(\tau) \sim \mathcal{N}(\tau_0,\, \Gamma^2). \tag{4.2}$$

with τ_0 the original threshold without CMN. Consequently, the distribution of the number of hits per event N_e is then a superposition of binomial distributions for varying thresholds.

At the symmetric 50% occupancy working point (e.g $\tau_0 = 0$), the common-mode fluctuation leaves the average occupancy equal to 50% while increasing the event-by-event hit-count variance beyond the pure binomial value of Eq. (4.1). This excess variance of the hit-count distribution relative to the binomial expectation is precisely the observable signature of common-mode noise.

The relation expressing the variance of N_e (see Ref.[35] for its expression) can be inverted to extract Γ directly from the measured hit-count distribution by working at the operating point where each channel is calibrated to fire with 50 % probability [35] ($\bar{N}_e = \frac{1}{2}n$, i.e. $\tau_0 = 0$, the same working point used for the binary-search pedestal calibration of Section 4.3):

$$\tilde{\alpha} = \frac{2\pi\,\big(\mathrm{Var}(N_e) - \bar{N}_e\big(1 - \bar{N}_e/n\big)\big)}{n(n-1)}, \tag{4.3}$$

$$\Gamma^2 = \frac{\sin\tilde{\alpha}}{1 - \sin\tilde{\alpha}}. \tag{4.4}$$

In practice, $\bar{N}_e$ and $\mathrm{Var}(N_e)$ are simply the sample mean and sample variance of the measured number-of-hits-per-event histogram, taken at the 50 %-occupancy working point over the full population of channels of interest (a CBC, a hybrid, or a full module, depending on the granularity of the study). Γ is then reported either directly, or converted back into the same VCth DAC units as the single-channel noise σ by multiplying it with the average single-channel noise of the group σ_i, a quantity referred to throughout Chapter 5 as $\mathrm{CM}_{\mathrm{VcTh}} = \Gamma \times \sigma_i$. This estimator only requires the first two moments of the 50 % occupancy distribution and is therefore, by construction, insensitive to any higher-order (non-Gaussian) structure in the shape of that distribution. This limitation is directly probed in Chapter 5 by comparing it against an independent, shape-sensitive estimator built from the rare, high-threshold tail of the same channels.

Tail-based estimator and toy Monte Carlo model. While Γ (equivalently $\mathrm{CM}_{\mathrm{VcTh}}$) is a robust, closed-form estimator, it is only sensitive to the width of the common-mode fluctuation and blind to its shape: any two distributions of $g(\tau)$ with the same variance give the same Γ, whether or not $g(\tau)$ is actually Gaussian. To test the Gaussian assumption itself, a second, independent estimate of the common-mode noise was obtained from the rare-occupancy tail of the hit-count distribution,

recorded at a much higher threshold (typically 3σ above pedestal, i.e. deep in the regime where only a common-mode fluctuation, not ordinary per-channel noise, can push a sizeable number of channels above threshold simultaneously).

This tail-based estimate was obtained with a toy Monte Carlo model rather than a closed-form formula. For a chosen common-mode width σ_{cm} (expressed in the same VCth DAC units as the per-channel noise σ_i), a large ensemble of simulated events was generated. In each event, every channel received independent Gaussian random noise with its own measured σ_i, on top of the shared, Gaussian common-mode fluctuation drawn once per event and common to the whole group of channels considered. Each channel is said to register a hit if its simulated fluctuation from the pedestal exceeds the fixed 3σ threshold. Repeating this procedure over many toy events produces a simulated hit-count histogram for a given trial value of σ_{cm}, which can be compared bin-by-bin against the real, measured hit-count histogram using a Pearson χ^2 statistic. The remainder of this procedure is solely about scanning all values of σ_{cm} and finding the one that minimizes the χ^2 which then yields the best-fit common-mode width $\sigma_{cm,\mathrm{best}}$. Because it is fitted directly on the shape of the rare tail rather than on the bulk mean/variance, $\sigma_{cm,\mathrm{best}}$ is sensitive to non-Gaussian features of the common-mode fluctuation that Γ cannot see. Since both estimators quantify the "same" common-mode noise under a Gaussian hypothesis, comparing $\mathrm{CM}_{\mathrm{VcTh}}$ (bulk, moment-based) to $\sigma_{cm,\mathrm{best}}$ (tail, shape-based) for the same group of channels provides a direct, purely data-driven test of whether the common-mode fluctuation is really Gaussian. As a matter of fact, an agreement between the two supports the Gaussian hypothesis, while a systematic mismatch is evidence of a non-Gaussian common-mode component that the Gaussian model cannot fully capture.

Goodness of fit. Because the number of entries per bin of the hit-count histogram falls quickly with occupancy, many of its bins hold only a few of counts, well outside the regime where the asymptotic χ^2 distribution with a fixed number of degrees of freedom is a reliable reference [36]. Therefore, the significance of a given χ^2 value was assessed with a "toy" procedure as follows. After fitting the 50% occupancy distribution, a large number of independent pseudo-datasets were generated from the toy model itself (with the same number of simulated events as the real run). Then, the same χ^2 statistic was recomputed for each of them and the resulting distribution of toy χ^2 values was used to obtain a p-value for the real data's χ^2. This procedure was implemented twice for every fit: once using the full range of the hit-count histogram, and once restricted to a "core" region in which the expected bin content is at least 5 counts. Both the obtained p-values make explicit whether the toy model is being rejected because it fails to describe the bulk of the distribution (visible in the core-region test) or only because of the handful of very low statistics bins in the extreme tail (visible only in the full-range test).

Correlated common-mode models. The simple model above assumes a single shared common-mode fluctuation for the whole group of channels under study. Two refinements of this model were used in Chapter 5 whenever the data showed that this single-value assumption was too crude:

- At the level of a full 2S module, the two hybrids (referred to as Left (L) and Right (R)) were found to have anti-correlated common-mode fluctuations

rather than independent or identical ones. This was modeled by drawing the common-mode value of one hybrid, and combining it with an independent Gaussian component for the other hybrid weighted so as to reproduce a target correlation coefficient ρ measured directly from the data, rather than assuming either full correlation ($\rho = 1$) or full independence ($\rho = 0$).

- At the level of a single hybrid, the eight CBCs that compose it were found to be only partially correlated with each other; not the "all CBCs share exactly the same common-mode value" limit implicit in the simple model above. This was captured with a one-factor model in which each CBC's common-mode fluctuation is written as a fixed, per-CBC correlation coefficient ρ_i (measured directly from data) times a hybrid-wide common factor, plus an independent per-CBC residual scaled so that the total variance of each CBC's common-mode fluctuation is preserved. The per-CBC correlation coefficients ρ_i entering this model were measured directly from the correlation between each CBC's own hit count and the hit count of the rest of its hybrid, and are reported throughout Chapter 5 as $\rho_{\text{per-CBC}}$.

In both cases, σ_{cm} (or, at module level, one σ_{cm} per hybrid side) remains the only free parameter of the fit, tuned by the same χ^2-minimization procedure described above; the correlation coefficients themselves are always measured quantities, not fit parameters. Whenever a more structured (anti-correlated or CBC-correlated) model is compared to the simple, single-shared-value model for the same data, the improvement or degradation of the resulting χ^2 and of the toy-to-data mean and standard-deviation ratios is used in Chapter 5 as direct evidence of how much correlation structure is actually needed to describe the data realistically, and of how that need differs between the left and right sides of a module.

Moreover, higher moments of the raw, bin-by-bin hit-count and pedestal distributions (in particular the third standardized moment, or skewness γ_3) were also computed directly from the histograms wherever the 50 % occupancy distributions showed a visibly asymmetric shape. Since a non-zero skewness is itself a purely non-Gaussian feature, this constitutes another, complementary, way of quantifying how far a given distribution departs from the ideal Gaussian common-mode picture of Eqs. (4.2)–(4.4).

It is worth stressing that the $\Gamma/\mathrm{CM}_{\mathrm{VcTh}}$ estimator of Eqs. (4.3)–(4.4) is only valid under the assumption of a Gaussian common-mode fluctuation, and moreover assumes a group of channels with a common, well-defined single-channel noise. While these conditions hold reasonably well for 2S modules, they break down for PS modules, where the mixed pixel/strip geometry and the distinct MPA and SSA front-ends violate the underlying assumptions. For this reason, the Γ-based analysis presented in this chapter and applied throughout Chapter 5 is specific to 2S modules.

Module-Level Noise Combination and Coherence Factors The reasoning to calculate $\sigma_{cm,\text{module}}$ is described in the following paragraph.

The total number of 3σ hits recorded by a module in a given event is simply the sum of the hits recorded by its right (R) and left (L) hybrids, so the module-level common-mode fluctuation is itself the sum of the two hybrids' own common-mode fluctuations, of width $\sigma_{cm,R}$ and $\sigma_{cm,L}$ respectively, correlated according to the

module's own measured correlation ρ. To be precise, this correlated pair is built in three steps:

- Two statistically independent, standard Gaussian random draws $Z_1, Z_2 \sim \mathcal{N}(0,1)$ are combined into a unit-variance correlated pair:

$$X = Z_1\,, \qquad Y = \rho Z_1 + \sqrt{1-\rho^2}\, Z_2\,. \tag{4.5}$$

  It can be proven that both X and Y retain unit variance for any value of ρ, while their correlation works out to be exactly ρ [37].

- Each of X and Y is then fed as a shared, hybrid-wide seed into an additional, per-CBC decomposition on its own side: each of a side's eight CBCs combines that side's seed (X for R, Y for L) with its own statistically independent noise, weighted by that specific CBC's own measured correlation to the rest of its hybrid, namely $\rho_{\text{per-CBC}}$.

- The result of this per-CBC combination is scaled by $\sigma_{cm,R}$ or $\sigma_{cm,L}$ to give the actual, strip-level common-mode contribution used in the toy model. In the end, $\sigma_{cm,R}$ and $\sigma_{cm,L}$ are the widths of each side's own per-CBC-correlated common-mode structure as a whole, rather than the amplitude of a single value shared identically by every strip of that side.

Physically, this construction makes explicit how much of each hybrid's common-mode fluctuation is shared with the other side and how much is its own: the fraction ρ that ties Y to X represents the part of the L-side fluctuation tied to the R side through their shared power infrastructure, while the remaining, independent part of Y, weighted by $\sqrt{1-\rho^2}$, together with each side's own per-CBC noise, represents whatever in each hybrid's fluctuation is genuinely its own and unrelated to the other side.

Finally, the resulting variance for the sum of two correlated random variables is the sum of the individual variances plus a cross term set by their correlation ρ. However, this rule only holds if each side's contribution were a single, fully-coherent value shared identically by all eight of its CBCs. In reality, a side's eight CBCs are only partially correlated with each other ($\rho_{\text{per-CBC}} < 1$) which explains why summing them does not reinforce as strongly as if they moved exactly together. Accounting for this, the variance of a side's eight-CBC sum comes out to $\left(\sum_i \rho_{\text{per-CBC},i}\right)^2 + \sum_i \left(1 - \rho^2_{\text{per-CBC},i}\right)$ rather than n^2_{cbc} (the value it would take if every $\rho_{\text{per-CBC},i} = 1$ with n_{cbc} the number of CBC), and the cross term between the two sides is reduced in the same way. Normalizing both by n^2_{cbc} to keep them on the same per-CBC scale as $\sigma_{cm,R}$ and $\sigma_{cm,L}$ gives three coherence factors:

$$\kappa_R = \frac{\left(\sum_i \rho_{\text{per-CBC},R,i}\right)^2 + \sum_i \left(1 - \rho^2_{\text{per-CBC},R,i}\right)}{n^2_{\text{cbc}}}\,, \tag{4.6}$$

with κ_L defined identically for the L side, and a cross-coherence factor

$$\kappa_\times = \frac{\left(\sum_i \rho_{\text{per-CBC},R,i}\right)\left(\sum_j \rho_{\text{per-CBC},L,j}\right)}{n^2_{\text{cbc}}}\,. \tag{4.7}$$

The module-level width is then defined as:

$$\sigma_{cm,\text{module}} = \sqrt{\sigma^2_{cm,R}\,\kappa_R + \sigma^2_{cm,L}\,\kappa_L + 2\rho\,\kappa_\times\,\sigma_{cm,R}\,\sigma_{cm,L}}\,. \tag{4.8}$$

4.3.3 Common mode noise measurements for PS modules

The $\Gamma/\mathrm{CM}_{\mathrm{V_{cTh}}}$ framework of Section 4.3.2 relies on every channel of the group under study sharing the same per-channel noise character and the same binary threshold-crossing statistics. This assumption holds well for the strip channels of a 2S module, but not across a PS module as a whole, whose strip (SSA) and macro-pixel (MPA) sensors have different geometries, different capacitances, and, as shown below, qualitatively different noise statistics. For this reason, the 3σ tail toy-model methodology was extended separately for the two sensor types that make up a PS module.

Strip (SSA) sensors: an effective-threshold correction. Repeating the naive 3σ-tail toy model of Section 4.3.2 directly on PS strip data (e.g independent per-channel Gaussian noise plus a single shared common-mode width σ_{cm}) revealed a problem specific to PS: the 3σ threshold already over-predicts how often a channel should fire purely by chance e.g. for the 960 strips of a PS hybrid at 3σ, pure per-strip Gaussian noise gives P(hits= 0) $\approx$ 27%, while the real PS module data show 42-75% across all 6 hybrids. σ_{cm} alone cannot close this gap because adding common-mode variance only ever pushes P(hits=0) to higher values. So the model cannot reproduce the real fraction of events with zero hits no matter how σ_{cm} is tuned.

This issue was resolved by promoting the threshold itself, rather than only σ_{cm}, to a fitted parameter: the effective threshold actually used in the toy model is $k_{\mathrm{thresh}} \times n_\sigma$ (VCth DAC units), with k_{thresh} and σ_{cm} tuned jointly by the same Pearson-χ^2 minimization used throughout my thesis. A value of $k_{\mathrm{thresh}} = 1$ would mean the 3σ threshold is already correct. Values of k_{thresh} found in practice are discussed in Section 5.2.1. No such correction was ever needed for 2S strips, where the effective threshold required by the data stayed within about 0.06σ of the 3σ everywhere. The mechanism that could explain the need of such a correction is explained in the following paragraph.

As said previously, the pedestal and single channel random noise used in the toy model come from the S-curve calibration, run at a busy 50% S-curve working point. The 3σ data being fit, by contrast, has very few channels ever fired. Therefore, if voltage droop on the hybrid's local supply line is activity-dependent, as previously observed in [38], then the channel's real analog baseline during the quiet 3σ run is not the same as during the busy calibration run. A fixed 3σ threshold would then systematically mispredict the real crossing rate at the different operating point, and k_{thresh} is exactly the kind of free parameter that would absorb that mismatch. If this activity-dependent droop also differs between L and R hybrids, one would get an L/R asymmetric k_{thresh} too.

Macro-pixel (MPA) sensors: from Gaussian common-mode to a Negative Binomial mixture. The single shared-Gaussian model was tested first on MPA data and rejected outright because the real macro-pixel hit count distributions are substantially over-dispersed relative to a Poisson distribution with the same mean, combined with a pronounced floor effect e.g essentially none of 10 000 real events register fewer than about 16 hits for a pixel sensor. A model built from independent Gaussian channel noise plus one additive, fixed-width common-mode term can only

ever widen a distribution relative to its own no-common-mode baseline; it cannot reproduce this specific combination of a hard bulk floor with a heavy tail, whatever width is chosen.

The model used throughout Section 5.2.1 is a Negative Binomial Distribution (NBD), fit directly and in closed form to the real hit-count histogram, with no toy Monte Carlo simulation involved. A negative binomial distribution arises naturally from a Poisson process whose own rate fluctuates event-to-event according to a Gamma distribution [39]. A single MPA, hybrid, or module is fit with a mixture of two (burn-in) or three (dee, Section 5.2.2) negative binomial components, each characterized by its own mean and its own dispersion parameter n (a smaller n means a more over-dispersed, heavier-tailed component), plus mixing weights.

Because two mixtures with identical overall mean and variance can still have visibly different shapes, the fitted mixture is reported throughout my thesis via its own parameters (weights, per-component means and n) rather than being collapsed into a single derived "common-mode" number the way σ_{cm} or Γ are for strips. Goodness of fit is still the same Pearson-χ^2 used everywhere else in my thesis, and the toy-to-data mean and standard deviation ratios are reported alongside it.

A structural limitation at PS module level. The data-taking software sums the hit counts of its two hybrids (R and L) into a single histogram exactly as for 2S. For 2S, this was not an issue because the ROOT files also contain a histogram of the two hybrids' cross-correlation (Section 5.1.1), which let the anti-correlation between them be measured and built into the module-level toy model. However, no equivalent cross-hybrid correlation histogram exists for PS modules in the datasets used in my thesis. A module-level fit using one shared $(k_{\text{thresh}}, \sigma_{cm})$ pair for strips is therefore forced to compromise between two hybrids that, as Section 5.2.1 shows, can differ substantially from one another, and correspondingly underpredicts the real module-level variance. Conversely, treating the two hybrids as statistically independent instead overpredicts it by a similar margin. Since Section 5.2 shows that this independent model already reproduces each hybrid's own variance correctly on its own, and independent variances simply add ($\text{Var}(X+Y) = \text{Var}(X) + \text{Var}(Y)$ when $\text{Cov}(X,Y) = 0$), its systematic overshoot of the true module-level variance can be explained by a missing, negative covariance term. This is itself indirect evidence that the two PS hybrids are anti-correlated in the same qualitative way as for 2S, without being able to measure that correlation directly, for lack of the relevant histogram. Module-level PS strip results should accordingly be read as a coarser, secondary aggregate where the hybrid-level results remain the more reliable granularity of this analysis.

Chapter 5

Results

This chapter presents the common mode noise results, obtained with the methodology of Chapter 4, in four parts. Section 5.1 covers 2S modules, first under burn-in (Section 5.1.1) and then on the Dee test structure (Section 5.1.2); Section 5.2 repeats the same structure for PS modules, whose strip and pixel sensors require the extended, sensor-specific toy models introduced in Section 4.3.3. Within each of these four sections, the same two working points are analyzed in turn: the 50 % occupancy point, which gives the value of Γ (or $\mathrm{CM}_{\mathrm{V_{cTh}}}$) but is blind to the shape of the common-mode fluctuation, and the rare 3σ-tail point, which is instead fit with a shape-sensitive toy model and is where most of the non-Gaussian common mode noise becomes visible. Each analysis is further repeated at module, hybrid, and CBC (for 2S) or SSA/MPA (for PS) granularity, since common-mode noise is expected, and found, to depend on how much shared electrical environment a given group of channels has in common. Note that the hybrid-level results are generally the most reliable of the three, as module-level aggregation is shown throughout this chapter to hide some of the underlying hybrid level behaviour.

5.1 2S modules

5.1.1 Burn-in tests CMN

The first tests of CMN measurements were carried out on five 2S modules mounted individually on the burn-in test bench described in Section 4.1, i.e. with each module operated on its own, outside of any larger mechanical support structure. This isolates the CMN intrinsic to a single module and its own services from any additional noise sources that only appear once several modules share a common mechanical and electrical environment. The comparison relevant to this second effect is the subject of Section 5.1.2.

As introduced in Section 4.3.2, the 50 % and 3σ working points probe fundamentally different features of the common-mode fluctuation (its width versus its shape) and are analyzed in turn below at every granularity level. For example, a small number of excessively noisy strips that continuously register hits will shift the 50 % occupancy hit distribution rather than broadening its core. As a result, this noise remains undetected by Γ, whereas it noticeably extends the distribution's tail and is readily identified at the 3σ level by σ_{cm_best}. Studying the two thresholds in separate subsections lets each estimator's own results be presented on its own terms before

the two are confronted with one another: it is precisely the comparison between the bulk width (CM_{VcTh}) and the tail shape ($\sigma_{cm,best}$) that turns them into a genuine, data-driven test of whether the underlying common-mode fluctuation is really Gaussian which is one of the central questions this chapter addresses.

50% occupancy distributions at module level

For every module, the per-strip pedestal and noise were first calibrated and the threshold of every channel of the module was then set to its own 50 % occupancy working point. A dedicated run was taken at this working point and, for every trigger, the total number of channels firing anywhere on the module was recorded, producing the module-level 50 % occupancy hit-count histograms of Figures 5.1 to 5.4.

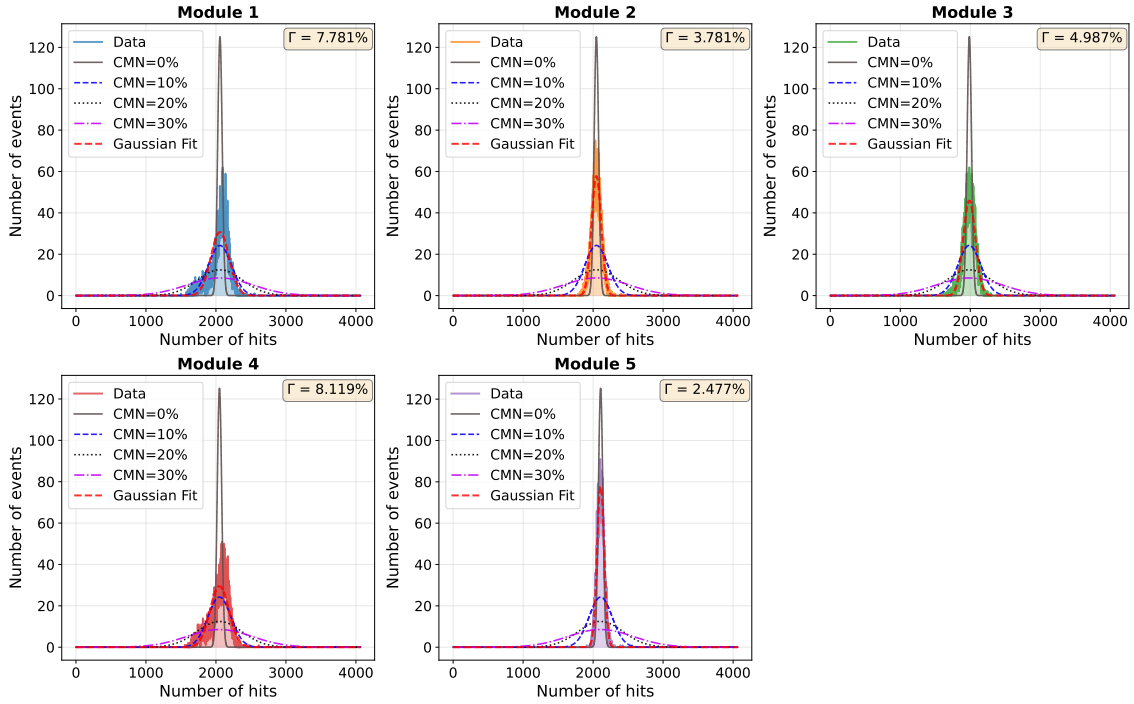


Figure 5.1: Measured hit-count distribution at the 50 % occupancy working point, for all five burn-in 2S module side by side. For each module, the narrow gray curves overlaid is the purely binomial (no CMN) expectation of Eq. (4.1) at its own measured mean occupancy; the measured distributions are visibly broader, which is the direct signature of a non-zero common-mode fluctuation. The excess width relative to the binomial expectation has a magnitude that differs from module to module, summarized by the per-module Γ values on the top right of each plot.

Figure 5.1 shows maximum Poisson log-likelihood Gaussian fits of the data distributions for all 5 modules which combines the uncorrelated binomial hit noise with the correlated noise introduced by baseline shifts across the strips. The aim of this figure is to compare the width of the fit to the ones of the theoretical curves describing the percentage of CMN in a specific module. Γ ranges from about 2.5 % to 8.1 % across the five modules, confirming that CMN is a real, module-dependent effect and not a fixed detector-wide constant. Specifically, modules 2, 3 and 5 exhibit smaller values for Γ (around 2 to 4 %) compared to modules 1 and 4 clearly showing a larger

deviation from the 0% CMN curve. Therefore, it would be interesting to see if this difference of CMN between modules could be explained at the hybrid level.

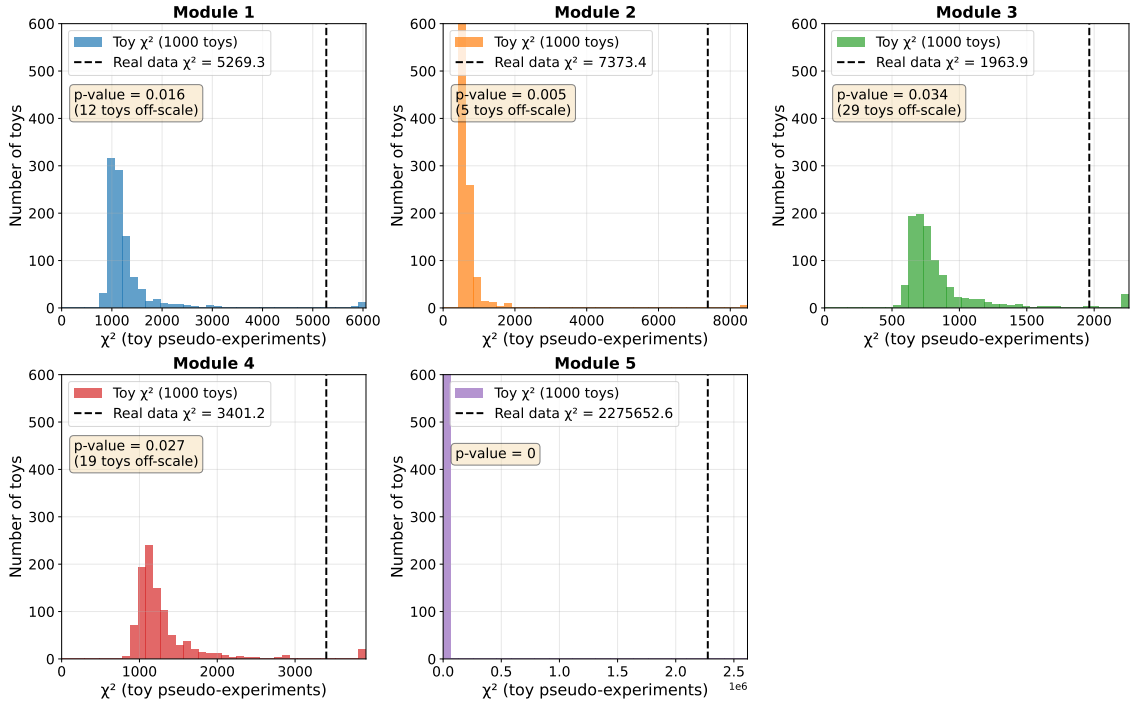


Figure 5.2: Toy-model comparison at the 50 %-occupancy working point: the observed module-level χ^2 overlaid with the sampling distribution of χ^2 under the null hypothesis "the Gaussian fit is correct" of 1000 toy Poisson pseudo datasets drawn from the fitted model's expected counts described in Section 4.3.2. This distribution of toy χ^2 is used to convert the observed χ^2 into a p-value against the pure-binomial (no-CMN) hypothesis.

All 2S modules analysis are done based on the assumption that the CMN follows a binomial distribution. However, the toy p-value obtained over the full range of the histogram (Figure 5.2) firmly rejects the pure-binomial hypothesis for every module ($p < 0.05$, even impossible for module 5). This behavior is expected, and not on its own a strong statement about how non-Gaussian the common-mode fluctuation is: with $\mathcal{O}(4000)$ channels and correspondingly large per-bin statistics, a χ^2 test has enormous statistical power, so even a genuinely real but small departure from a perfect Gaussian shape is enough to drive the full-range p-value to essentially zero. A vanishing full-range p-value therefore mainly confirms that the fit is not exactly perfect somewhere in the histogram, which is expected of any finite-parameter model applied to this much data, rather than indicating a large or physically significant departure from a Gaussian behavior.

Figure 5.3 instead restricts the comparison to the core region defined in Section 4.3.2 (bins with an expected content of at least 5 counts). Analysis revealed that all fits do not describe well the core region of the occupancy distribution except for module 5, which has a larger χ^2 p-value.

Even more informative is the residual structure of Figure 5.4, whose systematic, non-random "W" shape indicates that a purely Gaussian common-mode model, however well its width Γ is tuned, cannot absorb the fluctuations of the data histograms especially for module 1 and 4.

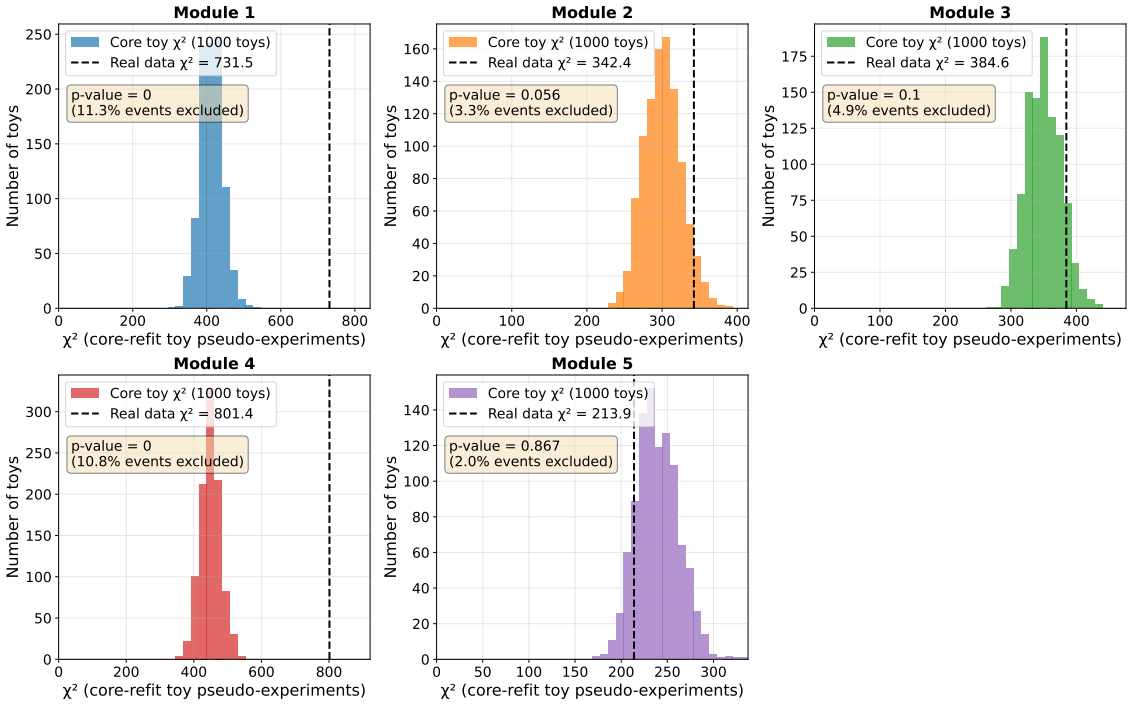


Figure 5.3: Same comparison as in Figure 5.2, now only considering the core data located near the peak of the Gaussian and not the rare events happening at the tail of the distribution.

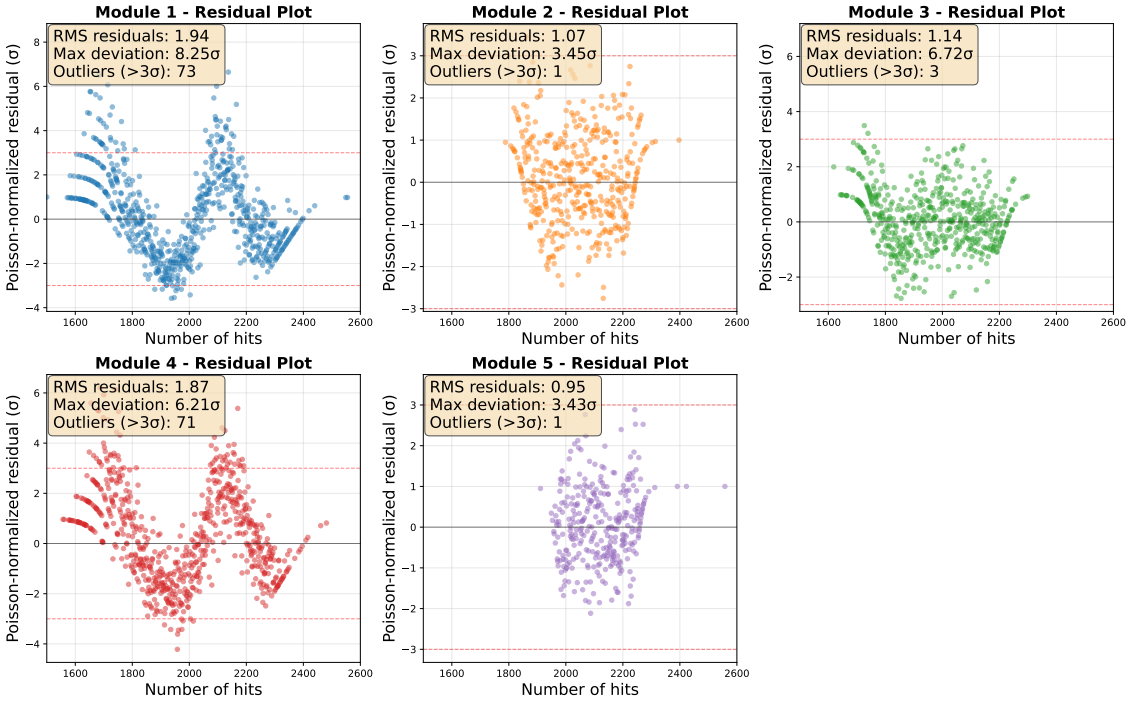


Figure 5.4: Bin-by-bin normalized residuals between the measured module-level 50 % occupancy histogram and the Gaussian common-mode corrected expectation. The residuals show a systematic, roughly "W"-shaped pattern rather than fluctuating randomly around zero. This behavior of the residuals indicates that even a Gaussian-common-mode correction to the binomial model does not fully capture the shape of the data.

50% occupancy distributions at hybrid level

Repeating the 50 % occupancy analysis separately for each of the ten hybrids (Figure 5.5 and 5.6) rather than for the whole module isolates whether the common-mode fluctuation, and any departure from a Gaussian behavior, is already present at the hybrid level or only emerges once the two hybrids of a module are put together. A systematic asymmetry is already visible on these figures: L-hybrid histograms are, on average, shifted on the left and shaped differently compared to R-hybrid histograms of the same modules.

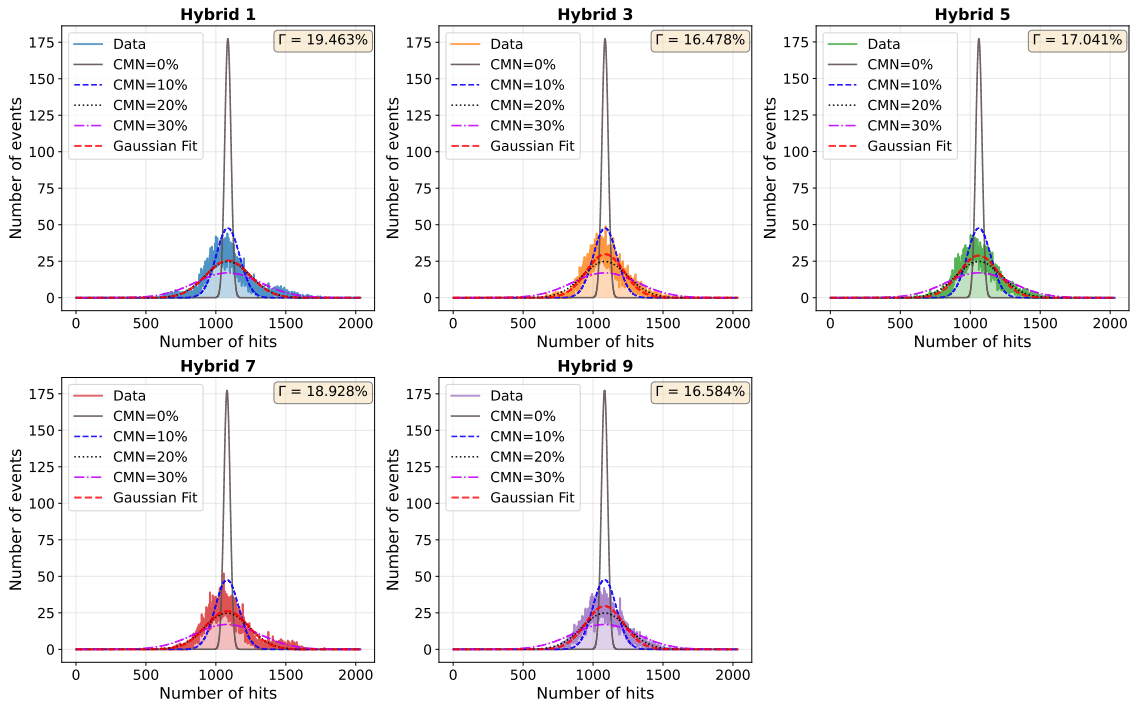


Figure 5.5: 50 % occupancy hit-count histograms for the 5 left burn-in hybrids, each with its own binomial (no-CMN) reference curve and 0, 10, 20 and 30% CMN curves. The value of Γ for each hybrid is displayed on the top right.

Figure 5.7 quantifies this asymmetry by computing the skewness γ_3 of each hybrid's 50 % occupancy distribution and shows a clean sign pattern. Indeed, L-hybrids have, on average, a positive skewness (mean $\gamma_3 \approx +0.475$, a right-leaning tail), while R-hybrids have, on average, a negative skewness (mean $\gamma_3 \approx -0.544$, a left-leaning tail). A purely Gaussian common-mode fluctuation, by construction, produces zero skewness. Therefore, a consistent, opposite-sign skewness on the two sides of every module is additional evidence that the common-mode fluctuation is not simply Gaussian, and also that whatever breaks the Gaussian assumption does so in a way that depends on which side of the module a hybrid sits on.

This L/R asymmetry is attributed to a voltage-drop mechanism on the Service Hybrid (SEH) that distributes low voltage power to the two Front-End Hybrids (FEH). Because the physical wire length from the DC-DC power converter to the left side FEH differs from the wire length to the right side FEH, the two sides experience slightly different voltage drops. This results in a shift of their effective baseline, which leads to a shift in their occupancy at a specific identical threshold setting, in opposite directions. Figure 5.8, which compares the $\mathrm{CM}_{\mathrm{VcTh}}$ of the top-row and

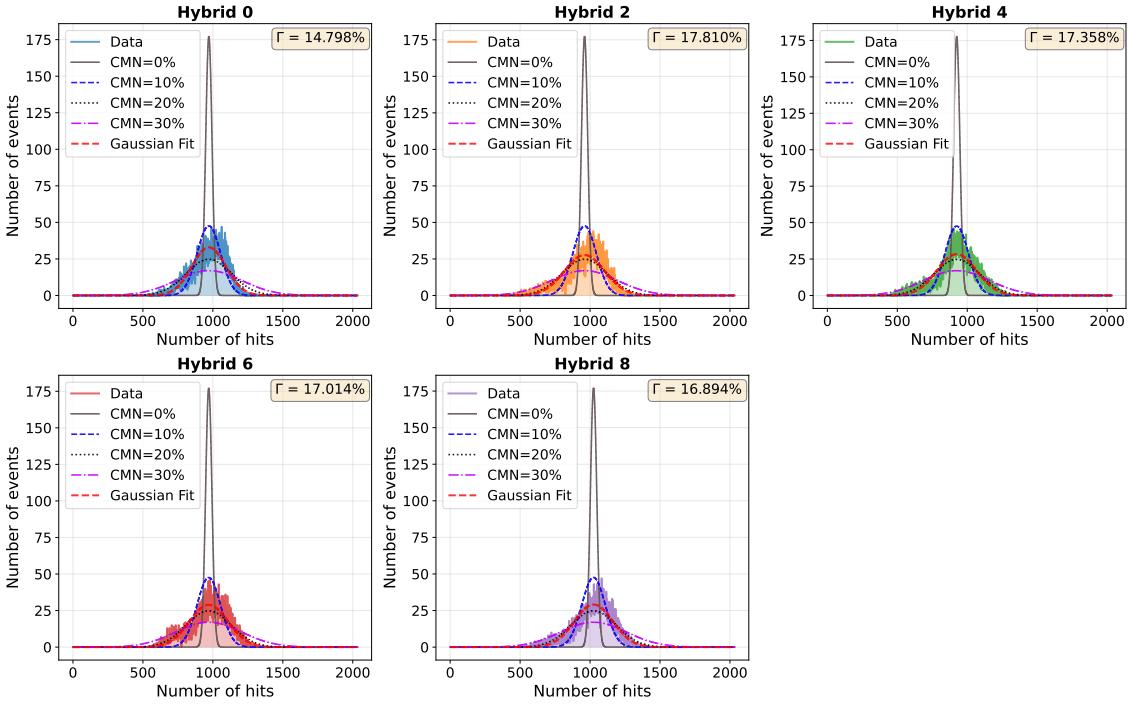


Figure 5.6: 50 % occupancy hit-count histograms for the 5 right burn-in hybrids, each with its own binomial (no-CMN) reference curve and 0, 10, 20 and 30% CMN curves. The value of Γ for each hybrid is displayed on the top right.

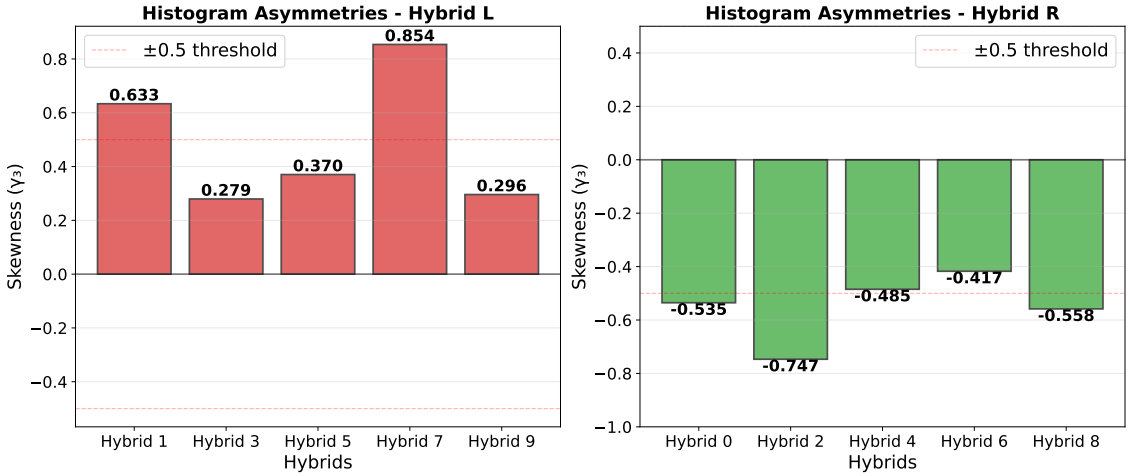


Figure 5.7: Skewness γ_3 of the 50 %-occupancy hit-count distribution, one bar per burn-in hybrid, grouped by side (L, R). L hybrids cluster at positive skewness, R hybrids at negative skewness, with no overlap between the two groups.

bottom-row strips within the same hybrid, is consistent with this interpretation. Indeed, the two strip rows, being again at slightly different distances along the same power distribution path, shows a smaller but qualitatively similar systematic offset. As a result, it reinforces the idea that the position of a group of channels along the power-distribution network, not just which hybrid or which CBC it belongs to, measurably affects its common-mode behaviour.

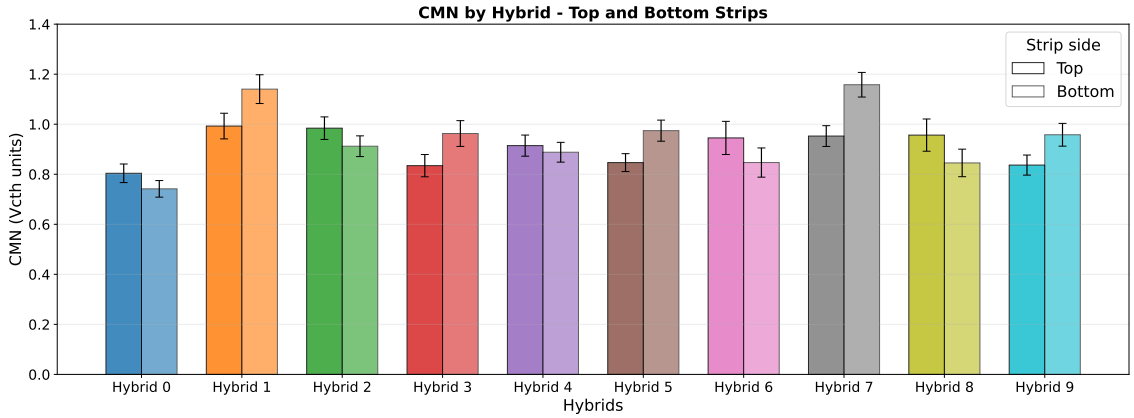


Figure 5.8: $\mathrm{CM}_{\mathrm{VcTh}}$ compared between the top and bottom strip rows of each burn-in hybrid. A small but consistent offset between the two rows is visible within most hybrids, in the same direction across hybrids of the same side.

50% occupancy distributions at CBC level

Analyzing the 50 % occupancy even further, by looking at the CBC-level, shows a monotonic trend of the CMN with respect to the position of the CBC's across a hybrid (Figures 5.9–5.10). Indeed, CMN decreases from CBC 0 to CBC 7 on L-hybrids and increases from CBC 0 to CBC 7 on R-hybrids. This is consistent with the voltage-drop mechanism already mentioned in Section 5.1.1: the CBC index runs along the physical length of the hybrid, so a CBC's distance from the DC-DC converter on the SEH maps directly onto its position in this monotonic trend, with the two hybrid sides sloping in opposite directions because they are serviced from opposite ends of the shared power-distribution path (Figure 5.11).

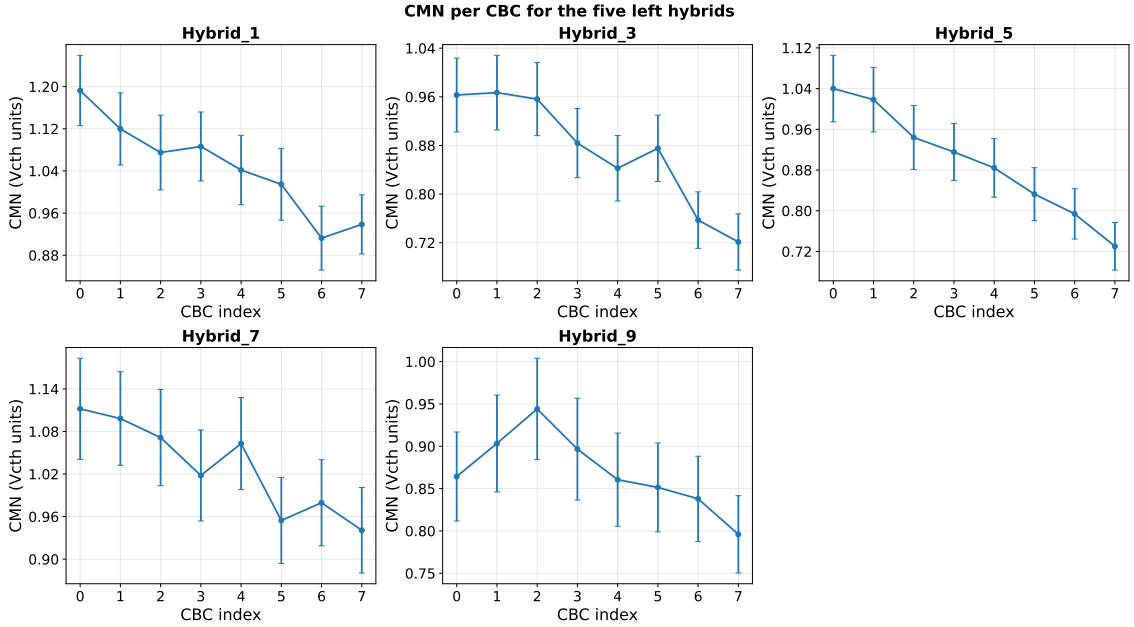


Figure 5.9: $\mathrm{CM}_{\mathrm{VcTh}}$ as a function of CBC index (0 to 7), for each of the five burn-in L hybrids. A decreasing trend from CBC 0 to CBC 7 is visible for every L hybrid.

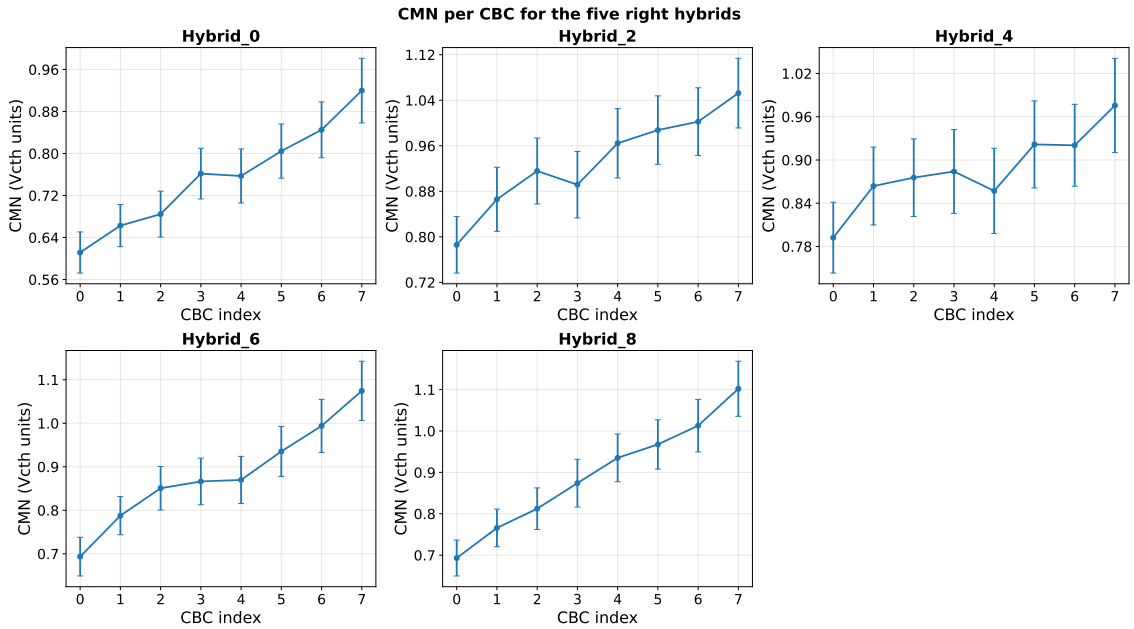


Figure 5.10: Same as Figure 5.9, for the five R hybrids. An increasing trend from CBC 0 to CBC 7 is visible for every R hybrid, opposite in direction to the L-hybrid trend.

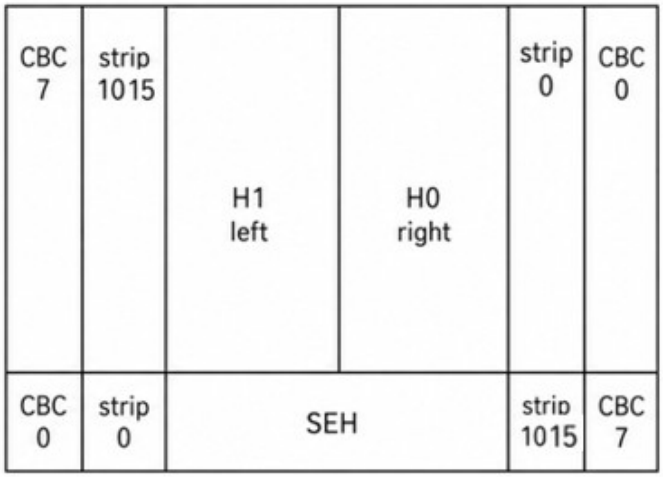


Figure 5.11: Illustration of the position of the CBCs with respect to the service hybrid (SEH) in a 2S module.

3σ hit distributions at module level

The 50 % occupancy analysis of the previous section is, by construction of the Γ estimator, only sensitive to the width of the common-mode fluctuation and blind to its shape. In order to probe the shape directly, every channel's threshold was instead raised to 3σ above its own pedestal: the operating regime described in Section 4.3.2, where essentially only a common-mode fluctuation, not ordinary per-channel noise, can push several channels above threshold in the same event. The resulting rare occupancy hit count histograms were recorded at module level (Figure 5.12). The long high-occupancy tail of the data represented in this figure, far in excess of the no-CMN Poisson prediction, can only be reproduced once a common-mode term is added to the toy model as will be shown.

Figure 5.13 compares the real 3σ-tail hit-count distribution of module 3 to the null-hypothesis toy model of Section 4.3.2 with no common-mode contribution at all ($\sigma_{cm} = 0$), using only each strip's own measured noise σ_i from the binary-search calibration (Section 4.3). Both distributions are shown as a rate (fraction of events) on a logarithmic scale, with their mean, standard deviation, and ratios reported directly on the plot. This isolates how much of the module's 3σ tail behaviour can be explained by ordinary, independent per-channel noise alone.

For module 3, the no common-mode toy model predicts a mean of only 5.5 hits per event with a standard deviation of 2.3, against a real measured mean of 9.1 hits and a standard deviation of 3.6. The toy underestimates the mean by close to 40 % and the spread by about 35 %, and its distribution falls off almost an order of magnitude faster in the tail, effectively vanishing beyond about 15 hits where the real data still extends out to nearly 30. Since this toy model contains no free parameter that could be tuned to close this gap, this systematic, large-scale mismatch is direct, model-independent evidence that independent per-channel noise alone cannot explain the module's rare-tail behaviour, and that a genuine common-mode contribution must be present. This is precisely the comparison that motivates introducing the common-mode width σ_{cm} as a free parameter in the toy model presented in Figure 5.14.

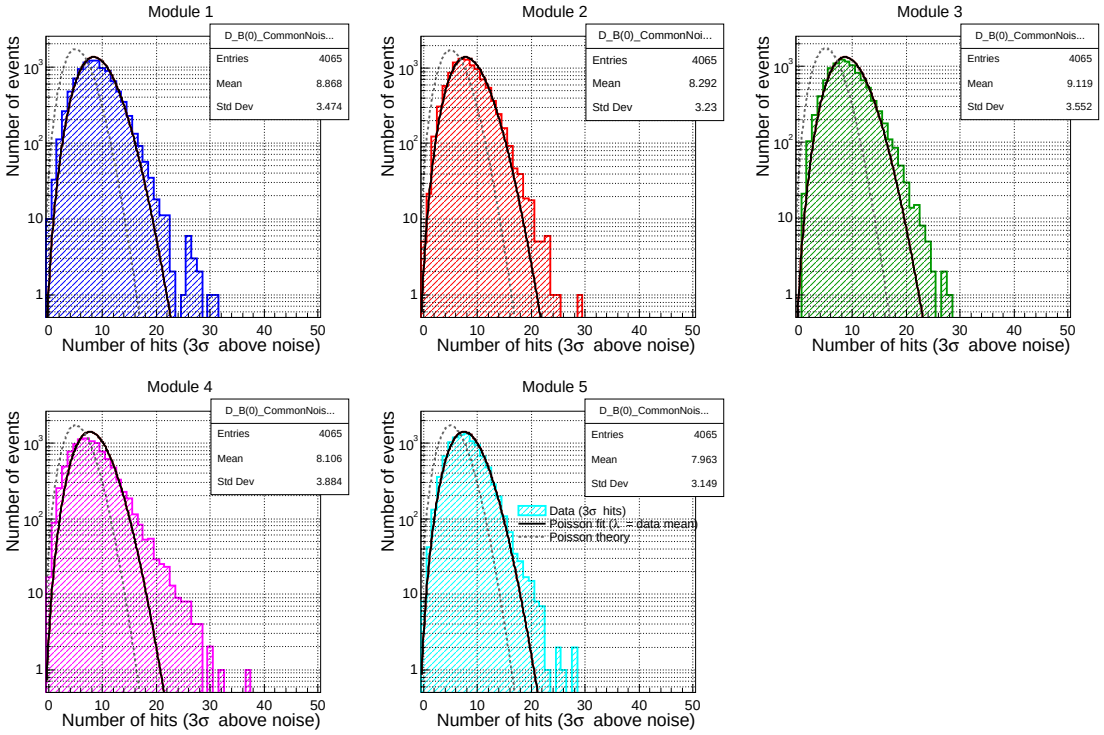


Figure 5.12: Rare occupancy hit-count distribution at the 3σ threshold, module level, for all 5 burn-in modules, together with a Poisson fit of the data and a theoretical Poisson expectation for the no CMN case: if every strip's noise were an independent Gaussian, the number of channels crossing a pure nσ threshold would be Poisson with $\lambda = n_{\text{channels}} \cdot P(Z > n_{\sigma,\text{th}})$ where Z represents a standard normal random variable and $\text{n}_{\text{channels}}$ is the number of channels.

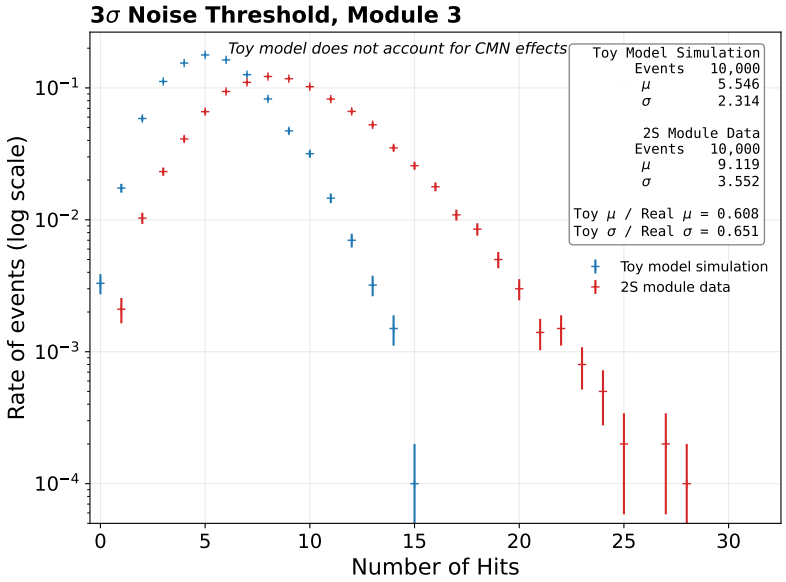


Figure 5.13: 3σ-tail hit-count distribution for Module 3: real data (red) compared to a toy-model simulation with independent per-channel noise only and no common-mode term (blue). The toy model underestimates both the mean ($\mu_{\text{toy}}/\mu_{\text{real}} = 0.61$) and the standard deviation ($\sigma_{\text{toy}}/\sigma_{\text{real}} = 0.65$) of the real distribution, and its tail dies off far faster than the data's which is direct evidence that common-mode noise, not accounted for in this toy model, must be present.

Introducing a single common-mode term shared by every channel of the module across both hybrids and all sixteen CBCs, as in Figure 5.14 (following the same toy-model and Pearson-χ^2-minimization procedure introduced in Section 4.3.2), improves the fit, but not enough to resolve it. For module 3, this gives a best-fit common-mode width of $\sigma_{cm,\text{best}} = 0.901$, in the same VCth units as σ_i.

Table 5.1: Summary of module noise fit parameters ($\sigma_{\text{cm,best}}$), goodness-of-fit metrics (χ^2), and moment ratio comparisons between the shared CMN toy simulation and empirical data.

Module	Real μ	Real σ	σ_i (per-strip)	$\sigma_{\text{cm,best}}$	χ^2	Toy μ / Real μ	Toy σ / Real σ
1	8.87	3.47	5.33	0.89	6597	0.72	1.26
2	8.29	3.23	5.37	0.80	5127	0.75	1.22
3	9.12	3.55	5.25	0.90	7137	0.70	1.25
4	8.11	3.88	5.39	0.87	3003	0.78	1.08
5	7.96	3.15	5.34	0.75	4335	0.77	1.20

Figure 5.14 shows that adding this single, module-wide common-mode term clearly helps the model. For instance, the toy model's mean for module 3 rises from 5.5 to 6.4 hits closing part of the gap to the real mean of 9.1. But the improvement is only partial since the toy's standard deviation now overshoots the real data. In other words, the single value of σ_{cm} that best matches the overall shape of the data cannot simultaneously get both the mean and the spread right. Table 5.1 confirms the results for the 4 other modules. This mean/variance tension is itself informative: it shows that treating the whole module as if they all shared exactly the same common-mode fluctuation is too crude a picture of the real noise struc-

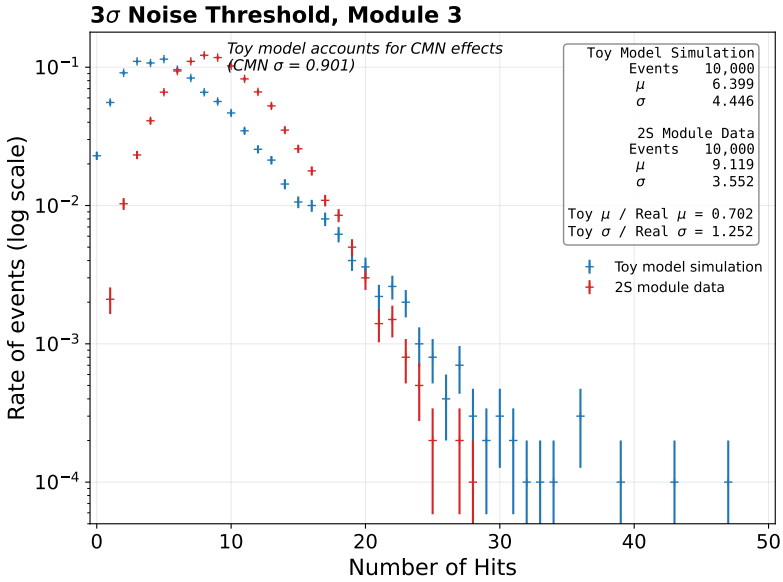


Figure 5.14: Same comparison as Figure 5.13, now with a single shared common-mode width $\sigma_{cm,\mathrm{best}}$ fitted to the data by χ^2 minimization. The bulk of the distribution is reproduced much better, though residual mismatch remains in the extreme tail and, for some modules, in the low-occupancy region.

ture, and motivates the more structured, correlation-aware toy models developed in Figure 5.16.

The anti-correlation between the two hybrids of a module, introduced in Section 4.3.2, was measured directly from the 3σ dataset by correlating, event by event, the number of hits on the L-hybrid against the number of hits on the R-hybrid of the same module (Figure 5.15). Across the five burn-in modules, the measured correlation coefficient ρ between L- and R-hybrids ranges from -0.57 to -0.93: a clear, strongly negative correlation rather than the independence that the naive single shared common-mode toy model implied. Physically, this anti-correlation is best understood as a consequence of the left and right hybrids sharing a single, finite power source on the module. Given that the voltage budget available from that shared source is limited, a fluctuation that momentarily increases the current drawn by one hybrid leaves correspondingly less available for the other hybrid on that same event, shifting its effective baseline in the opposite direction. Unlike the static L/R asymmetry discussed before, this anti-correlation is a dynamic, event-by-event effect: it concerns how the two hybrids' fluctuations around their own baselines move together, not how their average baselines differ. Therefore, the static asymmetry and the anti-correlation are two distinct consequences of the same shared power infrastructure: the asymmetric wiring sets the average operating point of each side, while the shared, finite instantaneous supply couples their fluctuations together independently of that average difference.

Applying the anti-correlated hybrid and per-CBC one-factor model of Section 4.3.2 (Eqs. (4.5)–(4.8)) resolves most of the mean/variance tension left by the previous toy models, using the module's own measured L/R correlation ρ (Figure 5.15) and per-CBC correlation $\rho_{\text{per-CBC}}$ (Table 5.2) rather than assuming full correlation on either side. The common-mode widths, $\sigma_{cm,R}$ and $\sigma_{cm,L}$, are still tuned by the same χ^2-minimization procedure used for the single shared value above.

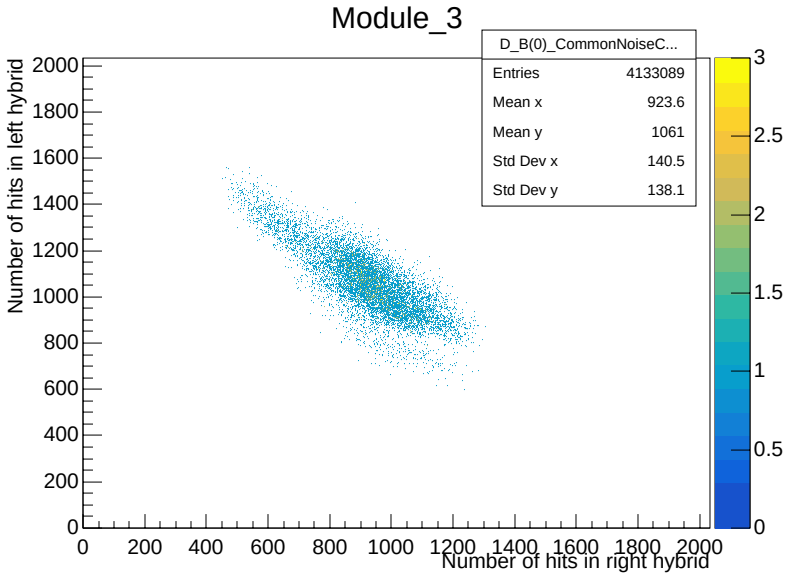


Figure 5.15: Event-by-event number of 3σ hits on the L-hybrid versus the R-hybrid of the same module, for one representative burn-in module. The negative slope of the point cloud is the direct, model-independent evidence of the L/R anti-correlation reported in the text.

Table 5.2: Mean per-CBC-to-hybrid correlation ($\rho_{\text{per-CBC}}$), averaged over the eight CBCs of each hybrid, across the five R and five L hybrids of the 2S burn-in dataset.

	Hybrid 0	**Hybrid 2**	**Hybrid 4**	**Hybrid 6**	**Hybrid 8**	**Mean (R)**
Mean $\rho_{\text{per-CBC}}$	0.351	0.377	0.385	0.377	0.394	0.377
	Hybrid 1	**Hybrid 3**	**Hybrid 5**	**Hybrid 7**	**Hybrid 9**	**Mean (L)**
Mean $\rho_{\text{per-CBC}}$	0.475	0.425	0.44	0.561	0.411	0.463

Figure 5.16 shows that the new toy model applied for the data of module 3 brings a $\mu_{\text{toy}}/\mu_{\text{real}}$ ratio of 0.98, essentially matching the data, compared to 0.70 for the single shared-value model. The standard deviation also improves, from an overshoot of 1.25 down to 1.15, though it is not fully resolved: the toy model still predicts a slightly broader distribution than the data shows, most visibly in the tail beyond about 25 hits, where the real data drops off faster than the simulation does. Table 5.3 confirms the results for the 4 other modules. The fact that the mean/variance tension observed with the single shared common-mode toy model is largely absorbed once the module's anti-correlation and per-CBC structure are accounted for, indicates that most of the mismatch was caused by the oversimplified assumption of one shared value across all sixteen CBCs and both hybrids. The small residual overshoot that remains points to the additional, non-Gaussian component of the CMN discussed further in this section.

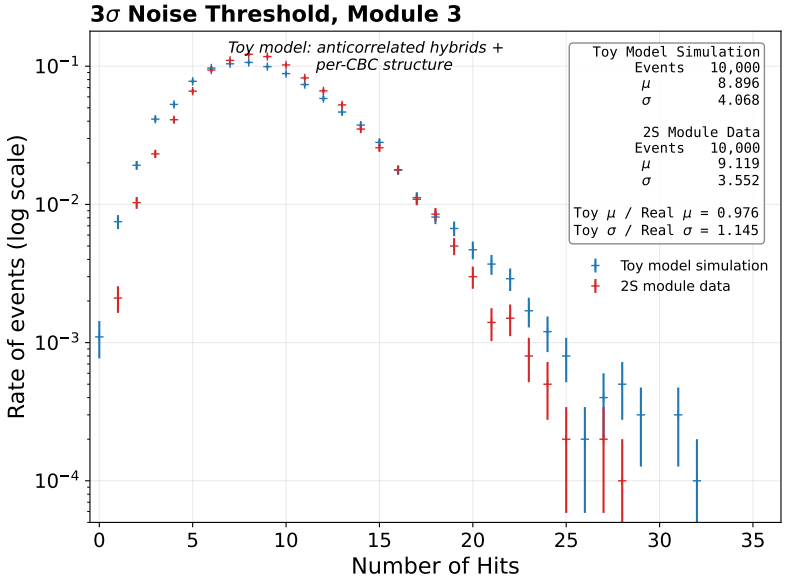


Figure 5.16: Same comparison, now with the anti-correlated two-hybrid + correlated CBC-hybrid toy model of Section 4.3.2, using the module's own measured L/R correlation coefficient ρ and CBC's own measured correlation to the rest of its hybrid $\rho_{\text{per-CBC}}$. This further improves the agreement in the tail relative to Figure 5.14 for modules, particularly for the ones with the strongest measured anti-correlation.

Table 5.3: Same as Table 5.1 but also reports, per module, the best-fit common-mode width obtained independently for each hybrid side, $\sigma_{cm,R}$ and $\sigma_{cm,L}$, together with a module-level combination $\sigma_{cm,\text{module}}$ that folds in the measured anti-correlation ρ between the two sides and the CBC-to-hybrid correlation.

Module	Real mean	Real σ	σ_i (per-strip)	$\sigma_{\text{cm,R}}$	$\sigma_{\text{cm,L}}$	$\sigma_{\text{cm,module}}$	χ^2	Toy μ/Real μ	Toy σ/Real σ
1	8.87	3.47	5.33	1.69	1.65	1.00	605	0.97	1.17
2	8.29	3.23	5.37	1.57	1.54	0.77	365	0.98	1.10
3	9.12	3.55	5.25	1.69	1.74	0.90	438	0.98	1.15
4	8.11	3.88	5.39	1.84	1.05	0.90	69	0.99	1.00
5	7.96	3.15	5.34	1.50	1.43	0.71	254	0.98	1.07

As was shown previously, ρ is consistently strongly negative for every 2S module studied. Therefore, Eq (4.8) produces genuine cancellation rather than a simple combination of the two hybrid widths: when the R-hybrid common-mode fluctuation pushes the module's occupancy up, the L-hybrid fluctuation tends to pull it back down by a similar amount, so a large part of each side's individual fluctuation is absorbed once both hybrids are considered together as a single module. This is why the common-mode noise measured at the level of the complete module is not representative of the common-mode noise actually present within each of its hybrids. This is directly visible in Table 5.3: for all modules, $\sigma_{cm,\text{module}}$ comes out smaller than either $\sigma_{cm,R}$ or $\sigma_{cm,L}$ individually, rather than lying between them as a naive average would. Table 5.3 also shows that modules whose R and L widths are the most mismatched are generally the ones with higher value of $\sigma_{cm,\text{module}}$. This is consistent with the cancellation mechanism itself since cancellation is most effective when the two sides' widths are comparable, and correspondingly weaker when one side dominates the other.

It is precisely the same cancellation mechanism, applied to the pedestal measurement rather than to the 3σ tail, that explains why Γ measured directly on the combined module data comes out so much smaller than Γ measured on either hybrid alone (2.5 % to 8.1 % at module level, against 14.8 % to 19.5 % at hybrid level, for the same five burn-in modules). The module's combined variance is itself reduced by the anti-correlation between its two hybrids before Γ is ever extracted from it. So a measurement made only at module level substantially underestimate how much CMN each hybrid individually contributes.

3σ occupancy distributions at hybrid level

At the single-hybrid level, the eight CBCs that compose a hybrid were found not to share one single, fully common-mode value but instead to be only partially correlated with the rest of their hybrid. The per-CBC correlation coefficient $\rho_{\text{per-CBC}}$ averages to about 0.377 on the R side and about 0.463 on the L side (Table 5.2): a real but far-from-unity correlation on both sides, somewhat weaker on R-hybrid than on L-hybrid. As a result, replacing the naive fully-correlated toy model with this measured $\rho_{\text{per-CBC}}$ model changes the fit quality very differently on the two sides (Tables 5.4).

On R-hybrids, the naive model systematically undershoots the mean of the measured hit-count distribution (the toy-to-data mean ratio is typically around 0.7) because assuming perfect CBC-to-hybrid correlation underestimates how often several CBCs happen to fire a hit together. Therefore, introducing the measured, weaker correlation corrects this undershoot (the toy-to-data mean ratio rises to about 0.95) and improves the χ^2 by a factor of roughly 5 to 14.

On L-hybrids, in contrast, the naive model already reproduces the data's mean and standard deviation almost exactly because L-hybrids operate at a systematically lower mean occupancy than R-hybrids at the same 3σ threshold as a consequence of the same baseline-shift mechanism already discussed before. At low occupancy, the exact value of the CBC-to-hybrid correlation matters much less to the shape of the resulting hit-count distribution, so introducing the measured correlation actually makes the toy model fit to the data worse in four of the five L-hybrids. The one L-hybrid that does improve under the correlated model (Figure 5.17) is also the L hybrid with the highest mean occupancy of the five, supporting an occupancy-driven

(instead of a strictly L/R-driven) explanation for which model variant is preferable for a given hybrid. From this analysis, one can deduce that the correlated model helps most whenever the mean occupancy is high enough that the exact CBC-to-hybrid correlation shapes the tail of the distribution, and R hybrids happen to sit in that regime far more often than L hybrids do.

Table 5.4: Comparison of CMN fit parameters, χ^2 goodness-of-fit, and moment ratios at the best-fit $\sigma_{cm,best}$ between the naive fully-correlated model and the measured $\rho_{\text{per-CBC}}$ model + anti-correlated hybrid's side, for all ten burn-in hybrids grouped by side. The mean ratio is what moves markedly closer to 1 for R hybrids under the correlated model (from about 0.73 to about 0.97, averaged over the five R-hybrids); the standard-deviation ratio stays overshot at around 1.2 under both models and is not meaningfully improved by the correlated fit. Most L-hybrids have a mean ratio already close to 1 under the naive model.

		Naive Model				**Corr Model**			
Hybrid	**Side**	$\sigma_{\text{cm,best}}$	χ^2	**Toy** μ / **Real** μ	**Toy** σ / **Real** σ	$\sigma_{\text{cm,best}}$	χ^2	**Toy** μ / **Real** μ	**Toy** σ / **Real** σ
0	R	1.381	6305	0.698	1.364	1.927	1188	0.931	1.272
2	R	1.257	2718	0.767	1.159	1.782	223	0.984	1.111
4	R	1.349	4682	0.708	1.215	1.897	679	0.939	1.175
6	R	1.214	3166	0.746	1.149	1.757	316	0.967	1.131
8	R	1.252	3034	0.747	1.145	1.781	299	0.962	1.114
1	L	1.057	287	0.993	0.981	1.378	1053	1.131	0.907
3	L	1.035	307	0.917	0.989	1.387	481	1.061	0.914
5	L	1.038	526	0.885	0.986	1.456	292	1.055	0.955
7	L	1.160	471	1.026	0.930	1.592	1430	1.230	0.966
9	L	0.921	46	1.011	1.022	1.159	799	1.104	0.902

Moreover, comparing CM_{VcTh} (the moment-based estimator obtained from the 50 % distributions) to $\sigma_{cm,\text{best}}$ (the shape-based, 3σ-tail estimator) for the same group of channels is one of the main point of this chapter. As a matter of fact, the two estimators do not agree, since they are built from different regions of the hit-count distribution and are only guaranteed to coincide if the common-mode fluctuation is exactly Gaussian. Table 5.5 summarises the ratio $\sigma_{cm,\text{best}}/\text{CM}_{\text{VcTh}}$ per hybrid: the tail-fitted width $\sigma_{cm,\text{best}}$ is systematically larger than the bulk, moment-based estimate CM_{VcTh}, and even more so for R-hybrids (a ratio between about 1.9 and 2.5) than for L-hybrids (between about 1.3 and 1.6). This direction of the mismatch is itself informative since a ratio $\sigma_{cm,\text{best}}/\text{CM}_{\text{VcTh}} > 1$ is precisely the signature expected if the true common-mode fluctuation has heavier-than-Gaussian tails. This means that rare, unusually large common-mode excursions occur more often than a Gaussian of the width measured from the bulk would predict. Since this departure is consistently larger on the R-hybrids than on the L-hybrids across all five modules, the non-Gaussian component of the CMN is obviously not a random, module-to-module fluctuation but a systematic, side-dependent feature of the module design. This pattern will be tied together with the other diagnostics of this section into a single physical picture.

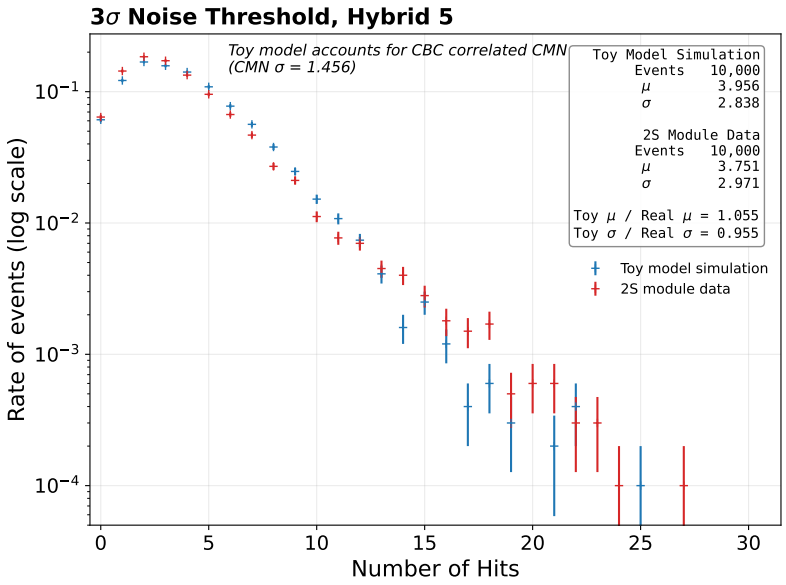


Figure 5.17: Hybrid-level 3σ hit-count data compared to the toy model using the measured per-CBC correlation $\rho_{\text{per-CBC}}$ of Table 5.2, for hybrid 5 of the left side. This situation is not representative of the L side as a whole: hybrid 5 is the only L-hybrid on which the toy model accounting for the per-CBC correlation performs better than the shared, fully-correlated CMN model. On the R-hybrid, accounting for this correlation actually improves the fit for every hybrid.

Table 5.5: Comparison of common-mode noise estimators ($\text{CM}_{V_{\text{cTh}}}$ vs. $\sigma_{\text{cm,best}}$) and their ratio across Right (R) and Left (L) sensor hybrids.

Hybrid	**Side**	$\text{CM}_{V_{\text{cTh}}}$ **(Γ method)**	$\sigma_{\text{cm,best}}$ **(3σ toy fit)**	**Ratio**
0	R	0.773	1.927	2.49×
2	R	0.951	1.782	1.87×
4	R	0.901	1.897	2.10×
6	R	0.897	1.757	1.96×
8	R	0.904	1.781	1.97×
1	L	1.057	1.378	1.30×
3	L	0.890	1.387	1.56×
5	L	0.905	1.456	1.61×
7	L	1.042	1.592	1.53×
9	L	0.883	1.159	1.31×

5.1.2 Sector tests CMN

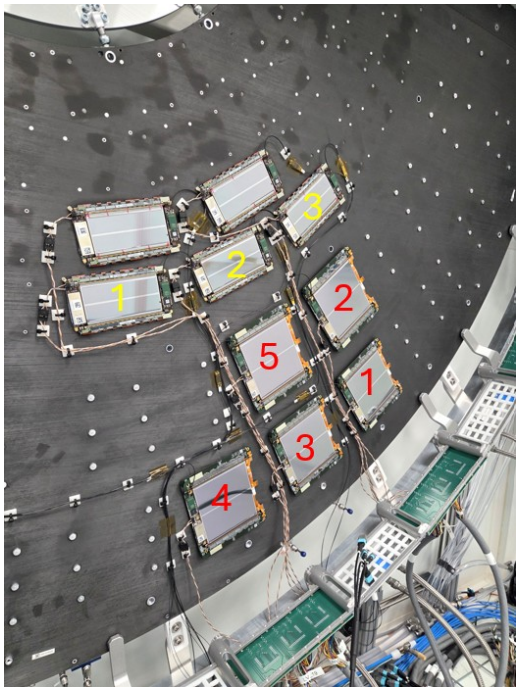


Figure 5.18: Photograph of the five 2S modules, labeled in red, and five PS modules, labeled in yellow, mounted together onto the dee test structure used for the sector common-mode noise measurements described in this section (3 out of those 5 PS modules are used in this study).

As explained in Section 4.1, studying individual modules through burn-in tests is not sufficient on its own since modules will ultimately be operated together within a large mechanical support structure known as a dee in the final detector. Several modules mounted together onto a dee share services (cooling, mechanical support and parts of the low-voltage power distribution) that a single module on its own burn-in bench does not experience. Therefore, testing modules mounted onto a dee reproduces conditions much closer to those of the final detector and can reveal common-mode noise contributions that burn-in testing alone cannot (Figure 5.18). The same five 2S modules and the same $\Gamma/\mathrm{CM}_{\mathrm{VcTh}}$ and $\sigma_{cm,\mathrm{best}}$ methodology of Section 4.3.2 were applied again to a dataset taken with the modules mounted together onto a dee test structure, so that the two configurations (burn-in and dee test) can be compared directly.

50% occupancy distributions at module level

Figure 5.19 summaries the module-level 50 % occupancy analysis for the five dee modules, obtained in exactly the same way as for burn-in: a ROOT binned Poisson-likelihood Gaussian fit to the module-level hit-count histogram (combining both hybrids, $n \approx 4064$ channels) gives the mean and standard deviation entering Eqs. (4.3)–(4.4). The range of Γ across the five dee modules (2.4 %–3.1 %) is narrower, and sits toward the lower end, than the wider 2.5 %–8.1 % range observed module-to-module in burn-in: on the dee, the five modules behave noticeably more uniformly with respect to each other than they did individually on their own burn-in benches. This is consistent with the dee's shared services acting, to some extent, as a common denominator that evens out module-to-module differences in common-mode behaviour that were still visible when each module was tested in isolation.

As for dee tests, the pure-binomial (no-CMN) hypothesis is essentially always rejected once the full range of the histogram is used (every full-range toy χ^2 p-value obtained for these five modules rounds to 0). Restricting the comparison to the well-populated core of the histogram gives a much more informative, module-dependent picture, ranging from a good core-region description (Modules 1, 3 and 4, χ^2 toy p

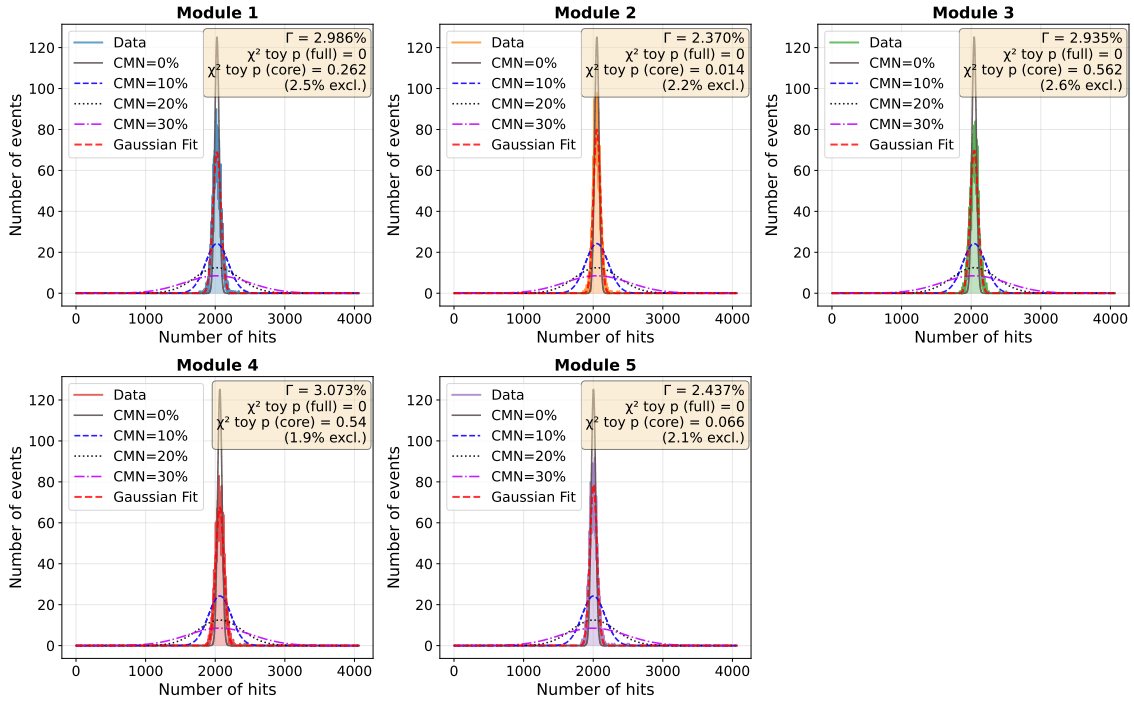


Figure 5.19: Module-level 50 % occupancy histograms for the five 2S modules on the Dee test structure: the expected no-common-mode (pure binomial) curve at the same mean as the data, the resulting common-mode fraction Γ, and the Poisson-toy p-value of the pure-binomial hypothesis for the full dataset and restricted to the well-populated core region of the histogram.

(core)>0.05) to a marginal one (Module 2, χ^2 toy p (core)$=0.014$). The same "the model can describe the bulk of the distribution reasonably well, but the extreme tail is where it is stressed" pattern already identified for burn-in modules in Section 5.1.1 is observed here as well.

50% occupancy distributions at hybrid level

Splitting the same 50 % occupancy analysis by hybrid side reproduces, on the dee dataset, the same qualitative L/R asymmetry already established for burn-in hybrids in Section 5.1.1, but more pronounced. The skewness of the individual hybrid 50 % occupancy distributions (Figure 5.20) shows the same sign pattern as burn-in: L-hybrids positively skewed, R-hybrids negatively skewed but with a substantially larger magnitude. Indeed, on the dee, L-hybrid skewness averages $\gamma_3 \approx +0.71$ (ranging 0.56 to 0.83 across the five L-hybrids) and R-hybrid skewness averages $\gamma_3 \approx -0.84$ (ranging -0.67 to -0.97), compared to averages of about $+0.475$ and -0.544 respectively for the same quantity in burn-in. Since the sign and general mechanism are unchanged but the magnitude grows once several modules share services on the dee, this is consistent with the same wire-length/voltage-drop asymmetry between the L- and R-hybrids power-distribution paths being present on burn-in already. It is simply amplified by the additional shared power and grounding infrastructure of the dee structure.

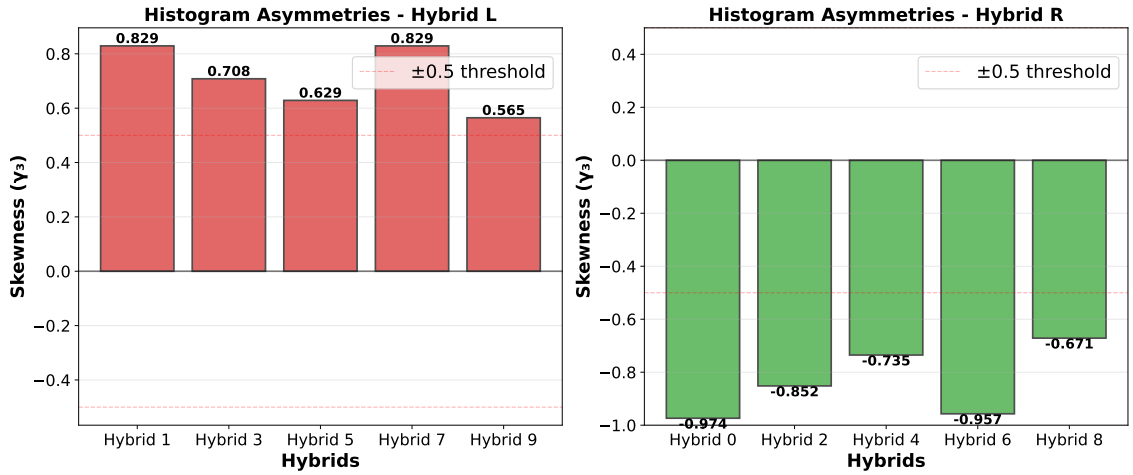


Figure 5.20: Skewness γ_3 of the 50 % occupancy hit-count distribution, one bar per dee hybrid, grouped by side (L, R). As for burn-in (Figure 5.7), L-hybrids are consistently positively skewed and R-hybrids consistently negatively skewed, here with a larger average magnitude on both sides.

50% occupancy distributions at CBC level

As for burn-in (Section 5.1.1), $\mathrm{CM}_{\mathrm{VcTh}}$ was also computed per individual CBC on the dee dataset (Figure 5.21–5.22). The same monotonic trend with CBC index across a hybrid is recovered (decreasing along L-hybrids, increasing along R-hybrids) and is consistent with the same DC-DC-converter-proximity mechanism [40] identified for burn-in persisting once the modules are mounted together onto the Dee.

3σ hit distributions at module level

Raising every channel's threshold to 3σ above pedestal and repeating the module-level toy model fit of Section 5.1.1 i.e. the anti-correlated two-hybrid model, using the L/R correlation coefficient ρ measured directly on the dee data, and, the CBC-correlated refinement within each hybrid side, improves the description substantially (Table 5.6). This table shows that the module-level fit converges to distinct $\sigma_{cm,R}$ and $\sigma_{cm,L}$ values for the two sides, with $\sigma_{cm,R}$ systematically larger than $\sigma_{cm,L}$ for all five modules, and a combined module-level $\sigma_{cm,\mathrm{module}}$ that folds in the measured L/R anti-correlation and the CBC-to-hybrid correlation.

Table 5.6: Module-level 3σ-tail toy-model results for the five 2S dee modules, using the anti-correlated two-hybrid, CBC-correlated model: best-fit common-mode width per side ($\sigma_{cm,R}$, $\sigma_{cm,L}$), the resulting module-level combination $\sigma_{cm,\mathrm{module}}$, the best-fit χ^2, and the toy-to-data mean and standard-deviation ratios at the best fit.

Module	Real μ	Real σ	σ_i (per-strip)	$\sigma_{\mathrm{cm,R}}$	$\sigma_{\mathrm{cm,L}}$	$\sigma_{\mathrm{cm,module}}$	χ^2	Toy μ / Real μ	Toy σ / Real σ
1	8.86	3.50	5.39	1.81	1.55	0.83	441	0.98	1.12
2	8.46	3.47	5.42	1.76	1.47	0.77	280	0.99	1.07
3	8.48	3.33	5.32	1.61	1.60	0.79	405	0.99	1.11
4	8.30	3.59	5.47	1.87	1.12	0.74	274	0.98	1.04
5	8.79	3.37	5.39	1.80	1.55	0.80	377	0.98	1.14

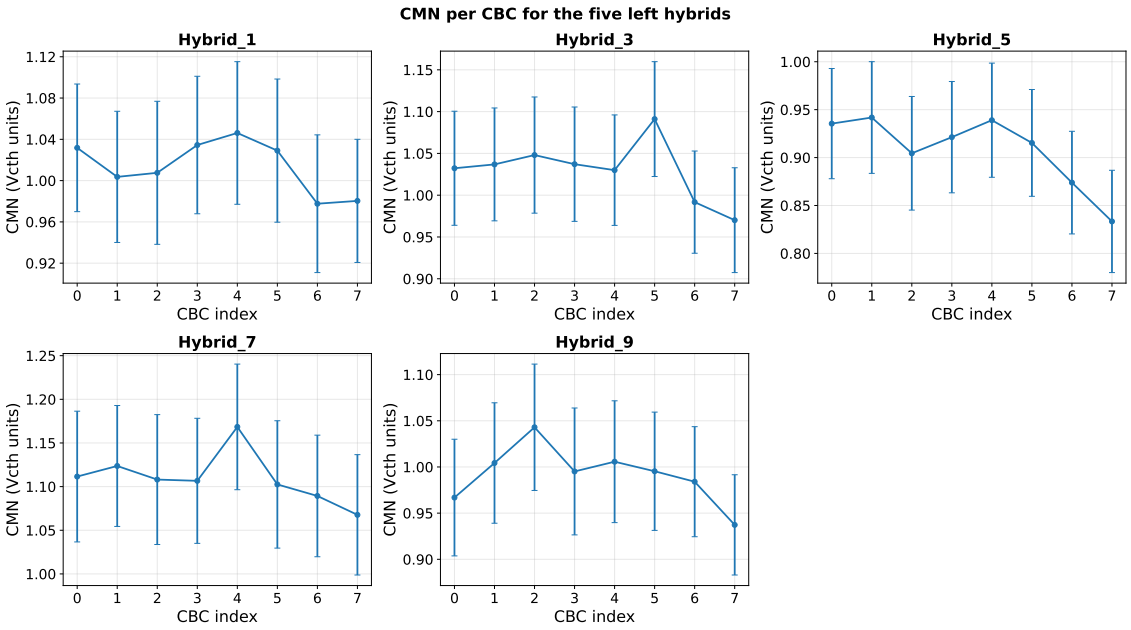


Figure 5.21: CM_{VcTh} as a function of CBC index (0 to 7), one panel per dee hybrid, grouped by module. As on burn-in (Figures 5.9), a monotonic trend across the CBC index is visible within every hybrid, decreasing for L-hybrids.

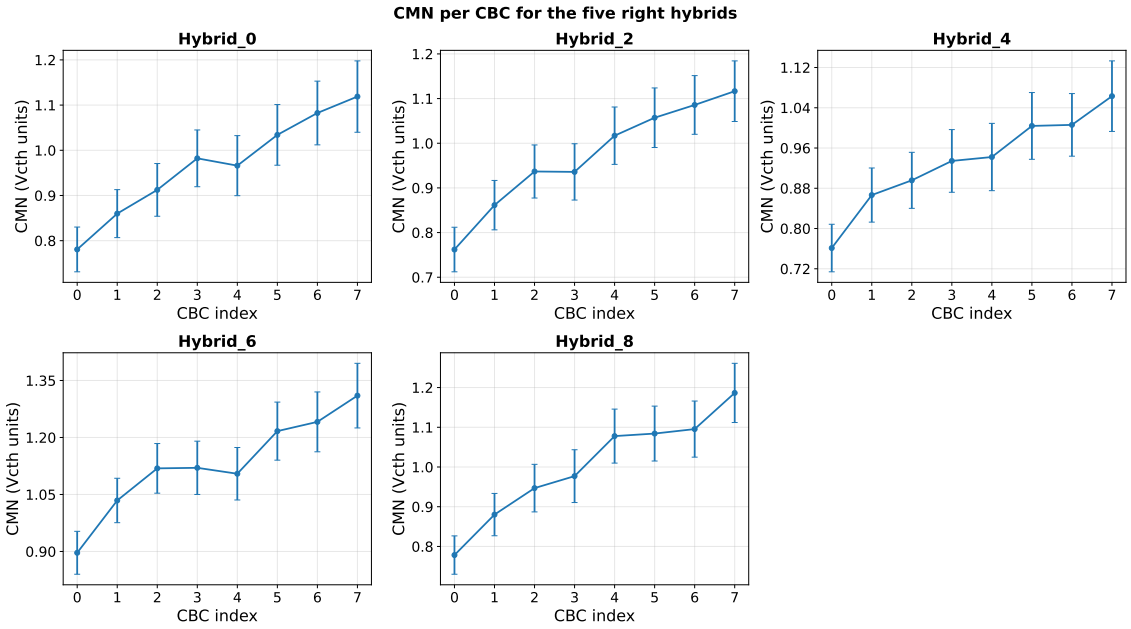


Figure 5.22: CM_{VcTh} as a function of CBC index (0 to 7) for all left hybrids. As on burn-in (Figures 5.10), a monotonic trend across the CBC index is visible within every hybrid, increasing for R-hybrids.

The mean occupancy is reproduced to within 2 % for every module (toy/data mean ratio between 0.98 and 0.99), while the standard deviation is systematically overestimated by 4 % to 14 %: the same qualitative mean/variance tension already seen in the burn-in analysis persists here at the combined-module level. This again points toward a residual non-Gaussian component of the common-mode fluctuation that a Gaussian-width toy model, however well-tuned its σ_{cm}, systematically compensates for by slightly over-broadening the bulk of the simulated distribution.

3σ occupancy distributions at hybrid level

The per-CBC-to-hybrid correlation coefficient $\rho_{\text{per-CBC}}$ was remeasured directly on the dee dataset for all ten hybrids, giving a mean of about 0.366 on the R side and about 0.514 on the L side (Table 5.7). One can essentially observe the same magnitude and the same L-above-R ordering already found for burn-in hybrids ($\rho_{\text{per-CBC}} \approx 0.377$ on R, ≈ 0.463 on L, Section 5.1.1). This reproducibility across two independent test and mounting configurations supports the interpretation that $\rho_{\text{per-CBC}}$ reflects an intrinsic property of the module and hybrid design (most likely the internal layout and grounding of the CBC-to-hybrid connections) rather than a feature specific to a particular test bench or mechanical environment.

Table 5.7: Mean per-CBC-to-hybrid correlation coefficient $\rho_{\text{per-CBC}}$, averaged over the eight CBCs of each hybrid, for the five R and five L hybrids of the 2S dee dataset.

	Hybrid 0	**Hybrid 2**	**Hybrid 4**	**Hybrid 6**	**Hybrid 8**	**Mean (R)**
Mean $\rho_{\text{per-CBC}}$	0.369	0.344	0.386	0.375	0.355	**0.366**
	Hybrid 1	**Hybrid 3**	**Hybrid 5**	**Hybrid 7**	**Hybrid 9**	**Mean (L)**
Mean $\rho_{\text{per-CBC}}$	0.525	0.540	0.466	0.477	0.561	**0.514**

Table 5.8 gives the corresponding hybrid-level 3σ-tail toy-model fit, using the measured $\rho_{\text{per-CBC}}$ of each hybrid. The same R/L split already identified for burn-in hybrids in Section 5.1.1 reappears clearly here: for the five R-hybrids, the model reproduces the mean almost exactly (toy/data mean ratio 0.938–1.003) but overshoots the standard deviation by 15 % to 32 %; for the five L-hybrids, the model instead overshoots the mean by 10 % to 21 % while reproducing the standard deviation to within about 10 % (ratio 0.906–0.962). This is the mirror image of the mean-undershoot/variance-overshoot tension already described for burn-in R-hybrids and the good naive-model agreement already described for burn-in L-hybrids in Section 5.1.1: R hybrids operate, on the dee as on burn-in, at a higher 3σ mean occupancy (about 4.7–5.5 hits) than L-hybrids (about 3.3–3.6 hits), and it is precisely in this higher-occupancy regime that the exact CBC-to-hybrid correlation structure has the largest effect on the shape of the resulting hit-count distribution.

The burn-in diagnostic of Section 5.1.1 (the ratio $\sigma_{cm,\text{best}}/\text{CM}_{\text{VcTh}}$) can be recomputed independently on the dee dataset, combining the hybrid-level CM_{VcTh} of Section 5.1.2 with the $\sigma_{cm,\text{best}}$ of Table 5.8 (Table 5.9). The result reproduces the burn-in pattern essentially unchanged: $\sigma_{cm,\text{best}}/\text{CM}_{\text{VcTh}} > 1$ for every hybrid on both sides, averaging about 1.90 on R side (range 1.64–2.18) and about 1.53 on L side (range 1.31–1.63) compared to about 1.9–2.5 on R side and 1.3–1.6 on L side for burn-in. That two fully independent mounting configurations and datasets return

the same side-dependent magnitude for this ratio, to within the module-to-module scatter already present within each test, is strong evidence that the heavy-tailed component of the common-mode noise identified in Section 5.1.1 is a genuine property of the module and hybrid design, rather than an artefact specific to the mounting configuration or dataset.

Table 5.8: Performance metrics and moment ratios comparing the naive uncoupled model ($\rho = 1$) against the correlated model across Right (R) and Left (L) front-end hybrids.

		Naive Model				Corr Model			
Hybrid	**Side**	$\sigma_{\text{cm,best}}$	χ^2	**Toy** μ / **Real** μ	**Toy** σ / **Real** σ	$\sigma_{\text{cm,best}}$	χ^2	**Toy** μ / **Real** μ	**Toy** σ / **Real** σ
0	R	1.285	3555	0.737	1.168	1.863	459	0.974	1.147
2	R	1.403	5559	0.719	1.340	1.918	897	0.938	1.202
4	R	1.461	4866	0.707	1.258	2.118	709	0.997	1.301
6	R	1.233	3721	0.721	1.136	1.824	482	0.963	1.158
8	R	1.275	2961	0.771	1.190	1.824	377	1.003	1.137
1	L	1.237	397	0.999	0.978	1.620	1128	1.184	0.942
3	L	1.224	442	1.004	0.959	1.644	1252	1.210	0.962
5	L	1.048	135	1.015	0.969	1.451	1316	1.199	0.945
7	L	1.050	282	0.947	0.943	1.424	852	1.103	0.906
9	L	1.414	341	1.036	1.041	1.692	1242	1.178	0.925

Table 5.9: Hybrid-level $\sigma_{cm,\text{best}}/\text{CM}_{V_{cTh}}$ ratio for the ten 2S dee hybrids, combining the $\text{CM}_{V_{cTh}}$ of Section 5.1.2 with the σ_{cm} of Table 5.8.

Hybrid	**Side**	$\text{CM}_{V_{cTh}}$ (Γ method)	$\sigma_{cm,\text{best}}$ (3σ toy fit)	**Ratio**
0	R	0.993	1.863	1.88
2	R	0.973	1.918	1.97
4	R	0.974	2.118	2.18
6	R	1.114	1.824	1.64
8	R	1.003	1.824	1.82
1	L	1.028	1.620	1.58
3	L	1.048	1.644	1.57
5	L	0.926	1.451	1.57
7	L	1.091	1.424	1.31
9	L	1.040	1.692	1.63

Comparison burn-in/dee

As a summary of the hybrid-level CM_{VcTh} values discussed throughout this section, Figures 5.23– 5.26 collect them into a single bar chart per side, with each bar annotated by its corresponding Γ. The single-channel noise σ_i needed to convert Γ into CM_{VcTh} was estimated, per hybrid, from the average bin content of the per-strip pedestal calibration across all channels of that hybrid. These summary charts are the natural point of comparison between the burn-in and module-integrated dee/sector measurement.

Taken together, the dee/sector measurements reproduce every qualitative feature already established on individual modules during burn-in testing: the L/R anti-correlation, the opposite-sign skewness of the two hybrid sides, the partial (rather than full) CBC-to-hybrid correlation with the same L-above-R ordering, the

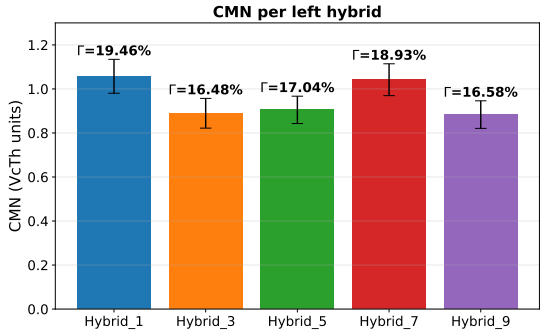


Figure 5.23: CM_{VcTh} per L hybrid, burn-in modules, each bar annotated with the corresponding Γ (in %).

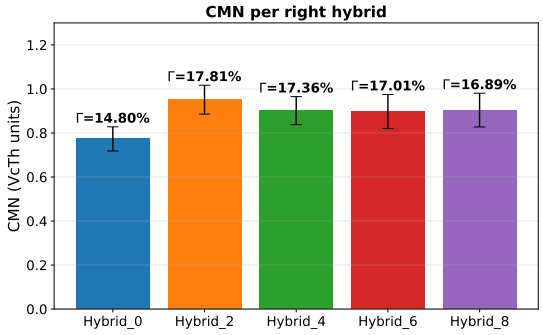


Figure 5.24: Same as Figure 5.23, for the five R hybrids.

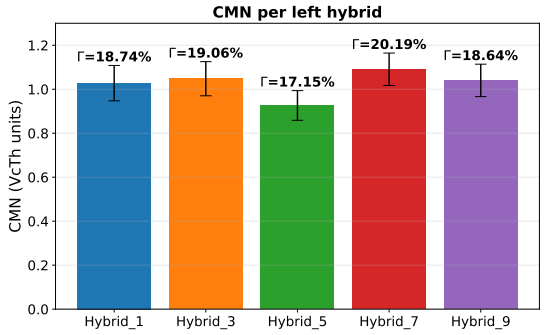


Figure 5.25: CM_{VcTh} per L hybrid on the Dee test structure, each bar annotated with the corresponding Γ (in %).

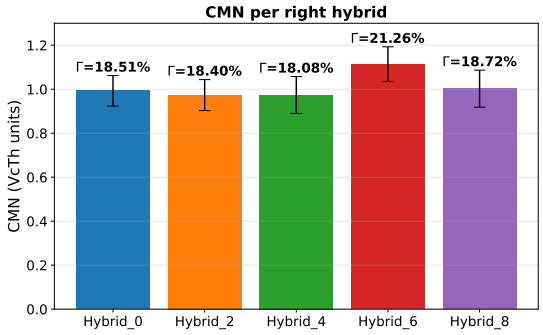


Figure 5.26: Same as Figure 5.25, for the five R hybrids.

CBC-index-dependent $CM_{V_{cTh}}$ trend, and the R-side mean-undershoot/variance-overshoot versus L-side mean-overshoot/variance-match pattern of the CBC correlated toy fit. This consistency indicates that the dominant common-mode noise mechanisms identified from single-module burn-in testing are properties of the module and hybrid design itself, not disrupted by mounting several modules together onto a dee.

A closer look at the hybrid-level summary charts (Figures 5.23–5.26), however, complicates this picture rather than simply confirming it. At the level of an individual hybrid, Γ increases on the dee relative to burn-in for nine of the ten hybrids, from a mean of about 17.2 % on burn-in (17.7 % on L-side, 16.8 % on R-side) to about 18.9 % on the dee (18.8 % on L-side, 19.0 % on R-side). This is exactly the signature expected if mounting modules together on a dee adds a genuine, shared common-mode contribution on top of each hybrid's own intrinsic noise. This sits in apparent tension with the module-level Γ, which, as shown above, instead decreases and narrows on the dee. This observation would stay consistent with the theoretical expectation only if the L/R anti-correlation ρ is itself more negative on the dee than in burn-in. Indeed, strong enough anti-correlation would more than compensate for each hybrid individually carrying more noise but still pull the combined module-level number down. After remeasuring ρ directly on the dee dataset, it yields -0.921 to -0.946 across the five modules (mean ≈ -0.93), against -0.57 to -0.93 (mean ≈ -0.75) for burn-in. This indicates that every dee module sits at or beyond the strongest anti-correlation seen on any burn-in module. Therefore, the apparent tension is resolved by understanding the two independent effects at play: a shared, additive contribution that raises each hybrid's own CMN once modules share dee services, and a genuine strengthening of the L/R anti-correlation that more than absorbs the previous effect once the two hybrids are combined into a module.

This still leaves open the origin of the shared, additive contribution itself. It could be genuinely common across different modules in which case different modules' 3σ hit counts should show a real, positive correlation with each other, event by event. Or it could be that each module's own internal dynamics simply shift in the same way under dee conditions, without any actual coupling between modules. The cross-module correlation test proposed above remains the way to distinguish these two possibilities. This test cannot, however, be carried out with the dataset used in this thesis.

5.2 PS modules

5.2.1 Burn-in tests CMN

The same burn-in procedure used for 2S modules (Section 5.1.1), i.e. individual modules, each on its own dedicated bench, isolating each module's intrinsic behaviour from any effect specific to sharing services with other modules, was applied to three PS modules (serials 10024, 10025 and 10026, referred to below as modules 1, 2 and 4). Because the strip and pixel sensors of a PS module have essentially decoupled readout chains and, as shown below, qualitatively different noise statistics, they are analyzed separately throughout this section, following the methodology of Section 4.3.3.

Strip sensors

Table 5.10 summaries the joint $(k_{\text{thresh}}, \sigma_{cm})$ fit at hybrid level. A clear R/L asymmetry is visible: R-hybrids only need a small correction to the 3σ threshold (an effective threshold of 3.12–3.28σ), while L-hybrids need a much larger one (3.66–3.97σ). For comparison, the equivalent correction for 2S strips never departed from the nominal 3σ by more than about 0.06σ on either side, for both R- and L-hybrids. So 2S shows essentially no R/L difference in this diagnostic at all. The fitted common-mode width follows the same pattern, at 0.94–1.50 on the R side against 2.55–3.26 on the L side, which means that the model requires a larger CMN for L-hybrids. Both diagnostics point the same way and are specific to PS modules: since 2S shows no comparable threshold-correction asymmetry, this is not simply a larger version of an effect already known from 2S, but a distinct R/L asymmetry particular to PS strips.

Table 5.10: Hybrid-level 3σ-tail joint fit of the effective threshold scale k_{thresh}, the common-mode width σ_{cm} and moment ratios across Right and Left front-end hybrids for the six PS burn-in strip hybrids.

Hybrid	Side	Real μ	Real σ	σ_i (per-strip)	k_{thresh}	eff. N_σ	$\sigma_{\text{cm,best}}$	χ^2	Toy μ / Real μ	Toy σ / Real σ
0	R	1.65	3.39	2.94	1.085	3.26	1.50	84	1.15	1.32
2	R	1.56	2.34	2.67	1.041	3.12	0.94	56	1.03	1.09
14	R	1.33	2.65	2.73	1.093	3.28	1.22	60	1.05	1.13
1	L	2.91	10.50	3.05	1.219	3.66	2.75	217	1.26	1.47
3	L	1.95	7.64	3.23	1.323	3.97	3.26	188	1.49	1.97
15	L	2.48	9.10	3.19	1.225	3.68	2.55	233	0.90	1.04

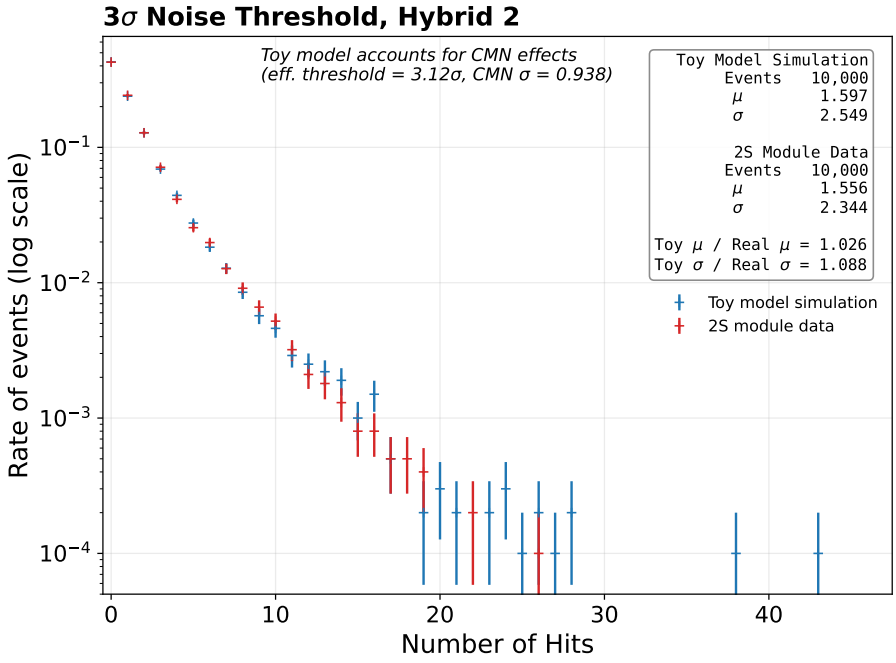


Figure 5.27: Hybrid-level 3σ strip hit-count data compared to the joint $(k_{\text{thresh}}, \sigma_{cm})$ toy model, for one R side PS hybrid. As in the equivalent 2S figures, the toy model's mean and standard deviation, and their ratios to the real data, are reported directly on the plot.

At module level (Table 5.11), the shared-CMN model ($\rho = 1$) consistently undershoots the real variance by a wide margin (toy-to-data standard-deviation ratio 0.43–0.64), exactly the structural limitation described in Section 4.3.3. As discussed there, treating the two hybrids as independent instead overshoots the variance by a comparable margin (ratio $\sigma \approx 1.5$–1.6, not tabulated here); this is itself indirect evidence, per the same argument, that the two PS hybrids are anti-correlated in the same qualitative way as for 2S.

Table 5.11: Module-level 3σ-tail joint fit for the three PS burn-in modules, using one shared $(k_{\text{thresh}}, \sigma_{cm,best})$ pair across both hybrids.

Module	**Real** μ	**Real** σ	σ_i **(per-strip)**	k_{thresh}	**eff.** N_σ	$\sigma_{\text{cm,best}}$	χ^2	**Toy** μ **/ Real** μ	**Toy** σ **/ Real** σ
1	4.56	10.62	3.00	1.041	3.12	1.20	417	0.79	0.57
2	3.50	7.66	2.95	1.057	3.17	0.91	402	0.68	0.43
4	3.81	9.15	2.96	1.056	3.17	1.21	464	0.87	0.64

Pixel sensors

Table 5.12 gives the two-component NBD mixture fit at hybrid level and Table 5.14 gives the same at module level. Two features stand out. First, the real Fano factor is large and highly variable confirming that macro-pixel hit counts are substantially over-dispersed relative to a Poisson process at every granularity studied. Second, the mixture weight w (the fraction of events drawn from the first, typically lower-mean component) varies widely between L-hybrids, from as low as 0.06 (Hybrid 15) to as high as 0.90 (Hybrid 1), indicating that the balance between the mixture's two populations is itself a module and hybrid specific property rather than a fixed detector-wide constant (Table 5.20–5.22). The toy-to-data ratios confirm the fit quality already anticipated in Section 4.3.3: the mean is reproduced to within a few percent in almost every case, and the standard deviation, while typically overshot by a modest 1–18 %, is far closer to the data than a single-Gaussian model could achieve given the same underlying over-dispersion. Note that, unlike the module-level strip fit, the module-level pixel fit does not show the same systematic variance undershoot. The absence of this strip-style undershoot for pixels is a fact about this model's flexibility, not evidence that PS pixel hybrids are more homogeneous than PS strip hybrids.

Table 5.12: Hybrid-level two-component Negative Binomial mixture fit for the six PS burn-in pixel hybrids. Fano is the Fano factor and is expressed as (Real σ)^2/Real μ.

Hybrid	**Side**	**Real** μ	**Real** σ	σ_i **(per-pixel)**	**Fano**	χ^2	**Toy** μ **/ Real** μ	**Toy** σ **/ Real** σ
0	R	50.25	33.96	2.30	22.95	1545	1.08	1.18
2	R	32.77	18.07	2.25	9.97	328	1.01	1.02
14	R	47.16	23.23	2.29	11.44	427	1.00	1.03
1	L	59.26	32.63	2.24	17.97	1390	1.02	1.05
3	L	82.11	16.83	2.23	3.45	183	1.00	1.01
15	L	92.90	19.24	2.24	3.98	1215	0.99	1.18

Table 5.13: Two-component mixture parameters (w, μ_1, μ_2, n_1, and n_2) fitted across Right and Left hybrids. w is the weight of the first component, n_1, n_2 are the two components' dispersion parameters (smaller n means more over-dispersed) and μ_1, μ_2 are the two means of the mixture.

Hybrid	Side	w	μ_1	n_1	μ_2	n_2
0	R	0.57	29.71	17.98	87.87	4.63
2	R	0.58	22.78	143.66	47.27	5.91
14	R	0.58	33.40	139.78	66.65	7.60
1	L	0.90	51.84	4.65	132.16	3.67×10^{11}
3	L	0.33	68.14	1.12×10^{11}	89.18	48.00
15	L	0.06	53.08	1.32	94.58	44.18

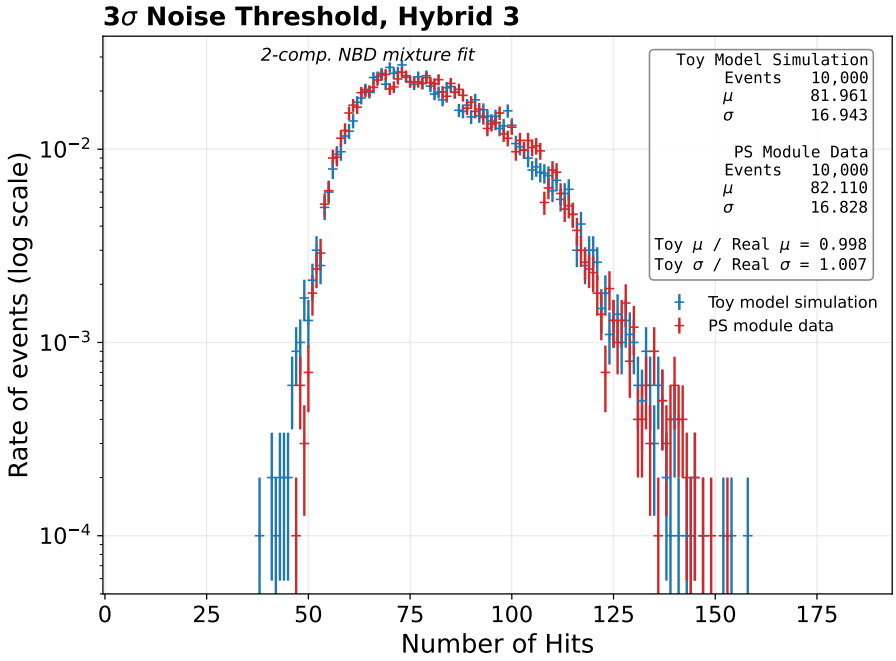


Figure 5.28: Hybrid-level 3σ pixel hit-count data compared to the fitted two-component negative binomial mixture, for one representative hybrid. Unlike the strip and 2S plots, no single common-mode width is reported on this plot since it is the mixture's shape, not a single scalar width, that characterizes the pixel over-dispersion.

Table 5.14: Module-level two-component NBD mixture fit for the three PS burn-in modules.

Module	Real μ	Real σ	σ_i (per-pixel)	Fano	χ^2	Toy μ / Real μ	Toy σ / Real σ
1	109.51	32.30	2.27	9.53	604	1.01	1.03
2	114.88	16.55	2.24	2.38	164	1.00	1.02
4	140.06	22.32	2.27	3.56	645	1.00	1.14

Table 5.15: Two-component mixture parameters (w, μ_1, μ_2, n_1, and n_2) fitted across the three PS burn-in modules.

Module	w	μ_1	n_1	μ_2	n_2
1	0.70	93.18	25.71	150.72	109.28
2	0.63	108.39	363.75	126.30	77.45
4	0.11	138.42	6.11	140.54	99.73

An independent finding: strip/pixel correlation

The ROOT files also contain a direct measurement of the correlation between a hybrid's strip hit count and its pixel hit count on the same event, which was checked across all six hybrids (Table 5.16). At the 3σ threshold, this correlation is weak (0.03–0.11): whatever drives a rare strip fluctuation and a rare pixel fluctuation on the same hybrid are, at this operating point, largely independent. Therefore, building this correlation into a joint generative toy model was not pursued. Two aspects of the measurement are nonetheless worth recording. First, the correlation is 3 to 8 times stronger at the lower occupancy-driven threshold than at 3σ (i.e Hybrid 2: $0.029 \rightarrow 0.404$), consistent with a genuine shared driver (i.e. power supply) having more opportunity to show up across many ordinary events than in the rare tail. Indeed, at the rare 3σ tail, events are likely dominated by threshold-specific, ASIC-specific idiosyncrasies (different amplification chains for strips vs pixels) that dilute the shared signal. Second, L-hybrids consistently show a higher 3σ correlation than R-hybrids (0.106–0.109 against 0.029–0.053). This L/R difference runs in the same direction as the effective threshold correction and the CMN σ_{cm} above, though nothing in this dataset establishes whether these are the same underlying effect or three independent consequences of whatever R/L hardware difference is present.

Table 5.16: Correlation coefficient ρ between a hybrid's strip hit count and its pixel hit count on the same event, at the 3σ threshold used by the toy models above and at a lower, occupancy-driven threshold, for the six PS burn-in hybrids.

Hybrid	Side	ρ (3σ)	ρ (occupancy-driven)
0	R	0.037	0.247
2	R	0.029	0.404
14	R	0.053	0.281
1	L	0.109	0.208
3	L	0.106	0.267
15	L	0.106	0.297

5.2.2 Sector tests CMN

The same three PS modules were also studied mounted together on the dee test structure, following exactly the methodology of Section 5.2.1. Two of the three modules (serials 10024 and 10025, referred to as Modules 1 and 2) are the same physical units tested during burn-in but the third module measured on the dee (serial 10011, referred to as Module 3) is a different physical module. Only Modules 1 and

2 are therefore directly comparable between the two tests. Module 3 is reported alongside them for completeness but does not have a burn-in counterpart in this dataset.

Strip sensors

Table 5.17 gives the hybrid-level joint fit on the dee dataset. The same qualitative R/L pattern already established for burn-in reproduces here: R-hybrids need only a small correction to the threshold (effective 2.99–3.41σ) and a modest common-mode width (1.02–1.49), while L-hybrids need a substantially larger correction (3.59–3.91σ) and a larger width (2.36–2.43). The effective-threshold ranges are close to the corresponding burn-in ranges on both sides, and R-hybrids' common-mode width is similarly stable; L-hybrids, however, show a narrower common-mode width on the dee than in burn-in (2.36–2.43 here, against 2.55–3.26 in burn-in). The reproducibility of the effective-threshold correction specifically, across two independent mounting configurations, supports the same kind of interpretation already reached for 2S, namely that this R/L difference is a property of the module and hybrid design rather than an artefact of a particular test bench. This is inferred here from a different diagnostic (the fitted threshold and common-mode-width asymmetry) than the skewness-based effect established for 2S, and unlike the threshold correction, the common-mode-width magnitude itself is not fully stable between the two tests.

Table 5.17: Hybrid-level 3σ-tail joint fit for the six PS dee strip hybrids.

Hybrid	Side	Real μ	Real σ	σ_i (per-strip)	k_{thresh}	eff. N_σ	$\sigma_{\text{cm,best}}$	χ^2	Toy μ / Real μ	Toy σ / Real σ
0	R	1.99	4.19	2.87	1.075	3.23	1.49	106	1.08	1.20
2	R	2.55	3.81	2.63	0.996	2.99	1.02	84	1.03	1.08
6	R	1.06	2.44	2.99	1.135	3.41	1.47	49	1.08	1.19
1	L	2.22	8.89	2.95	1.236	3.71	2.43	153	1.02	1.11
3	L	0.99	4.02	3.02	1.303	3.91	2.36	112	1.14	1.37
7	L	2.46	7.93	2.99	1.196	3.59	2.41	166	1.17	1.44

At module level (Table 5.18), the same shared CMN fit undershoot already identified for burn-in is present again, though somewhat less severe (toy-to-data standard-deviation ratio 0.71–0.91 on the dee, against 0.43–0.64 on burn-in). Since this undershoot is a direct consequence of forcing one shared $(k_{\text{thresh}}, \sigma_{cm})$ pair across two hybrids whose own fits differ, a smaller gap is consistent with the two hybrids' individual fits being less different from each other on the dee than on burn-in for these particular modules. It does not indicate that the underlying structural limitation described in Section 4.3.3 has been resolved.

Table 5.18: Module-level 3σ-tail joint fit for the three PS dee modules, using one shared $(k_{\text{thresh}}, \sigma_{cm})$ pair across both hybrids.

Module	Real μ	Real σ	σ_i (per-strip)	k_{thresh}	eff. N_σ	$\sigma_{\text{cm,best}}$	χ^2	Toy μ / Real μ	Toy σ / Real σ
1	4.21	9.42	2.91	1.037	3.11	1.21	254	0.92	0.71
2	3.54	5.16	2.83	1.023	3.07	0.94	89	0.98	0.91
3	3.52	8.00	2.99	1.074	3.22	1.36	353	0.91	0.81

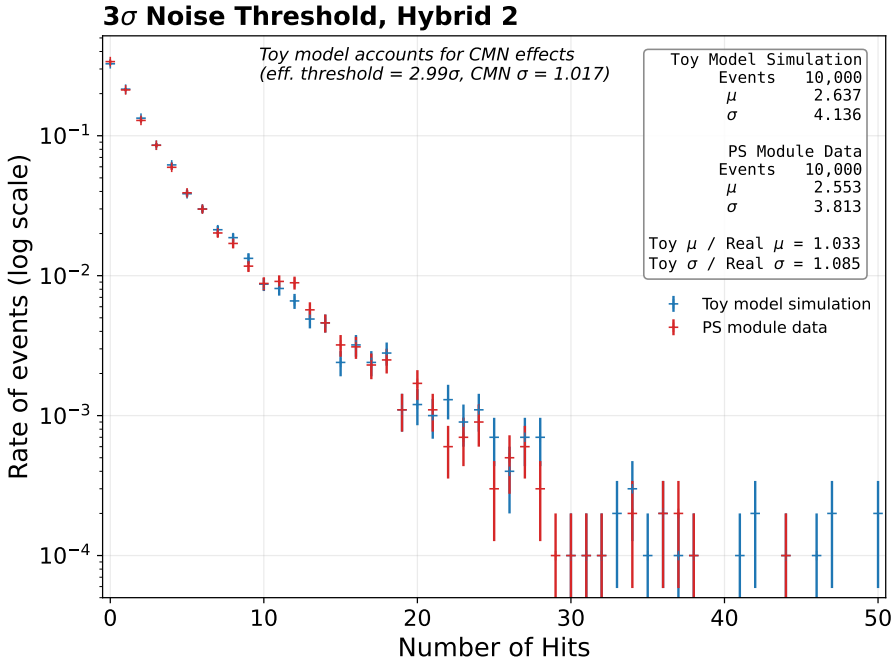


Figure 5.29: Hybrid-level 3σ strip hit-count data compared to the joint $(k_{\text{thresh}}, \sigma_{cm})$ toy model, for one R side PS hybrid. As in the equivalent 2S plots, the toy model's mean and standard deviation, and their ratios to the real data, are reported directly on the plot. The optimized values for the effective threshold and CMN width are also displayed.

Pixel sensors

Fitting the same two-component NBD mixture used for burn-in to the dee pixel data left a visibly worse residual shape at every granularity, and a three-component mixture was used instead throughout this section (Table 5.19), consistent with the general methodology of Section 4.3.3. The middle of the three components is fixed to a plain Poisson rather than left as a free negative binomial (Table 5.20): in the unconstrained fit, that component's own dispersion parameter already converged to values of order 10^6–10^{12} for most hybrids and MPAs checked. Therefore, it was already behaving as a Poisson term, so fixing it explicitly removes a degenerate, physically uninformative free parameter, without ever fitting worse than the unconstrained model and in several cases noticeably better.

Table 5.19: Hybrid-level three-component negative binomial mixture fit for the six PS dee pixel hybrids.

Hybrid	Side	Real μ	Real σ	σ_i (per-pixel)	Fano	χ^2	Toy μ / Real μ	Toy σ / Real σ
0	R	89.03	48.60	2.25	26.53	3961	1.07	1.01
2	R	61.60	25.54	2.22	10.59	363	1.01	1.04
6	R	89.98	38.57	2.22	16.54	4554	1.07	1.05
1	L	89.28	46.04	2.22	23.74	3800	1.07	1.00
3	L	99.85	62.01	2.20	38.51	2414	0.96	0.98
7	L	92.40	49.55	2.44	26.57	1983	1.02	1.03

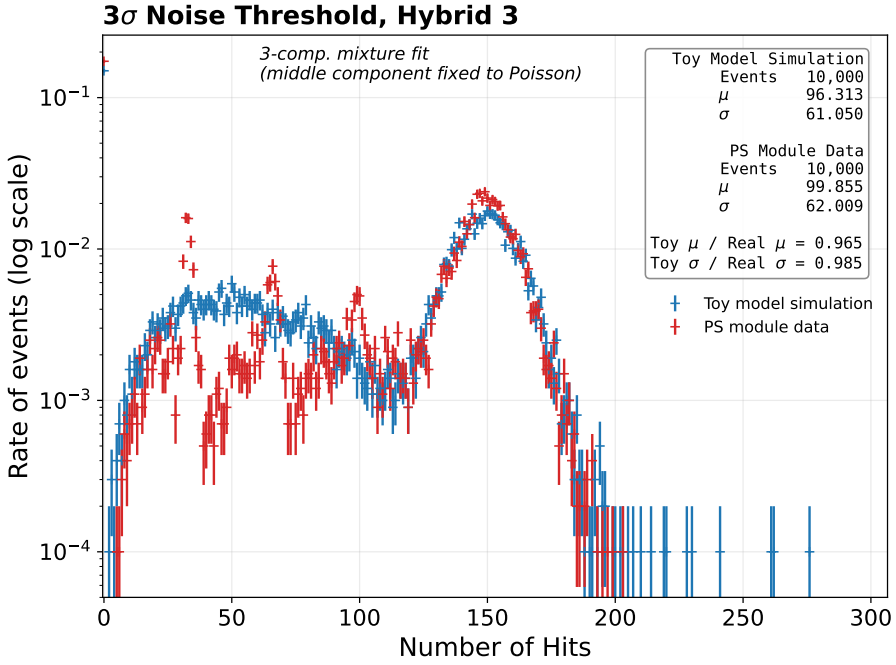


Figure 5.30: Hybrid-level 3σ pixel hit-count data compared to the fitted two-component Negative Binomial mixture, for one representative hybrid.

Table 5.20: Parameters of the three-component mixture model (w_k, μ_k, and n_k, including the Poisson component) fitted across Right and Left hybrids.

Hybrid	**Side**	w_1	w_2	w_3	μ_1	n_1	μ_2 (Poisson)	μ_3	n_3
0	R	0.0825	0.5181	0.3994	1.26	0.018	132.79	65.49	3.225
2	R	0.4686	0.0657	0.4657	43.89	32.681	127.23	72.21	16.565
6	R	0.0428	0.4493	0.5079	15.84	0.265	129.51	73.35	5.608
1	L	0.0546	0.5149	0.4305	4.40	0.077	132.44	61.67	4.078
3	L	0.1560	0.4911	0.3529	23.91	0.000	149.89	63.63	3.275
7	L	0.0931	0.4341	0.4728	5.02	0.068	136.76	73.42	3.409

The fact that a third component is needed on the dee but not on burn-in is itself a finding: every chip, hybrid and module checked on the dee improved by a factor of 1.04–3.06 in χ^2 when the third component was added, whereas the same addition gave essentially no benefit on the burn-in data. This is consistent with the dee's different environment genuinely introducing a third pixel-occupancy regime that the modules studied in burn-in did not show.

However, this toy model should not be read as a fully resolved model. Indeed, there remains substantial residual χ^2 for every hybrid indicating a genuine structural shape mismatch rather than small statistical noise. For instance, hybrid 3 still drives its fit to a degenerate boundary (a dispersion parameter n_1 pinned at $\sim 10^{-6}$) and the resulting toy model (Figure 5.30) visibly overpredicts the tail of the distribution. Precisely, for hybrid 3, the fitted mixture itself assigns roughly a 0.25% probability to hit counts above 190, corresponding to an expected 20-25 such events per 10,000 event toy sample. In contrast, the real data show no events above their own observed maximum of 203 hits across the same 10,000 events. This remaining discrepancy is reported as an open, unresolved limitation of the pixel sensor model on the dee dataset.

Table 5.21: Module-level three-component NBD mixture fit for the three PS dee modules.

Module	**Real μ**	**Real σ**	**σ_i (per-pixel)**	**Fano**	**χ^2**	**Toy μ / Real μ**	**Toy σ / Real σ**
1	178.31	52.24	2.24	15.31	1529	1.0	1.10
2	161.45	65.30	2.21	26.41	513	1.0	1.02
3	182.38	65.75	2.33	23.71	1209	1.0	1.07

Table 5.22: Parameters of the three-component mixture model (w_k, μ_k, and n_k, including the Poisson component) fitted across modules 1 to 3.

Module	w_1	w_2	w_3	μ_1	n_1	μ_2 (Poisson)	μ_3	n_3
1	0.1431	0.0925	0.7644	94.09	1.546	253.24	183.62	29.960
2	0.4102	0.2512	0.3385	94.21	4.323	203.26	210.45	45.157
3	0.1421	0.1859	0.6720	95.06	1.584	263.40	178.58	12.114

Comparison burn-in/dee

By restricting the comparison to Modules 1 and 2 (the only two PS modules measured in both tests), the strip data reveal a distinct left-to-right (L/R) asymmetry in the width of the 3σ-tail occupancy distributions. This behavior highlights a different type of L/R effect compared to the skewness asymmetry established for 2S modules, which was not evaluated for PS modules in this dataset. This width asymmetry persists on the dee for module 1, but largely disappears for module 2, where the real spread on the two sides becomes nearly equal (3.81 vs 4.02 from Table 5.17). So the PS strip effect is only partially reproducible between burn-in and dee. The pixel side shows an even less consistent pattern: module 1's Fano difference between the left and right hybrid is modest and shrinks between burn-in and dee (from a 28% to a 12% relative difference), while module 2's ordering reverses outright since the

Fano on the R side is higher than on the L side on burn-in (9.97 vs 3.45 from Table 5.12) but lower than on the L side on dee (10.59 vs 38.51 from Table 5.19).

Restricting to these same two modules also allows the common-mode amplitude itself to be compared directly, and here real differences emerge. For strips, the R-side fitted σ_{cm} is essentially unchanged between burn-in and dee (module 1: 1.50 to 1.49; module 2: 0.94 to 1.02), while the L-side value measurably narrows on the dee (module 1: 2.75 to 2.43; module 2: 3.26 to 2.36, a reduction of roughly 12–28%); the effective threshold correction itself stays essentially flat for both sides across the two tests, so this narrowing is specific to the common-mode amplitude rather than an artefact of a shifting threshold fit. This L side narrowing in σ_{cm} is consistent with the near-disappearance of module 2's real-spread asymmetry noted above. For pixels, the real data's own Fano factor increases from burn-in to dee for every one of the four hybrids, but unevenly: modestly on the R side (hybrid 0: 22.95 to 26.53; hybrid 2: 9.97 to 10.59) and sharply on the L side (hybrid 1: 17.97 to 23.74; hybrid 3: 3.45 to 38.51). This uneven growth is what drives module 2's R/L reversal: it is not that R-hybrids become less over-dispersed on the dee, but that L-hybrids become so much more over-dispersed that it overtakes R-hybrids.

Between the two tests, the strip and pixel sensors on the exact same hybrids behave in fundamentally different ways. For instance, common-mode width narrows for strips while over-dispersion grows unevenly for pixels, neither showing a consistent R/L pattern across both modules. This indicates that the dee's distinct operating environment does not act as a single, shared noise driver affecting both sensor types or both sides of a module in the same way. A further, related difference is that the pixel mixture itself needs a third component on the dee where two sufficed for burn-in. Unlike the amplitude-level differences above, this is a change in the pixel distribution's shape rather than its scale.

Chapter 6

Conclusion

The goal of this thesis was to characterize common-mode noise of the first CMS Phase-2 Outer Tracker modules assembled and tested at Louvain-la-Neuve, for both module technologies (2S and PS) and under both test conditions (burn-in, with a module operated alone, and the dee/sector test, with several modules integrated together and sharing services). The second objective was to build statistical toy models capable of reproducing the resulting occupancy distributions at the 50 % and 3σ working points.

2S modules. At the 50 % occupancy working point, every burn-in module showed a clear common-mode fraction Γ ranging from 2.5 % to 8.1 %, confirming CMN as a real, module-dependent effect rather than a fixed detector-wide constant. A parametric-bootstrap test also firmly rejected the pure-binomial (no-CMN) hypothesis for every module, and the residual, systematically-shaped bin-by-bin structure showed that even a single Gaussian common-mode term does not fully capture the data. Splitting the same analysis by hybrid revealed a clean, opposite-sign skewness pattern between the two sides of every module (L-hybrids positively skewed, R-hybrids negatively skewed, with no overlap between the two groups), and a monotonic trend of $\mathrm{CM}_{\mathrm{V_{cTh}}}$ with CBC index running in opposite directions on the two sides. Both observations are consistent with a voltage-drop asymmetry along the shared low-voltage distribution path on the service hybrid. At the 3σ tail working point, the two hybrids of a module were found to be strongly anti-correlated (ρ between -0.57 and -0.93), and the eight CBCs of a single hybrid only partially correlated with each other ($\rho_{\text{per-CBC}} \approx 0.38$ on R side, ≈ 0.46 on L side). Building both effects into the toy model resolved most, but not all, of a mean/variance tension left by the simplest, single-shared value model. A persistent ratio $\sigma_{cm,\text{best}}/\mathrm{CM}_{\mathrm{V_{cTh}}} > 1$ for every hybrid (larger on R side $\approx$ 1.9–2.5×, than on L side $\approx$ 1.3–1.6×) showed that the residual common-mode fluctuation itself has heavier-than-Gaussian tails. Repeating this full analysis on the dee test structure reproduced every one of these qualitative features (e.g the the hybrid anti-correlation, the opposite-sign skewness, the partial CBC correlation, and the side-dependent heavy-tail ratio) confirming that they are properties of the module and hybrid design itself rather than artefacts of a particular test bench. The module-level Γ nonetheless narrowed and shifted toward the lower end of the burn-in range on the dee, while the hybrid-level Γ instead grew slightly. Reconciling the two showed that the dee's shared services add a genuine common-mode contribution to each hybrid individually, while at the same

time strengthening the intrinsic L/R anti-correlation enough to more than absorb it once the two hybrids are combined into a module.

PS modules. The $\Gamma/\mathrm{CM}_{\mathrm{V_{cTh}}}$ framework developed for 2S strips does not carry over to PS, whose strip (SSA) and pixel (MPA) sensors required two distinct extensions. For strips, the 3σ threshold was found to systematically overpredict the real, quiet-state crossing rate. This mechanism was essentially absent for 2S, whose effective threshold never departed from the 3σ by more than about 0.06σ on either side. It was corrected by promoting the threshold itself to a jointly-fitted parameter alongside σ_{cm}. This revealed an R/L asymmetry substantially larger than anything seen for 2S (effective threshold 3.1–3.3σ on R side against 3.6–4.0σ on L side), together with a corresponding asymmetry in σ_{cm} itself. A candidate physical mechanism responsible for this asymmetry was proposed in Section 4.3.3: since the pedestal and per-channel noise entering the toy model are calibrated at the busy, 50 % occupancy S-curve working point, while the 3σ data itself is taken in a near-quiescent state, an activity-dependent droop of the local supply voltage would shift a channel's real baseline differently between the two regimes. k_{thresh} is exactly the kind of parameter that would absorb such a mismatch. If this droop differs between the two hybrid sides, an L/R asymmetric k_{thresh} would follow naturally. This explanation remains unconfirmed, however, since no direct measurement establishes that the SSA's threshold and pedestal-generation circuitry is actually sensitive to local supply level. At module level, the lack of any cross-hybrid correlation histogram in the PS datasets (unlike 2S, where this histogram made the L/R anti-correlation directly measurable) left a structural ceiling on this part of the analysis. In this analysis, a shared-parameter fit systematically underpredicts the real variance, while treating the two hybrids as independent overpredicts it by a similar margin. This is indirect but consistent evidence of an anti-correlation that could not be measured directly with the data available. For pixels, the real hit-count distributions combine over-dispersion (Fano factors between about 3 and 40 depending on hybrid and test) with a hard bulk floor that no Gaussian common-mode model can reproduce. As a result, a closed-form negative binomial mixture was adopted, using two components for burn-in data and three for dee data, indicating that the dee's different geometry introduces a genuine third pixel-occupancy regime absent in burn-in. Fixing the mixture's middle component to a plain Poisson (motivated by that component already converging there in the unconstrained fit for most hybrids) removed a degenerate free parameter without ever fitting worse than the original three-negative-binomial model. A residual, non-negligible χ^2 nonetheless remains for every dee pixel hybrid and reported here as an open, unresolved limitation of the pixel model. Comparing burn-in to dee for the two PS modules measured in both tests showed, again, that the dee does not act as a single, uniform driver of PS module noise. Indeed, the strip CMN measurably narrows on the L side while staying flat on the R side, whereas the real pixel Fano factor grows for every hybrid but far more sharply for L-hybrids than for R-hybrids.

Cross-cutting observations. Three themes recur across both module types and both test. First, an R/L asymmetry tied to the physical hybrid layout is present everywhere it was looked for, e.g. in 2S skewness, in PS strip threshold and common-mode width and in PS pixel Fano factor. Nothing in my thesis establishes that these

are all manifestations of one single underlying mechanism rather than several independent consequences of the same general R/L hardware or calibration difference; distinguishing between these possibilities remains open. Second, CMN is measurably non-Gaussian throughout my thesis. As a matter of fact, the $\sigma_{cm,\mathrm{best}}/\mathrm{CM}_{\mathrm{V_{cTh}}}$ mismatch for 2S and the complete inadequacy of a Gaussian model for PS pixels both show that a purely moment-based estimator such as Γ misses real structure in the shape of the fluctuation. Third, analysis at the module level can actively obscure the physics visible at hybrid level. Indeed, the 2S module-level Γ is reduced by hybrid anti-correlation well below what either hybrid shows on its own, and the PS module-level strip fit is structurally capped by R/L heterogeneity that the hybrid-level fits do not share. Consequently, hybrid-level results should be treated as the more reliable granularity throughout this analysis.

Limitations. Several questions raised in my thesis could not be resolved with the available datasets. It could not be established whether the PS R/L threshold asymmetry reflects a genuine hardware or environmental difference between the two hybrid positions. Whether the shared common-mode contribution observed on 2S dee hybrids reflects a genuine coupling between modules could not be settled since no cross-module correlation histogram exists in the 2S dee dataset that I used. Additionally, the residual shape mismatch in the PS dee pixel fits was substantially reduced but not eliminated. Finally, the PS analysis rests on only three modules per test, with only two of those measured in both tests, limiting how confidently the observed R/L and burn-in/dee patterns can be generalized beyond the specific modules studied.

Outlook. The most direct follow-up is instrumental. Adding a cross-hybrid correlation histogram to the PS modules would allow the PS module-level anti-correlation to be measured directly rather than inferred indirectly. It would also remove the structural ceiling currently limiting PS module-level results. Similarly, a dataset retaining a direct cross-module correlation histogram on the dee would finally make it possible to carry out the cross-module correlation test proposed in Section 5.1.2 and settle whether the observed dee/burn-in differences reflect real inter-module coupling. On the modeling side, the residual PS pixel mismatch on the dee could be revisited with explicit bounds on the negative binomial dispersion parameters to prevent the boundary-degenerate solutions observed for some hybrids. Most importantly, applying this methodology to a much larger sample of PS and 2S modules is a crucial next step. Such a study would determine if the observed R/L asymmetries and burn-in/dee differences are intrinsic to the hardware design (as the evidence gathered in my thesis suggests) or if they are driven by external factors, such as the module's position on the dee.

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